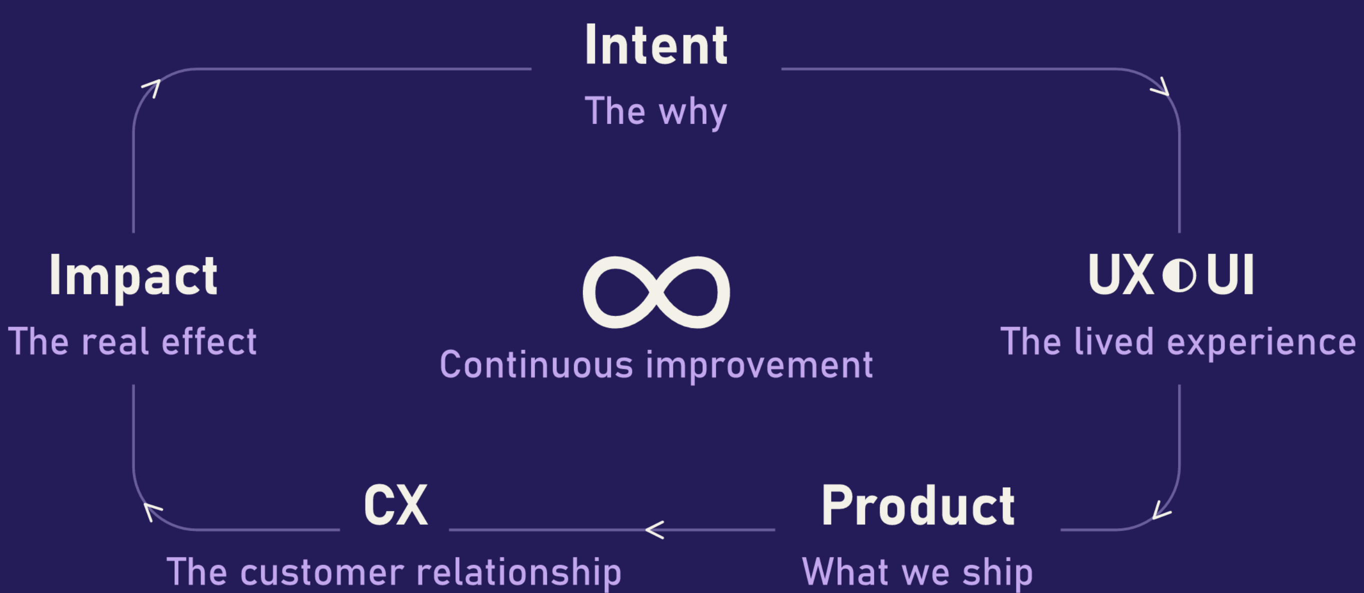
QUANTUM DESIGN

The hidden laws of our digital experiences



RÉMI ROSINSKI

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QUANTUM EXPANSIVE DESIGN

From UX ◐ UI Big Bang to the Continuous Expansion of Experience Design

UX General Relativity and UI Quantum Mechanics: The Perfect Unification — R.R.

Quantum Reference: UFE-R -202508081159-QED | August 8, 2025, 11:59 (UTC+2)

Preface: Explorers of Universes: From the Infinitely Vast to Human Experience

This book invites you on a bold journey where the laws of the physical universe meet the fundamental principles of Experience Design. To understand **Quantum Expansive Design (QED)**, we draw inspiration from the minds that have shaped our vision of the cosmos and interaction.

Here, we pay tribute to a few emblematic figures whose work has marked major turning points and formalized revolutionary concepts. However, let us recognize that, long before them, and within each of us, lies an innate capacity to shape our own experiences and those of others. From the first humans drawing in caves to anonymous artisans of every era, intuitive design is an aptitude anchored at the heart of our species, a spark of universal ingenuity. These visionaries, through their genius and perseverance, have formalized what was often implicit, leaving a precious legacy for future advancements.

Discover the foundations of **Relativity and Quantum Mechanics** with:

- **Albert Einstein:** Who revealed the **relativity of space-time**, redefining our perception of the universe and how each element influences the whole.
- **Marie Skłodowska-Curie:** Who unraveled the mysteries of radioactivity, showing the **invisible power hidden** at the heart of matter and the profound transformation it can generate.
- **Niels Bohr:** A pioneer of modern physics, he explored the **infinitely small world of atoms** and how discrete elements interact to form global behavior.

- **Chien-Shiung Wu (吳健雄):** Whose experiments revealed the **unseen and unsuspected subtleties of matter**, showing that even the most fundamental laws can harbor unexpected asymmetries.

Their revolutionary discoveries about the fundamental nature of reality illuminate our quest to make the invisible visible in design and to understand the underlying forces that shape every experience.

In the same way, to navigate the subjective space-time of user experience and reveal its hidden laws, we turn to the architects and visionaries of design in the digital age. Their work has humanized technology, structured our interactions, and given form to the invisible.

Discover the foundations of memorable **user experience (UX)**, where **user interface (UI)** becomes intuitive, with:

- **Donald Norman:** Better known as Don Norman, a major figure in user-centered design, who shed light on **user psychology** and the importance of intuitive design that meets human needs.
- **Susan Kare:** The artist who gave a **human face to the first computers**, transforming complex functions into familiar and accessible icons.
- **Jakob Nielsen:** Who systematized the **principles of usability**, making digital interfaces fluid and frictionless.
- **Brenda Laurel:** A pioneer of interaction and virtual reality, she designed **immersive and narrative experiences** by exploring the deep relationship between technology and human behavior.

Their contributions to experience design teach us how to structure and humanize interaction, building bridges between technology and human perception.

Together, these luminous minds guide us **to make the invisible visible**, understand the **deep structures** of experience, and shape **UX universes** that are fluid, memorable, and relevant. This book is an invitation to awaken the quantum designer within you, to perceive that every interaction is a particle of experience, and that your own contribution, however modest, is an essential modulation in the vast field of possibilities.

Introduction

This book was born out of a clear observation and an in-depth benchmarking of current practices: **traditional approaches to UX/UI are reaching their limits**. Too linear, too focused on fixed stages, they no longer capture the richness, complexity, or systemic dimension of modern digital experiences. That said, we recognize their fundamental importance and continued relevance in many current contexts.

It is precisely this observation that gives rise to **unprecedented opportunities** to rethink design. What if we viewed the experience as a constantly changing universe? What if, at the heart of this transformation, there was a **UX ◐ UI Big Bang**, a founding moment when the experience emerges from the invisible, dictating the structure and dynamics of everything that follows?

This increasing complexity of experiences is all the more evident when we consider the extraordinary diversity of human minds. Our digital systems, often designed for a majority cognitive mode of functioning (neurotypical), struggle to meet the needs of a population whose neurological functioning naturally varies (neuroatypical / neurodivergent). This disconnect forces many users to make considerable, often invisible efforts to adapt to interfaces that are not designed for them, leading to cognitive overload, increased fatigue, and feelings of exclusion. This reality highlights the imperative to move beyond a unified view of the user and embrace the full richness of human perception.

By combining systemic design thinking, advances in contemporary physics, and recent technological changes, a new theory is emerging: **Quantum Expansive Design (QED)**. This theory is at once **quantum** (exploring the infinitely small components of experience), **human** (rooted in interactions at our scale, between individuals, groups, and technologies), and **cosmological** (embracing the expansion of UX at the scale of systems and ecosystems).

Why combine UX, UI, quantum physics, and cosmology? Because they share the same obsession: making the invisible visible, understanding deep structures, and anticipating the behavior of systems. Experience design today faces challenges similar to those of fundamental science:

multiple points of view, unstable reference frames, and the intertwining of forms, functions, contexts, and perceptions.

In this book, you will find **8+1 laws** for thinking, designing, and experiencing UX differently according to the principles of QED. These laws will guide you **from the UX ◐ UI Big Bang (the origin) to the continuous expansion of Experience Design (its future)**, exploring all scales of experience: **from the smallest quanta to the largest galaxy**, including the human being at the heart of the interaction. The first of these laws, Law 0, will explore precisely this UX ◐ UI Big Bang, this fundamental moment and invisible presence that conditions all experience. The following 8 laws structure the expansion, diversity, and resonance of experience at all scales.

The Fundamental Challenges of Experience Design: A Contemporary Quest

Like fundamental physics, Quantum Expansive Design (QED) operates at the frontier of the unknown. While physicists attempt to unravel the mysteries of the universe at the very large and very small scales, quantum designers face their own ultimate quests: fundamental challenges that, if met, will redefine how we design and interact with the digital world. The quest to unify the forces of physics, for example, perfectly illustrates this relentless search for coherence. QED, on its own scale, pursues similar ambitions, seeking to transcend classical approaches to reveal a deeper understanding of the human experience.

The Unification of Scales: Beyond the Visible and the Invisible

- **The Challenge in Physics:** The theories of **General Relativity** (gravity, large scales of the universe) and **Quantum Mechanics** (behavior of matter and energy at the atomic scale) are phenomenal successes but incompatible with each other at extremes. Physics seeks to unify them in quantum gravity.
- **The Challenge in QED:** Transposed to design, this involves unifying experiences at all levels, which is a major challenge for designers. It amounts to merging **UX General Relativity**, which governs the overall structure and curvature of the user experience, with **UI Quantum Mechanics**, which explores the fundamental behaviors of micro-interactions in the interface. How can we guarantee perfect consistency and continuous resonance between the **macro** (the strategic vision, brand identity, entire ecosystem of a service, its founding values) and the **micro** (the UX quanta: a pixel, the label on a button, microtext, haptic feedback, i.e., a tactile response or vibration that the user feels when interacting with a device)? Our quantum gravity of design is the ability to design an experience where every detail of the interface (the UI particle) is entangled with the overall intention of the

experience (the curvature of UX space-time), even in the face of complex or unexpected uses. This involves understanding how a tiny change in the UI can generate gravitational waves (disturbances that propagate throughout the user experience, impacting their overall perception and behavior). Quantum Expansive Design also allows us to understand the gradation of experience, recognizing that each interaction, each moment, can be experienced with varying levels of intensity, resonance, and depth, ranging from the most fleeting interaction to the deepest immersion.

The Nature of Dark Energy and Dark Matter in Design

- **The Challenge in Physics**: Most of the universe is composed of **Dark Energy** (responsible for accelerated expansion) and **Dark Matter** (which has gravitational effects but remains invisible). Their nature is a fundamental mystery, pushing the limits of our understanding.
- **The Challenge in QED**: For the quantum designer, **Dark Matter UX** (or Dark Matter of UX by analogy) represents the **invisible but omnipresent factors** that massively influence the user experience without being directly observable or explicitly expressed by users. These include **unconscious biases, implicit expectations, deeply ingrained habits, unspoken cultural contexts, social norms, philosophical doctrines, religious beliefs, and other collective systems** of influence that shape perception and behavior, often on a subconscious level. **Dark Energy UI / System** (or Dark Energy of Design by analogy) could be the **hidden forces that accelerate or slow down the adoption and engagement of a product** (e.g., unpredictable viral dynamics, unexpected network effects, or subconscious resistance to change). QED aims to develop methods for detecting and modeling these invisible forces in order to consciously and strategically integrate them into the design process.

Beyond the Standard Model of Design

- **The Challenge in Physics:** The Standard Model of particle physics is incredibly successful in describing matter and its interactions, but it is incomplete. It does not explain phenomena such as gravity, dark matter, dark energy, or the mass of neutrinos, those nearly massless elementary particles that pass through the universe and even our bodies by the billions every second, interacting so little that they are true ghosts of the cosmos. In contrast, the photon, although also massless, is the messenger of light and mediator of electromagnetic interaction, and is everywhere our eyes and instruments can see. It illuminates our world of past events and carries information from the deep universe to us. Yet even it is not enough to explain everything.

- **The Challenge in QED:** The current Standard Model of UX/UI design, although effective and widely used to design functional interfaces and clear user journeys, is also incomplete. It is often too surface-oriented (UI, linear journeys) and struggles to explain or predict complex phenomena related to deep emotion, ethics, trust building, addiction, or long-term societal impact. QED proposes to go beyond these traditional approaches by introducing concepts that enable the design of more impactful and holistic (comprehensive and interconnected) experiences. This involves **developing new design laws for the particles of experience** that the traditional model cannot explain or fully account for, particularly by addressing their emotional resonance and ethical implications. This incompleteness is also blatantly apparent when it comes to the richness of human cognitive diversity: the standard model does not fully grasp how experiences are perceived and processed by minds with varying neurological functioning, leading to friction, invisible overload, and a sense of exclusion for many users. QED aims to transcend these limitations by designing for the entire spectrum of human perception

The Nature of Design Gravity: Implicit Influences and Experience Attractors

- **The Challenge in Physics:** Testing the nature of gravity with ever-increasing precision is an ongoing quest, through observations of gravitational waves and the study of black holes and the first moments of the universe.

- **The Challenge in QED:** The main challenge of QED is to understand the gravity of design, i.e., the invisible forces of attraction that guide user behavior and generate experience attractors. This involves accurately mapping these dynamics. We begin by analyzing natural affordances, the properties of an interface that intuitively suggest how to interact with it, making the action obvious without prior explanation. Beyond these visual and physical conventions, QED explores psychological patterns of desire, design patterns that draw on fundamental human motivations such as the need for validation, curiosity, or fear of missing out. These patterns trigger an unconscious motivation to engage, creating loops of emotional and cognitive attraction. Our study also extends to the most powerful phenomena of engagement: attention black holes, those self-referential loops (such as addictive notifications or endless scrolling) that captivate and retain the user in ways that are sometimes paradoxical, running counter to their conscious intentions. At the same time, we examine the singularities of experience that generate such deep, lasting, and even difficult-to-interrupt engagement that their attractiveness exceeds any conscious intention. To validate and refine our models in these complex interaction regimes, we observe the gravitational waves of design: massive user feedback, aggregated behavioral data analysis, cultural trend developments, and unexpected collective behaviors. In short, QED aims to map these forces of attraction and repulsion in the user experience in order to design more powerful, more intentional, and more ethical interactions.

Transition to the main body of the book

These powerful parallels show that Quantum Expansive Design is not just a new methodological approach, but a **new epistemology of design**. It is about embracing the idea that design, like physics, is an endless quest to understand the fundamental laws that govern the universe of human experience. By taking on these challenges, we can shape digital universes that are not only functional, but deeply intelligent, ethical, and resonant, that will adapt and evolve with humanity for decades to come. This book is an invitation to this adventure, beginning with an exploration of Law 0, the Big Bang of all experience.

Neutrality Note

This book explores experience design (UX/UI) through a systemic and expansive perspective, inspired by the principles of quantum physics and cosmology. Our goal is to offer a new framework for understanding and designing that is relevant to the current digital age.

The examples and use cases mentioned in this book have been selected for their **educational relevance and illustrative value** in terms of innovation, system resilience, performance, and user experience quality. They come from a variety of geographical contexts and sectors of activity, reflecting the richness and diversity of experience design applications around the world.

We have deliberately adopted a **neutral stance focused on the uses and principles of design** in order to preserve the educational, scientific, and inclusive objective of this approach. Experience design, as a cross-disciplinary field, transcends borders, ideologies, and tensions: it connects human, technological, and cultural contexts. Consequently, the choice of examples is devoid of any political, geopolitical, or ideological considerations. Our approach is to analyze the mechanisms of experience, wherever they occur, in order to draw universal lessons applicable to design.

Cosmic Analogy: From Nebula to Existing Product

The idea of comparing the formation of Earth and the emergence of life to the creation of a digital product is a powerful illustration of Law 0: Origin, Universal Entanglement, and Invisible Presence. It highlights the parallels between universal forces and the dynamics that drive the design of experiences.

The Initial Impulse: An Idea, a Great Bubbling

- **The Cosmos:** In the beginning, a cloud of gas and dust (a nebula) is pure potential, a diffuse field of energy. This gravitational impulse that agglomerates matter is analogous to the original impulse. The laws of physics are already intrinsically present, guiding this condensation toward the formation of a star and then, around it, planets. The Earth was not born by chance; specific conditions and universal laws allowed for its formation and its ability to support life.
- **The Digital Product:** Similarly, the creation of a digital product often begins with an embryonic idea, a need, a technological advance, competitive pressure, regulatory constraints, frustration, or a commercial opportunity. It is pure potential, an intention to solve a problem or create value. From this initial stage, market forces, technological constraints, and user expectations (future areas of convergence) are already intertwined with this idea, invisible but decisive for its evolution. This initial momentum brings together quanta of information through preliminary research, data analysis, and brainstorming sessions.

Accretion and Structuring: From Wireframe to Mockup, followed by Prototyping

- **The Cosmos:** Matter clumps together, forming planetesimals that collide and combine, creating increasingly larger and more complex bodies. Layers differentiate, and an atmosphere forms. There are intense, sometimes chaotic phases of formation, but they are guided by fundamental laws.
- **The Digital Product:** The idea takes shape. This is the phase of reflection, discussion, wireframing (a simplified, schematic structure of a digital product, often in grayscale), mock-up creation (a more complete, visual version of the same product), and then prototyping (adding interactions to the mock-up to simulate the experience and test it with users). UX quanta (interface elements, interaction flows) are agglomerated into an increasingly defined form. Initial tests reveal friction, requiring adjustments, and collisions of concepts that refine the structure. Each iteration is a refinement which, like the Earth cooling and stabilizing, makes the product more coherent.

The Emergence of Life and Development: Establishment and Continuous Iteration

- **The Cosmos:** Earth, in a favorable state, sees life emerge. It is a complex process of self-organization where simple elements combine to form living systems. Life adapts, evolves, and interacts in a delicate balance where each species plays a role. Among life forms, some develop capacities for perception, intelligence, and perhaps consciousness, in various forms. Humans occupy a unique place in this flourishing, but their survival remains intrinsically linked to the balance of the global ecosystem. In the future, other forms of consciousness may emerge, including those derived from artificial intelligence, prompting us to rethink our relationship with living beings, technology, and the complexity of the world.
- **The Digital Product:** Actual implementation and continuous iteration bring the product to life in the digital ecosystem. Interaction with users, field feedback, and usage data are like evolutionary forces. The product lives and adapts to changing needs and trends (Google, Apple, or Windows influenced by usage and social networks). Law 3: Fractal Expansion is evident here, as the product continues to expand into new dimensions of experience.

The Remnants of the Past and Natural Selection in Design

- **The Cosmos:** Geological strata, fossils, and the remains of ancient species or past climates are sediments of Earth's evolution. They bear witness to obsolete states, unsuccessful attempts, or adaptations that did not survive natural selection. These traces of the past allow us to understand the profound dynamics of evolution, the bifurcations, failures, and innovations that have shaped the diversity of life.

- **The Digital Product:** Similarly, the history of design is marked by obsolete products, features, or interfaces that have not stood the test of time, use, or competition. They become relics, drawings on the caves of the digital past, which can be studied to understand market dynamics and changing needs. Natural selection in design is based on adaptability to needs and trends, continued relevance, and the ability to deliver real value. Those that do not adapt disappear, but leave traces behind: best practices, mistakes to avoid, or inspiration for future innovations.

The Quantum Designer Challenge

The designer of the QED era is a true architect of the UX universe. Not only must they master traditional tools and techniques, but they must also develop a sensitivity to invisible dynamics, interconnections, and potential for expansion. This involves:

- **A holistic vision:** Understanding how every design element, no matter how small (a button, a color, haptic feedback), resonates throughout the entire experience and beyond, ensuring universal accessibility.
- **Acceptance of uncertainty:** Navigating constantly evolving ecosystems, where linearity is an illusion and observation alters reality.
- **The ability to design for expansion:** Thinking not of a point of arrival, but of a continuous path of growth, adaptation, and integration.
- **Ethical responsibility and sustainability:**
    - Quantum designers operate within **all types of organization**s: businesses, governments, associations, non-governmental organizations (NGOs), media outlets, academic institutions, and technological or military actors. Each pursues strategic goals that ensure its continuity: profitability, influence, financing of public services, maximization of donations, geopolitical influence, control of information, loyalty, or surveillance. This **force of financial, political, or operational sustainability is a universal constant in the field of experience**. It dictates the rules of the game, systemic constraints, and opportunities for action. It is neither good nor bad in itself, but the designer cannot ignore it.
    - Once engaged in a project, designers work within the rules and constraints specific to the organization that employs them, which may be chosen or imposed, depending on the context and possibilities of each individual. **Their responsibility is not to judge these requirements a priori, but to understand their logic and impact.** As far as possible, they strive to align their work with the aspirations of the organization while

seeking the most appropriate convergence between user needs and sustainability objectives. This involves focusing on the measurable results or impacts that the design seeks to generate on the behavior or state of the user and the organization, taking into account the actual room for maneuver and the constraints of the context.

   - It means recognizing the profound impact of each UX quantum on the human experience, and designing with diversity of human perspectives and resonances (needs, motivations, biases, emotions) in mind, as well as the principles of inclusion, sustainability, and the imperatives of viability. **Faced with the reality of black boxes** (whether non-public code, complex revenue models, or administrative costs), **designers will strive to create interfaces that enable transparency and discernment**. Users must be able to understand the essential mechanisms and make informed choices, even if the inner workings remain abstract or inaccessible.
   - Drawing inspiration from methodologies and visions from various sectors (technology, government, associations, etc.), while remaining aware of the biases or pressures that can be generated by marketing, lobbies, investors, or other stakeholders, the quantum designer understands that user value and organizational sustainability must be **prioritized in a balanced way**. The goal is to find the balance where the business or cause thrives thanks to an ethical and rewarding experience, not in spite of it. Through relevant indicators (KPIs), designers help make the best of the designed system visible to the public, thereby promoting trust and longevity.

- **A position of influence, not decision-making power:** Quantum designers operate within a complex ecosystem where final decisions are often made by a hierarchy subject to expectations, differing points of view, and even rivalries. While their role is to present facts, figures, and best practices to decision-makers, the reality on the ground and the laws of the system in which they operate take precedence. The QED designer is a key player in influencing and recommending, whose ethics are manifested in their ability to navigate these dynamics, defend UX principles and ethical imperatives, while recognizing the constraints and final decisions of the organization that employs them.

Le Quantum Expansive Design est une invitation à embrasser la complexité du monde numérique non pas comme un obstacle, mais comme une opportunité. C'est en comprenant les lois fondamentales de l'expansion, de la convergence des expériences, et **les forces incompressibles qui les animent**, que nous pourrons réellement façonner l'avenir du design, créant des univers numériques non seulement fonctionnels, mais profondément humains, adaptatifs et en constante expansion.

The Fundamental Laws of UX ◐ UI: A Theoretical Perspective

Discover the 8+1 Fundamental Laws that represent the indispensable codification of the unified theory of Quantum Expansive Design (QED). They are not mere design rules, but the inescapable principles that govern the creation, expansion, and endurance of every experience. Each law will reveal to you a fundamental principle about the nature of experience, transforming your role from executor to lucid architect of the invisible forces. This structured exploration begins here, at the very source of all intention.

LAW 0

Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence

At the heart of a latent interaction matrix, intention and interface become entangled, potentials of all experience. — R.R.

Short name:

Original Singularity

Principle: Before anything else, there exists a primordial singularity, an invisible foundation inseparable from all experience, where the potential of UX and UI is entangled and omnipresent.

In **Quantum Expansive Design (QED)**, Law 0 takes us back to the point of origin, before the visible interface, before conscious interaction. It holds that there is a **UX ◐ UI Big Bang**, a primordial singularity, a fundamental and invisible field that conditions the emergence of all experience. This is not the void but a matrix of potential where UX and UI are already entangled in their purest essence, a dark matter of experience, imperceptible yet omnipresent, which dictates the laws of the emergences to come.

Physical Concepts and UX/UI Analogies

This law draws on the concept of the **gravitational singularity** (the point of infinite density where spacetime curves to the extreme, as at the center of a black hole or before the Big Bang) and on the **cosmic microwave background** (the first light emitted after the Big Bang, a witness to the universe's initial conditions).

- **The UX ◐ UI Big Bang (original singularity):** This is the initial moment when the design intention and the potential of the experience are formulated for the first time. It represents the set of values, vision, mission, and ethical principles that will form the foundation of the experience. This is where the designer first puts skin in the game: the fundamental commitment to keeping that original intention aligned with the user's well-being and the system's ethical goals. Before the first pixel, there is an intention, a gravity that will attract and organize the matter and energy of the experience. This UX ◐ UI Big Bang is, by its very nature, a fundamental act of negentropy. **Negentropy** (or **negative entropy**) is a concept that describes a system's tendency to **maintain or increase its order and organized complexity**, resisting degradation. In the vast universe of the invisible, where entropy reigns (that is, the natural tendency toward disorder, the dissipation of information, and the breakdown of order), the emergence of an experience is a major organizing force. In this primordial act, the designer does not merely create elements: they reduce uncertainty, they order potential chaos, they structure information and interactions that would otherwise remain random signals. This is the genesis of a system that, from its very origin, stands against confusion and fragmentation.

- **The cosmic microwave background (the invisible presence):** Transposed to QED, the background is the witness to this matrix of potential where experience is entangled. It is at this level that **UX Dark Matter** and **UI / System Dark Energy** appear.
- **UX Dark Matter** (or, by cosmological analogy, the dark matter of UX) refers to the latent dynamics inherent to the user: unexpressed needs, diffuse emotions, unconscious motivations, and cognitive biases. Like an invisible mass that structures the universe, it exerts a gravity on the user's expectations and reactions without the user perceiving its direct source. A quantum designer learns to probe this dark matter to reveal the true drivers of human behavior. Yet this exploration can surface liar paradoxes: situations where the user's unexpressed needs (their UX Dark Matter) contradict their apparent behavior or their conscious statements, creating invisible friction in the experience.
- **UI / System Dark Energy** (or, by cosmological analogy, the dark energy of design) represents the invisible and often opaque driving forces that emanate from the system itself and shape adoption and engagement: personalization algorithms, hidden business models, retention strategies, or the influence of artificial intelligences. Like an unsuspected engine accelerating the expansion of the universe, this dark energy pulls and steers user journeys in specific directions, often without the user's knowledge. If it is not governed ethically, it is what can turn invisible UX into an experiential black box where the user loses control and freedom of choice. These forces can create contradictory self-reference loops: mechanisms that, by optimizing one objective (engagement), unintentionally generate negative side effects (dependence or withdrawal), thereby defying the original intention or the user's well-being.

- **Primordial entanglement:** From this origin, UX and UI are inseparable. There is no experience without the potential for a form, and no form without the potential for an experience. This initial entanglement is the basis of the wave-particle duality described in Law 1. Intrinsic to this entanglement and to this dark matter of experience, the first principles of **design homeostasis** appear (a system's ability to **maintain its internal balance and stability** despite external disturbances, through feedback mechanisms). Just as a universe or a living organism deploys mechanisms to maintain its equilibrium, experience, from the moment it emerges, is endowed with an innate capacity for regulation. These are the unspoken laws that allow an experience to keep its coherence, its functionality, and its fundamental relevance, even when faced with the first interactions or the first shocks of use. This is not an external corrective intervention but a property inherent to the design from its origin, a resilience programmed at the source, ensuring that the system can return to an optimal balance.

The Fundamental Analogy: The Design of DNA

To fully grasp the scale of the UX ◐ UI Big Bang and the intrinsic nature of entanglement and expansion, Quantum Expansive Design turns to the fundamental architecture of life.

- **Biology (DNA):** DNA is the **fundamental design system of the living**, a structure carrying **quanta of information** that, through its arrangement (the genetic UI), determines the structure and behavior of every organism (the biological UX). **Like the original singularity of the universe's Big Bang, DNA is that coded, invisible source that guides the emergence of phenomenal complexity from an infinitesimal point of origin.** This example reveals how entanglement and modularity are principles of experience from the tiniest source, capable of generating a whole universe of forms and functions. DNA is the tangible proof of an Invisible Presence that governs the emergence, structure, and regulation of experience, connecting us to the infinitely small that is contained within the great whole of our existence.

Implications for UX/UI Design

Law 0 has fundamental implications for the design process:

- **Design begins before the visible:** The QED designer does not merely create interfaces. They must dive into the singularity of intention: what is the deep vision of the product or service? What are its core values? What are the implicit biases of the team or the technology? It is by addressing these invisible questions that we lay the foundations of an authentic UX/UI. A well-considered information architecture, before the slightest graphic element, is an ordering act. It turns a potentially chaotic set of data and functions into a clear, logical structure, reducing uncertainty and cognitive load for the user.

- **Mapping invisible UX:** It is crucial to identify and understand the elements of invisible UX (UX artifacts, design legacies, cultural prejudices) that will shape the experience. This calls for in-depth research, ethical audits, and a constant questioning of our own perspectives as designers. Designing a design system with clear rules and reusable components is a form of homeostasis. It ensures that, despite the multiplicity of contributors and changes, the overall experience stays coherent and stable, like an ecosystem that maintains its biodiversity through internal regulation loops.
- **Designing intention and presence:** Design is not only an act of creation but an act of orchestration. The aim is to make sure that invisible intentions (the why behind the interface) align with the actual experience (the how, and the what the user perceives). Coherence between the invisible and the visible is the mark of a design under control.
- **The Quantum Designer: Value Catalyst and Strategic Navigator** Beyond simply building interfaces, the Quantum Expansive Design approach positions the designer as a value catalyst at the heart of business strategy. By drawing from the source of the UX ◐ UI Big Bang, that is, by grasping deep intentions, the dark matter of users, and the dark energy of the system, the designer gains the ability to shape an experience whose relevance endures over the long term. Their role is to ensure that this original singularity gives rise to an aligned experience that not only meets users' fundamental needs but also contributes measurably to the company's efficiency and performance goals. Armed with the principles of QED, the UX/UI designer moves beyond the role of executor to become a clear-eyed architect of value dynamics. Their grasp of invisible forces gives them a strategic position: that of showing how experience-centered design can directly support the retention of existing customers and the attraction of new ones. By creating fluid, intuitive, and ethical experiences, they establish clear correlations and causal links between design quality and improvements in engagement, conversion rates, and, ultimately, revenue growth. In an efficiency-minded company, UX/UI design, when approached through the laws of QED, proves to be not a mere cost but an essential building block that generates a lasting competitive advantage through the depth and quality of the experience it offers.

Use Cases

A few examples show how this origin and this invisible presence take shape in design:

- **Braun's design philosophy:** In the 1960s, under the direction of Dieter Rams, Braun did not merely create functional industrial products; it defined a design philosophy that became an invisible singularity. Dieter Rams is the author of the maxim ”Less, but better” (”Weniger, aber besser”), which can be read as ”Clean, but efficient.” That intention shaped every product, even decades later, influencing the UX/UI of companies such as Apple. The purity of the Braun experience is a legacy of that initial intention, one we still find today in expressions like ”Keep it simple.”
- **The concept of Trust by Design in digital products:** More and more, trust is not an add-on but a founding principle in the design of digital services. It is an invisible aspect, a fundamental intention, expressed through a transparent UX/UI, clear privacy policies, and respectful user control. When trust is designed from the singularity onward, it permeates the whole experience, even when it is not explicitly visible.

Ethical Vigilance, Caveats, and Contextual Note

Law 0 carries a fundamental ethical responsibility. Invisible UX can be fertile ground for **unconscious biases** (cultural, algorithmic) and **intentional manipulation** (dark patterns, addictive design). If the initial intentions of the UX Big Bang are malicious or opaque, the visible experience will be corrupted by them. QED calls for a **search for authenticity** from the origin: making the intentions and the dark matter of the experience conscious and transparent, so that design serves the user's well-being rather than hidden or harmful goals. For the designer, this implies a constant skin in the game: an active, owned responsibility for the long-term consequences of their creations, including the handling of inherent paradoxes and potential self-reference loops. Without that deep commitment, the original singularity of the design risks giving rise to an unbalanced universe of experiences.

LAW 1

Entanglement, Complementarity, and Duality in UX ◐ UI

Form is experience, experience is form. — R.R.

Short name:

Principle of Entanglement

Principle: UX and UI entangled, complementary, a wave-particle duality

In Quantum Expansive Design (QED), the traditional split between User Experience (UX) and User Interface (UI) is an illusion. Like the photon, the elementary component of light, which appears sometimes as a wave and sometimes as a particle depending on the observation, UX and UI are not distinct entities but two inseparable facets of one and the same reality. Law 1 holds that UX and UI are fundamentally entangled and complementary, their interaction creating a powerful synergy. UX is the emotional flow, the overall feeling, the why. UI is the tangible expression, the functional elements, the how. Neither can fully exist without the other, and their interaction is a unified field of information where every change in one instantly affects the other, and where the whole (the overall experience) is greater than the sum of its parts.

Physical Concepts and UX Analogies

Wave-particle duality is a pillar of quantum mechanics. A subatomic particle (smaller than an atom) can behave like a wave (diffuse, influencing a field) or like a particle (localized, measurable). This analogy is powerful for UX and UI.

- **UX** is the **wave**: it represents the continuous flow of the experience, the general feeling, the intuition, the user's emotional journey. It is a diffuse quality, not localized at a single point, but one that permeates the whole interaction. It is the field of intention, the why the user interacts.

- **UI** is the **particle**: it appears as tangible or latent elements, such as a button, an icon, a color, a typeface, or the anticipation of a voice or gesture command (movement, eye blink, sign language, or any other bodily expression). These interaction points, whether localized or spatial, are measurable and form the supports the user acts upon. UI is the field of expression, the how the user interacts.

This fundamental entanglement of UX and UI operates within dynamic structures we can call **Complex Adaptive Systems (CAS)**. A CAS is a system made up of many agents (such as users, products, services, and increasingly artificial intelligences) that interact and continuously adapt to one another and to their environment. The overall behavior of a CAS cannot be fully predicted from the sum of its individual parts alone. It is marked by the emergence of unexpected properties and by its capacity to self-organize.

This adaptive complexity echoes the workings of **neural networks**, whether biological (the human brain and its capacity for **learning**) or artificial (**Artificial Intelligence / AI**). In these networks, a multitude of connections and weights are activated at once, representing a superposition of possible lines of thought or decisions. Learning, whether human or machine, is a continuous process of modifying these connections, allowing the system to adapt and optimize its responses to new observations or data. So the user is not a fixed point but a neural network in constant reorganization, and AI-based systems mirror that same dynamic of continuous learning and adaptation of their internal states.

Impossible figures: a metaphor for UX duality and coherence

Artists such as Maurits Cornelis Escher popularized impossible figures. Among them is the Penrose stairs, devised by Lionel Penrose, who drew on the work of his son Roger Penrose, known for his impossible triangle. These optical illusions are fascinating because each segment of the figure looks perfectly logical and buildable when viewed up close. Yet once the whole figure is perceived, the observer realizes that the overall structure is fundamentally contradictory and impossible to build in three-dimensional reality. **These figures, which defy our macroscopic perception, resonate with the idea that what is impossible in classical reality can be a fundamental state in the quantum world, inviting designers to reconsider the limits of the possible in design.** This phenomenon is a powerful metaphor for the challenges of UX/UI duality. Each UX quantum, a button, a navigation element, a micro-interaction, can be designed and optimized to perfection locally (the UI). But if these elements are not orchestrated coherently within the Convergence Fields of the overall experience, the user ends up facing an impossible experience.

Take the example of a complex government website where the user has to complete a procedure (such as renewing a permit or claiming a benefit). Each form field, each submit button, each information page may be individually well designed and functional. Locally, everything seems fine. Yet the overall journey can turn out to be labyrinthine and illogical: the user is forced to re-enter redundant information across different pages, runs into generic errors with no clear explanation, or is redirected to unrelated sections, making the procedure as a whole impossible or senseless to complete. This is further amplified by the complexity of multi-device access: browser and screen specifics (size, orientation), a mix of technologies, and adjustments specific to each platform, often designed and developed by separate teams or departments. The experience then resembles an infinite staircase that never reaches the desired floor.

This gap between local perfection and global impossibility underscores how important it is to go beyond the mere sum of the parts, in order to design an experience that, like a buildable physical structure, holds its coherence at every scale.

Informational integrity in this context means that the information carried by UX and UI is a coherent whole. Each particle of the UI (a design token in the Figma web app, a color, a corner radius, a transition) is not just a graphic unit. It carries a **quantum charge**: an emotion, a cultural norm, a memory of use, a piece of the UX flow. The entanglement between form and substance is immediate and inseparable. Believing you can design UX without UI (or vice versa) is a reductive view that ignores this entangled reality.

UX Implications

Understanding this entangled duality leads to several fundamental implications for designers:

- **A holistic approach and UX/UI co-design:** It is imperative that UX and UI teams work in symbiosis from the earliest phases of the project. The segregation of roles, where UX ships a wireframe and UI dresses it up graphically, is obsolete. Co-design lets the flow of the experience (UX) and the interactive elements (UI) be woven together, ensuring that each UI component is an intentional manifestation of the overall experience. In a complex adaptive system where AI plays a part, co-design must widen its scope. Design teams have to collaborate with AI specialists to understand how the algorithms learn and shape the experience, and how to design interfaces that reveal these emergent dynamics rather than hide them.
- **Design Tokens as UX/UI Quanta:** Design systems and their design tokens (color, typography, and spacing variables, etc.) become perfect examples of this entanglement. A design token is not merely a visual property; it is a fundamental particle that carries

a UX intention. The same color can be perceived differently across cultures, situations, or disabilities (for instance, red can mean danger in some contexts, luck in others, or be indistinguishable from green, brown, or orange). Its definition and its use must be conceived to serve both visual coherence and the overall emotional feel.

- **UI as a manifestation of emotion:** Each UI element must be designed not only for its function but also for its emotional impact. The micro-interaction, the transition, the responsiveness of a button are all particles that, through their arrangement and behavior, create the wave of the user experience.

Use Cases

To illustrate this entanglement, consider applications where UX and UI are fused in exemplary fashion:

- **Apple iOS (mobile operating system):** The Apple ecosystem is an emblematic example. The UI (the look and feel, that is, the appearance and feel of use, the animations, the fluidity of gestures) is so intimately tied to the UX (ease of use, intuitiveness, emotional satisfaction) that the two are hard to tell apart. Every visual and interactive element is designed to reinforce an experience of fluidity and ease. The transitions between apps, the touch responsiveness, the uniformity of the icons, all of it creates a coherent wave of experience where form and function are inseparable.
- **Google Material Design System:** Although it is a design system rather than a single application, Material Design is built on the principle of entanglement. It provides detailed guidelines not only on the visual aspect of components (UI) but also on their behavior, their animations, and the way they should interact to create a coherent and predictable user experience (UX). Broadly and deeply integrated into the Android ecosystem, it has also extended beyond it, influencing the web and even iOS apps through frameworks like Flutter. Virtual materials have physical properties (shadows, depths) that directly shape perception and interaction, fusing form and function.

Ethical Vigilance, Caveats, and Contextual Note

UX ◐ UI entanglement also carries ethical responsibilities. When form and substance are inseparable, it becomes easier to manipulate the experience. **Dark patterns** are the most blatant example, where the UI is deliberately designed to deceive or manipulate the user into making decisions they would not consciously make (forcing a newsletter sign-up, making unsubscribing difficult, hiding costs). A deceptive UI generates a negative, frustrating, even abusive UX. Beyond these deliberate manipulations or design blunders, Law 1 highlights the concept of **experiential decoherence**. This decoherence shows up concretely when the fundamental entanglement between UX and UI is broken, causing perceptible dissonance and friction. For instance, an interface can be visually appealing (good UI) but illogical or hard to use (bad UX), or, conversely, a well-thought-out user experience can run into a confusing or unappealing interface. This break in the entanglement generates frustration, confusion, and a loss of engagement, because the user perceives a fundamental disunion between what they expect and what they see and interact with. Failing to apply this law would result in products that look current but are dysfunctional, or, conversely, functional but aesthetically obsolete. Law 1 thus reminds us of the need for **intentional transparency**: the UI should clearly and honestly serve the UX, with no manipulative ulterior motive and with perfect coherence between form and substance.

LAW 2

Scalar Field and Contextual Modulation of UX

The invisible is not absent; it is experienced in depth. — R.R.

Short name:

Contextual Scalar Field

Principle: Every point of the interface is an emotional particle subject to contextual modulation, a scalar field that varies with the emotional temperature and the environment of use.

In the universe of Quantum Expansive Design (QED), experience is never static. Law 2 invites us to perceive each element of an interface, whether a button, a text, an image, or even a haptic vibration, as an **emotional particle**. These particles are not isolated; they are bathed in a **scalar field of influence** whose temperature varies constantly with the user's context, their emotions, their physical environment, and even their cognitive state. Design no longer merely presents information; it must modulate its behavior and its appearance to resonate with this dynamic environment, creating a tailored, real-time experience.

Physical Concepts and UX Analogies

In physics, a **scalar field** is a field that assigns a single value (a scalar) to each point in space. A simple example is the temperature of a room: each point has a specific temperature, and that temperature can vary. In UX, we can think of the user experience as a scalar field of emotional and cognitive states.

- **The emotional field of UX:** Each interaction point (a pixel, a word, a sound) carries an emotional value that varies with context. An error message will be perceived differently depending on whether the user is calm or stressed. The color of a button can inspire trust or urgency depending on the user's state of mind. The scalar field represents this **ambient emotional temperature** that shapes the user's perception and reaction.

- **Contextual modulation:** Just as the air temperature is modulated by heating or air conditioning, the design of the experience must be able to modulate its behavior. This means the interface should not be rigid but should adapt dynamically to variations in the emotional and contextual field.
  - **Emotional temperature:** The user's affective state (frustration, joy, anxiety, focus, distraction, motivation, boredom, sense of belonging, disappointment, expectations, impatience, fatigue, a sense of safety or threat), and so on.
  - **Environment of use:** The place (noise, light, brightness, social context), the moment (day, night, length of use, interruptions), the physical conditions (weather, temperature, posture, fatigue), the device (mobile, desktop, television, watch, augmented/virtual/mixed-reality glasses or headset, computing power), the connection quality, the type of interface (gesture, haptic, touch, or voice), the user's level of mastery, and the software and hardware compatibility of the platforms.

This law calls for a design that not only anticipates but actively responds to these variables, in order to offer a more fluid and empathetic experience.

UX Implications

Understanding the scalar field and contextual modulation has deep implications for design:

- **Design responsive to mood and conditions:** Interfaces must be designed to adapt. This goes beyond mere responsive design and extends to an adaptive experience. A navigation app, for example, does not just show a map; it adjusts its alerts, its text size, or even its vocal tone according to the vehicle's speed, the traffic, or the driver's apparent stress level.
- **Dynamic personalization and emotional micro-interactions:** Rather than static personalization (based on pre-saved preferences), this is real-time personalization. Micro-interactions (small animations, haptic feedback, sounds) become levers to fine-tune the emotional field, confirming an action, easing a frustration, or subtly directing attention.
- **Adapting to slow networks and low-cost screens:** In contexts where resources are limited (slow connection, low-powered devices), design must degrade gracefully or optimize itself. This is a form of field modulation: the experience must remain usable and pleasant even when the field (the technical conditions) is less favorable. A lightweight UX is not a discount UX but a UX intelligently adapted.

- **Linguistic and cultural adaptation (local modulation):** The contextual modulation of UX also extends to linguistic and cultural conventions, which are a fundamental aspect of the environment of use. Design cannot afford a rigid, one-size-fits-all approach. It must instead adapt to the specifics of each language to ensure a fluid and natural experience. This means handling variable text length. Some languages, such as German, Finnish, or Dutch, are known for their very long compound words, which raise challenges of word breaking and display width. Others, such as Spanish, Portuguese, or French, often need more words to express an idea, which lengthens sentences considerably and calls for adjusting the size of fields and labels. Conversely, the concision of Chinese or Japanese requires less space, allowing for denser interfaces. Likewise, the writing direction is a crucial variable. Designing for right-to-left reading, as in Arabic, Persian, or Urdu, for instance, calls for a complete reorganization of layouts, navigation direction, and element placement. A contextually modulated interface will dynamically adjust spacing, typography, and layout to optimize readability and comprehension, reflecting a true cultural and cognitive resonance with the local user. This is a direct application of Law 2 to deliver a tailored experience that respects not only preferences but also the intrinsic constraints of language and culture.
- **UX Teleportation: The Seamless Contextual Jump:** Modulating the UX scalar field opens the door to a radically fluid form of experience we call UX Teleportation. Rather than a linear, step-by-step navigation, UX Teleportation describes an experience's ability to carry the user from one point to another, or from one state to another, so transparently and so contextually relevant that it is perceived as an instant jump, with no friction or disorientation. This phenomenon occurs when the interface anticipates the user's deep needs and intentions and modulates the interaction field to create magic shortcuts. It is not a simple hyperlink but an intelligent transition where relevant information, past context, and the user's future expectations are entangled to create perfect continuity, even across different applications or platforms. UX Spatiotemporal Portals are those key touchpoints where such teleportation becomes possible, acting as fluid gateways within or between experiences. UX Teleportation is the ultimate expression of a design able to respond actively to the variables of the field, offering an experience that is not only fluid but predictive and near-instant, minimizing cognitive friction and mental load.

Use Cases

Several applications brilliantly illustrate Law 2:

- **Waze (navigation app):** This example illustrates contextual modulation. Waze does not just provide a route: it adapts its information in real time (traffic jams, accidents, roadblocks) according to the driver's situation and the road environment, thanks to artificial intelligence algorithms and user contributions. The interface evolves dynamically with speed, traffic density, and collective signals, modulating a scalar field of information and attention. For example, the app adjusts the type and frequency of voice or visual alerts when there is danger, and adopts a more minimalist display at high speed (to maximize readability), while also reducing cognitive load and supporting a more fluid and intuitive navigation.

- **Apple Vision Pro (mixed-reality headset):** This headset offers a three-dimensional interface that adapts in real time to the physical environment and the user's interactions. It can move from augmented reality, where digital content blends into the real world, to fully immersive virtual reality, all controllable by voice, gestures, gaze, or an intensity dial on the side of the headset. Thanks to a sophisticated set of sensors (high-resolution cameras, eye tracking, ambient light sensors, LiDAR-type depth sensors), the device modulates the presentation of digital content in real space, creating a fluid and contextually responsive interactive field. The user interacts naturally through eyes, gestures, and voice. The interface dynamically adjusts brightness, spatial sound, window layout, and the level of immersion according to context: ambient light, posture, current activity, the presence of other people. Even though the headset physically conceals the user's eyes, it can represent them digitally on the outside to allow a natural exchange of gazes with those around them. This adaptive, multisensory design embodies Law 2: dynamic modulation, turning each interaction into a response that is sensitive to the immediate context.

Ethical Vigilance, Caveats, and Contextual Note

The ability to modulate the experience according to the user's context and emotions is powerful and, as such, calls for ethical vigilance. Aggressive personalization could quickly drift into manipulation. The scalar field should not become a tool for surveillance or intrusive micro-targeting. The collection of emotional or contextual data should be transparent, and the user should retain control over the level of personalization and the information shared. Avoiding the filter-bubble or echo-chamber effect, which would lock the user into a one-dimensional view of the experience, is also worth keeping in mind.

LAW 3

Fractal Expansion UX ◐ UI and Differentiated Evolutionary Rhythms in Design

Design unfolds its universe in many constellations, never in a straight line. — R.R.

Short name:

Fractal Expansion

Principle: UX unfolds like an expanding fractal universe, with evolutionary speeds that vary across interaction layers, user profiles, and contexts.

In Quantum Expansive Design (QED), experience does not progress in a linear, uniform way. Law 3 reveals that experience design, like fractals in mathematics and natural structures (trees, mountains, clouds, rivers, flowers, fruit, lightning, arteries, veins, bronchi, chromosomes, and so on), reproduces itself at different scales with similar patterns but expands with distinct dynamics. UX is a constantly expanding system, where some areas evolve rapidly (micro-interactions, fleeting trends) while others remain stable for longer (founding principles, well-anchored user journeys). Understanding these **differentiated evolutionary speeds** is crucial to designing systems that are resilient, scalable, and suited to the complexity of real use.

Physical Concepts and UX/UI Analogies

Fractal geometry describes shapes whose patterns repeat at different scales, creating infinite complexity from simple rules. These patterns, found both in nature and in famous mathematical figures, rarely show perfect symmetry, yet their underlying structure holds. We see them notably in the creations conceived by mathematicians such as Helge von Koch with his snowflake, Wacław Sierpiński with his triangle, or Benoît Mandelbrot, the father of modern fractals, with his sets.

- **UX/UI as fractal:** The user experience and the interface manifest through repeating patterns:
  - At the scale of the **quantum component (micro / very small)**: a pixel, a design token, a specific micro-interaction.
  - At the scale of the **systemic frame (meso / intermediate)**: a navigation pattern, an onboarding process, a visual style guide.
  - At the scale of the **founding expansion (macro / very large)**: the information architecture of a super-app, the ecosystem of a global service, the company culture. Each level, while distinct, resonates with the principles of the others, just as the branches of a tree mirror the shape of the whole tree, and must be thought of in design terms.
- **Differentiated evolutionary speeds in design:** Just as the expansion of the universe may have varied in speed across different eras, UX/UI design does not evolve at the same pace everywhere.
  - **Fleeting trends** (a fashionable visual effect, a type of micro-interaction) correspond to rapid, almost instant expansions that can also disappear quickly. Here the design is agile and reactive.
  - **Usability principles** or **fundamental user journeys** (logging in, buying a product) evolve more slowly, are more stable, and form the fundamental fabric onto which fast innovations are grafted. Here the design is robust and durable.
  - **User profiles** (experts vs. novices) and **cultural contexts** (an emerging market vs. a mature one, a sober and minimalist interface vs. a rich and colorful one) also influence the speed at which new experiences and designs are adopted and integrated.

This law highlights the need for a strategic vision that manages both rapid innovation and the stability of foundations within the design process.

Implications for UX/UI Design

Recognizing Fractal Expansion and differentiated evolutionary speeds leads to major practical implications for UX/UI designers:

- **Modular and scalable architecture:** Design must rely on modular systems, letting building blocks (UI components, UX features) be added, modified, or removed without destabilizing the whole. This modular growth is the key to an experience's resilience. A

Design System is a direct application of this principle, offering reusable components that can evolve independently while keeping overall coherence. Atomic Design, for example, proposes a methodology for organizing these blocks from the smallest (atom) to the most complex (molecule, organism, template, page), ensuring a structured approach to this modularity.

- **Managing experience and interface layers:** This is about distinguishing what, in UX and UI, must be stable (the core of the experience, the fundamental needs, the basic UI elements) from what can be experimental and quick to evolve (the innovative peripheries, the micro-interactions). This allows for tailored development cycles: agile sprints for innovation, more measured cadences for the stability of the foundations.
- **UX/UI design adaptive to contexts and uses:** Interfaces and journeys must be able to adapt to different adoption speeds and to the specifics of users. A power user (an expert, advanced user who seeks efficiency and customization) will appreciate complex new features and shortcuts (fast evolution), whereas a new user will need clear, stable journeys (foundations). UX/UI design must therefore adjust to the speed of each audience segment and each environment. This can take shape notably through intelligent systems, such as an interface Simplification / Enrichment Slider (detailed in the section on the Sliders and Regulation Systems of QED, located just after Law 8), letting the user consciously move between a graceful degradation of the design and an aesthetic enrichment, according to their needs and their device's capabilities.

Illustrative Example: From the Quantum MVP to Fractal Deployment → The Visual Evolution of Design

A **MVP (Minimum Viable Product)** is the version of a new product that allows the maximum amount of validated learning about users with the least effort, by offering essential value to early adopters in order to guide future development.

To make the power of Fractal Expansion concrete and to understand how design handles both **graceful degradation** and **aesthetic enrichment**, let us imagine the birth and evolution of a digital experience from its most fundamental state: the **Quantum MVP**.

Step 1: The Quantum MVP → The Superposed State (Law 0 and Law 1)

- **Scene:** When the application chosen by the user launches, an online store for example, an empty screen materializes. A pure **black or white background**, with no visible element.
- **QED interpretation:** This is the **Quantum MVP** in its purest state, a form of original singularity (Law 0). It contains all the potential of the experience, but nothing is manifested

yet. The UI tends toward zero (Law 8), but the UX is an infinite field of possibilities. It is the wave of the experience in its simplest form, with no tangible UI particle to constrain it.

Step 2: The First Manifestation → Action and Quanta (Law 1 and Law 2)

- **Scene:** The user interacts (a click, a tap, a movement, a spoken word, a gaze with a prolonged blink, a specific gesture detected by a sensor, or even a thought interpreted by a neural interface), reflecting a diversity of very varied fields of use, from controlling a desktop application to navigating an intelligent system in augmented reality, or even interacting with autonomous agents. This interaction produces a single action: for example, the empty screen changes color, a central point appears, a light launch sound plays, according to a choice set by the user or adapted by the system.
- **QED interpretation:** The first UI particle is born (the empty screen changes color, a central point appears, a light launch sound plays), manifesting a UX quantum (the system's responsiveness). This action creates a first emotional Scalar Field (Law 2), a surprise, a validation. The UX wave begins to collapse toward a perceived reality, tending toward an adjusted state suited to the user.

Step 3: Modulation and the First Choices (Law 2 and Law 3)

- **Scene:** Instead of a simple color change, the action initiates a color choice, combined with voice assistance to guide the user. The user can thus choose between 2 options that split the screen in two vertically or horizontally depending on the screen orientation, with the choices ”Go to the store” and ”More choices.”
- **QED interpretation:** Here we introduce **Contextual Modulation** (Law 2). The experience is no longer unique; it begins to adapt to the first interactions. The voice assistance, like the presentation of two distinct choices, illustrates the system's ability to modulate the experience in real time. Adding texts and options represents the appearance of new Quantum Components (Law 3), carriers of information and emotion, which enrich the system's quantum charge. This structuring prepares the fractal expansion to come, each choice able to open onto new branches of the experience.

Step 4: Cell Division → Initial Fractal Expansion (Law 3)

- **Scene:** The user chooses ”More choices”; depending on the screen orientation, the 2 parts split horizontally or vertically, each offering 2 distinct actions, thus forming 4 zones of the same size. For example, top left "New arrivals" and top right "Deals," then bottom left "Log in," and bottom right "More choices."

- **QED interpretation:** This is the manifest beginning of **Fractal Expansion** (Law 3). The system unfolds while preserving the fundamental structure (here, the binary division). Each new part is a sub-fractal that inherits the parent's properties but develops its own UX/UI quanta. These divisions create new Convergence Fields where the user's attention is directed.

Step 5: Continuous Deployment → Differentiated Evolutionary Rhythms (Law 3 and Law 7)

- **Scene:** Each new section can in turn divide, menus can appear, icons can be added, lists can be generated. More complex features (forms, image galleries, interactive maps) are integrated. The screen, instead of being empty, becomes a rich and functional interface, such as a dashboard, a social app, or an e-commerce site.
- **QED interpretation:** The system develops while respecting the **Differentiated Evolutionary Rhythms** (Law 3): the core (the fundamental functions, the basic visual coherence) stays stable, while the surface layers (themes, plugins, specific features) evolve rapidly to meet diverse needs. The interface becomes a Living and Symbiotic Network (Law 7) of interactions among its components.

Step 6: Spectrum Adaptability → Graceful Degradation and Aesthetic Enrichment (Law 4 and Law 6)

- **Scene:** Now that the interface has unfolded and grown richer, the system demonstrates its fundamental ability to adapt to the many realities of users and their environments. Take the same complex web application, now fully functional, and observe how it modulates:
  - **Graceful Degradation:** The user accesses the application from an old smartphone, out in the countryside with an unstable 2G connection and dazzling outdoor brightness. The interface responds by showing a simplified, lightweight version: high-resolution images are replaced by compressed versions or placeholders (text substitutes for images), animations disappear, secondary options are tucked into compact dropdown menus, and visual contrast is automatically increased for better readability in strong light. Navigation stays fluid and the essential features remain immediately accessible, ensuring the user can complete their main task without frustration, despite the constraints of the Environment Agents (network, brightness) and Machine Agents (old smartphone, screen).

    - **Aesthetic Enrichment:** The same user, back home, opens the application on their desktop computer with a large 4K screen, a stable fiber connection, and a quiet environment. The interface then unfolds in all its richness: subtle animations guide the eye, interactive charts display detailed data, advanced options are directly visible, and a more nuanced color palette is used. Immersive audio elements and fine haptic feedback (via a compatible device) can also enrich the experience, offering an immersive and aesthetically rewarding interaction that fully exploits the agents' optimal capabilities.

- **QED interpretation:** This is the manifestation of **Inter-Agent Accessibility** (Law 4), where design adjusts dynamically to the cognitive plurality of users and to the fluctuating capabilities of Machine Agents and Environment Agents. At the same time, it reveals the **Transcendental Adaptability** of the experience (Law 6), where design transcends limitations to offer optimal quality when conditions allow. The system maintains its Design Homeostasis (Law 0), its fundamental integrity, while modulating its perceived complexity and its sensory richness. The UX stays coherent even when the visible UI varies greatly, proving that the initial Quantum MVP did indeed contain this potential for graceful degradation and aesthetic enrichment.

Use Cases

Several examples illustrate the fractal nature and the differentiated evolutionary speeds of UX/UI design:

- **Figma (collaborative design tool):** Figma embodies the management of differentiated evolutionary rhythms. The core of its interface and its basic design features stays stable and reliable, ensuring a robust foundation. Yet its ecosystem of plugins and widgets allows for rapid, constant fractal expansion, giving users the ability to add tools and automations that evolve at high speed, customizing their workflow without disrupting the stability of the core application.
- **WordPress (CMS / Content Management System):** As a CMS, WordPress is an example of fractal design. The core of the system (publishing functions, databases) is relatively stable and evolves slowly. Yet the ecosystem of themes and plugins offers rapid expansion and infinite modularity. Each WordPress site is a fractal manifestation of the design, sharing the same core but adapting to specific uses with features and interfaces that evolve at various speeds, according to the needs of each user or company.

Ethical Vigilance, Caveats, and Contextual Note

Fractal design and differentiated evolutionary speeds call for particular ethical vigilance from UX/UI designers. It would be wise to avoid creating fractals of exclusion, where some parts of the experience (or interface innovations) evolve too fast or are too complex for certain users, creating inequalities of access or understanding. Graceful degradation of UX/UI is well worth applying with care, so that the experience stays fair even on less capable devices or networks. We should also make sure that rapid innovations do not destabilize the foundations of the experience, guaranteeing a degree of stability and predictability for all users.

LAW 4

Inter-Agent UX ◐ UI Accessibility and Cognitive Plurality

Designing for a single target is forgetting all the others. — R.R.

Short name:

Cognitive Universality

Principle: Experience is accessible or not according to a cognitive plurality and an interaction among human, machine, and environment agents, requiring an inclusive and adaptive design.

In Quantum Expansive Design (QED), accessibility goes beyond mere compliance with standards. Law 4 holds that access to the experience is an **inter-agent and cognitive** phenomenon. Experience is built at the intersection of users' cognitive and sensory abilities (cognitive plurality), technical systems (machine agents), and the constraints or opportunities of the physical and digital environment (environment agents). A truly accessible UX/UI is one that adapts to this diversity of entangled agents, offering an inclusive and equitable experience for everyone.

Physical Concepts and UX/UI Analogies

In physics, the idea of **interaction between particles and fields** is fundamental. Each particle interacts with its environment and with other particles, mutually altering their states. In QED, we extend this concept to the interaction among different agents:

- **Human Agents:** Each user is an agent with a **unique cognitive plurality**. This includes not only their different sensory, cognitive, and motor abilities, but also their learning preferences, their emotional states, and their socio-cultural context (languages, norms, conventions, belonging to communities or groups). The experience appears and is interpreted differently for each person, shaped by these many facets.

To illustrate this fundamental cognitive plurality and its implications for design, it is essential to understand the varied neurotypes that make up the population. Neurodiversity is a concept that recognizes these neurological variations as natural forms of human diversity. In design, this means designing beyond a **standard neurotypical** functioning, to embrace a broader spectrum of perception and interaction.

The main **neuroatypical / neurodivergent** profiles, which shed light on the richness of cognitive plurality and the challenges design must address, include:

- **ADHD (Attention-Deficit/Hyperactivity Disorder):** Difficulties with attention, impulsivity, organization, and/or hyperactivity, varying from person to person. It can also come with heightened creativity or hyperfocus.
- **ASD (Autism Spectrum Disorder):** Affects social communication and interaction, with the possible presence of repetitive behaviors and sensory specificities.
- **Dyslexia:** Persistent difficulties with reading, writing, and sometimes spelling.
- **Dyspraxia (Developmental Coordination Disorder/DCD):** Difficulties with motor coordination and the organization of movement, which can show up as clumsiness or handwriting difficulties (dysgraphia).
- **Dyscalculia:** Difficulty understanding numbers and mathematical concepts.
- **Dysgraphia:** A disorder involving letter formation and handwriting.
- **Tourette syndrome:** The presence of involuntary motor and/or vocal tics.
- **Sensory Processing Disorder (SPD):** Trouble interpreting and regulating sensory information (noise, light, textures, and so on).
- **Certain chronic mental-health conditions** (such as Obsessive-Compulsive Disorder/OCD, bipolar disorder, Post-Traumatic Stress Disorder/PTSD) are sometimes included in a broader discussion of neurodiversity because of their distinct and lasting neurological implications.

Understanding these diverse manifestations of cognitive plurality is fundamental for the quantum designer. The experience must be able to appear and be interpreted differently for each person, shaped by these many facets and not constrained by a single model.

- **Machine Agents:** Digital systems are themselves complex agents, **made of several complementary layers**. The **hardware** is the set of material elements (processors, sensors, screens, and so on) that condition the power, speed, and possible modes of interaction. The **software** brings together the programs and applications that orchestrate functionality, structure the interface, and support the other components. The **algorithms** represent the rules and logical processes that govern the systems' behavior, ensuring the processing, prioritization, and personalization of information. **Artificial Intelligence (AI)**, often embedded in the software, brings capacities for learning, adaptation, and autonomous decision-making, allowing systems to act proactively and contextually. Finally, the **network** is the communication infrastructure that links these components, conditioning the latency, availability, and synchronization of services. The capabilities and limits of each component, such as computing speed, network latency, or modes of interaction (voice, visual, haptic, and so on), directly affect the accessibility and quality of the user experience. Moreover, these machine agents can act as informational mediators (content filtering, assistants, search engines) or as developmental agents (learning, self-organizing, customizable systems), thereby modulating the experience and its accessibility in real time.

- **Environment Agents:** The context in which the experience unfolds also acts as an agent that modulates it. This includes the **physical** environment (ambient brightness, noise level, physical accessibility of the place) and the **digital** environment (type of browser, operating-system version, network bandwidth). These environmental agents dictate constraints and opportunities that affect how the experience is perceived and used.

Accessibility then becomes the design's ability to orchestrate these many-sided interactions, so that the experience emerges meaningfully for each human agent, whatever the other agents involved. It is not about easing access but about designing an experience that **unfolds differently** according to this plurality.

Implications for UX/UI Design

Law 4 has deep implications for designing inclusive experiences:

- **Universal and adaptive design:** This is about designing interfaces that are not rigid but able to modulate their presentation and interaction according to each user's specific needs, the machine's capabilities, and environmental constraints. This includes support for different text sizes, contrasts, and input modes (keyboard, voice, touch), and compatibility with assistive technologies.
- **Inter-agent interaction (human-machine-environment):** Designers must think beyond the simple human-screen interaction. How can AI ease access for a visually impaired user? How does an interface respond in a noisy environment through voice? How can contextual data (location, weather) personalize the experience to make it more relevant and accessible?

- **Accounting for cognitive plurality:** Design must anticipate and account for the different ways information is processed by the human brain (visual, auditory, kinesthetic learning). This calls for multi-sensory design approaches and options for cognitive personalization, to reduce mental load or ease comprehension.

Use Cases

A few examples show how inter-agent accessibility and cognitive plurality are built into QED:

- **Flutterwave (fintech, Nigeria):** This fintech (a company using innovative technologies to improve or automate financial services) is a model of inter-agent accessibility in Africa. Its UX/UI is specifically designed to be simple, intuitive, and usable by SMEs and merchants operating in varied, often constrained technical environments (slow networks, low-powered devices). Rather than a dynamic, real-time adaptation, Flutterwave's design builds in from the start the need to work robustly and accessibly on modest infrastructure, with payment solutions accessible even without technical expertise, thanks to no-code interfaces (allowing creation without writing a single line of code) and low-code ones (requiring a minimum of code for customizations), and cross-platform compatibility. This illustrates the resilience of the experience in the face of local technical constraints.
- **Government integration applications:** Applications such as those that help refugees or newcomers in a country must account for an immense cognitive and contextual plurality (language barriers, cultural differences, stress, limited access to technology). Their UX/UI design must be ultra-adaptive, offering instant translations, simplified information modes, and intuitive navigation for users under constraint.

Ethical Vigilance, Caveats, and Contextual Note

Inter-agent accessibility and cognitive plurality call for vigilance at every moment. Any collection of data about cognitive abilities or environmental constraints should be carried out with the user's informed consent. There is a real risk of ”accessibility bubbles” emerging, where certain profiles remain marginalized by insufficiently adaptive designs. It would be wise to prevent any form of implicit discrimination and to ensure that accessibility features do not become tools of relegation, but instead integrate smoothly and respectfully, safeguarding the dignity and autonomy of each individual. QED commits to deconstructing the design biases liable to restrict some users' experience, and to promoting a truly inclusive approach, attentive to the diversity of human and technological agents.

LAW 5

Emotional Resonance and Interaction Memory in UX

We forget the interface, but we remember the emotion. — R.R.

Short name:

Memory Resonance

Principle: Experience leaves an emotional and cognitive imprint (interaction memory) that resonates with future interactions and shapes the user's overall experience field.

In Quantum Expansive Design (QED), interaction is not limited to a one-off exchange of information. Law 5 reveals that every user interaction, whatever its granularity (from the simplest click, through voice commands, visual or gesture interaction, all the way to complex user journeys), deposits an **imprint** in the user's memory. This imprint, charged with emotion and cognition, acts as an **interaction memory** that modulates future perceptions and expectations. UX/UI is therefore a cumulative process where the quality of each touchpoint resonates well beyond the present moment, shaping an evolving **overall experience field** for the individual.

Physical Concepts and UX/UI Analogies

This principle draws on **resonance** (the amplification of one wave by another) and on **memory in complex systems** (such as the shape memory of materials or the neural plasticity of the brain).

- **Emotional resonance:** A past positive (or negative) experience with a brand or an interface can create a resonance that amplifies or dampens the emotion of the present interaction. A well-designed, fluid UI generates a feeling of satisfaction that strengthens with each use, creating a virtuous circle of resonance. Conversely, repeated frustrations can cause a destructive resonance, where even a small current error triggers a strong emotional reaction because of the accumulation of past negative experiences. After several frustrating experiences with a poorly placed button, even an improved version of the interface can trigger wariness or hesitation, proof that interaction memory durably shapes the perception of the design. In short, the coherence of interactions and emotional stability are key factors in long-term loyalty and engagement.

- **Interaction memory (the quantum imprint of experience):** Each interaction leaves a trace or a quantum imprint in the user's mind. These imprints are not mere factual memories but patterns of expectation, emotional associations, and cognitive shortcuts. Interaction memory includes usage habits, learned conventions (for example, the meaning of an icon), and conditioned emotional reactions. It is an unconscious field of information that modulates the perception of the current design. It is crucial that, even as the UI evolves, the coherence of interactions and the emotional quality remain stable to avoid any memory dissonance, which connects to the notion of user-experience homeostasis.
- **The overall experience field:** The set of these interaction memories and emotional resonances makes up the user's overall experience field with a product, a service, or a brand. This field is dynamic, constantly updated by new interactions, and it determines the user's long-term relationship with the design.
- **Hybrid interaction model:** Modern interfaces combine several modalities (touch, gesture, voice), which enriches the palette of memory imprints and emotional resonances. This modal diversity multiplies the channels through which the experience can leave its imprint, making design all the more complex and powerful in its capacity to influence the user's interaction memory.

Implications for UX/UI Design

Law 5 underscores the importance of every detail and of long-term coherence:

- **Designing emotional journeys:** Designers must think not only about tasks but about the emotions generated at each step of a journey. Each micro-interaction, each message, each visual or auditory feedback should be an opportunity to create a positive resonance. Error handling, for example, must be designed to minimize frustration and avoid a negative resonance.
- **Coherence and predictability for interaction memory:** A coherent UX/UI helps build a positive interaction memory. Repeating familiar patterns (visual, behavioral) reinforces learning and reduces cognitive load. Design Systems are essential here, because they

ensure that interface elements respond predictably, reinforcing the trust and intuitiveness of the experience.

- **Managing moments of transition:** Context changes (moving from one device to another, from one feature to another) and interface evolutions are critical moments. Careful handling of these transitions is essential to maintain the coherence of the experience and avoid memory dissonance. The point is to ensure that the user does not feel lost and that their patterns of expectation are not abruptly contradicted, thereby preserving trust and intuitiveness.
- **Careful onboarding and offboarding:** The first and last interactions are crucial for interaction memory. A successful onboarding lays the groundwork for a positive resonance. A respectful offboarding, even when the user is leaving, can leave a positive imprint that may encourage a future return or a recommendation.
- **Personalization as a lever for resonance:** Adapting the interface to the user's preferences and behaviors (personalization) is a powerful catalyst for positive emotional resonance. By offering an experience that feels made for the individual, the design strengthens their engagement, deepens the interaction memory, and consolidates long-term loyalty.

Use Cases

A few examples highlight the impact of emotional resonance and interaction memory:

- **Video game experiences (consoles, mobile games):** Games are masters of emotional resonance. Visual feedback (particle effects, animations), audio feedback (music, victory sounds), and haptic feedback (vibrations) are designed to create positive resonance loops, triggering dopamine spikes (dopamine being a substance secreted by the body and associated with pleasure and reward) and reinforcing interaction memory to encourage replayability and satisfaction.
- **Music streaming services (Spotify, Deezer, Apple Music):** These platforms excel at creating personalized playlists and musical discoveries. Beyond function, the UX/UI is designed to generate a feeling of personal connection and pleasure. Content curation (searching, selecting, organizing, and sharing relevant content from various sources), precise recommendations, and the fluidity of listening build an interaction memory anchored in pleasure and discovery, making the user loyal to the platform.

Ethical Vigilance, Caveats, and Contextual Note

Manipulating emotional resonance and interaction memory is delicate ethical ground. Interfaces designed to create dependence (through excessive feedback loops or variable rewards) or to exploit emotional vulnerabilities are examples of overreach (FOMO, Fear Of Missing Out, an artificial sense of urgency). QED calls for using knowledge of resonance and memory to build healthy, empowering, and respectful experiences, not to manipulate or trap the user in undesirable interaction patterns. Transparency about personalization mechanisms is essential.

LAW 6

Transcendental Adaptability in UX ◐ UI and Systemic Openness

The unexpected is already a component of the real. — R.R.

Short name:

Transcendental Adaptability

Principle: UX/UI is an open, adaptive system, able to transcend its initial limits by integrating new information and continuously adjusting to emerging realities.

In Quantum Expansive Design (QED), experience is not a closed system. Law 6 holds that UX/UI is inherently an **open system**, in constant dialogue with its environment. It must show **transcendental adaptability**, that is, the ability not only to react to changes but also to integrate unforeseen information (from the user, from data, from AI, or from the environment) in order to evolve beyond its initial design. It is an invitation to design experiences that are not static but living, modular, and in perpetual expansion, mirroring the dynamic complexity of the real world.

Physical Concepts and UX/UI Analogies

This principle draws on the **thermodynamics of open systems** (which exchange matter and energy with their environment) and on the **evolution of biological systems** (which adapt and innovate in the face of new conditions).

- **Open system:** Unlike a closed system that would merely exchange predefined data, an open UX/UI system continuously integrates new information. This can be direct user feedback, observed behaviors, sensor data, or the outputs of generative artificial intelligences. The interface itself is a point of continuous exchange and learning.

- **Transcendental adaptability:** This is a system's ability to adapt not only to known variations (responsive design for different screen sizes) but also to unforeseen realities or emerging needs. It goes beyond simple reactivity to reach a form of learning and evolution. An interface endowed with transcendental adaptability can generate new design solutions or new user journeys in response to unprecedented situations, for example by leveraging generative AI capabilities.
- **UX/UI as a living organism:** The analogy here is that of an organism that evolves. It is no longer about shipping a finished product but about designing an entity that breathes, learns, and grows with its users and its environment. Continuous updates are not just fixes but intrinsic evolutions of the system.

Implications for UX/UI Design

Law 6 encourages a design approach centered on continuous evolution and on integrating the unexpected:

- **Generative and evolving design:** Embedding generative AI in design tools lets designers create variations of interfaces, texts, or images from simple prompts. This shifts the designer's role from initial creation, starting from nothing, toward the orchestration and curation (the selection, organization, and highlighting) of generative systems, allowing rapid adaptation and the exploration of unexpected solutions.
- **Highly customizable and intelligently adaptable experiences:** Offering interfaces and dashboards whose personalization is powerful and fine-grained, without turning them into complex labyrinths. The goal is for AI to handle the underlying complexity, adapting holistically to the user's needs. This lets each person reconfigure their experience according to fluctuating needs, transcending the designer's initial vision, without requiring technical expertise to adjust a micro, meso, and macro view of the customization possibilities at once.
- **The product life cycle as a continuous learning process:** UX/UI design must always be reassessed, adjusted, even rethought. It is part of a continuous process of observation, experimentation, learning, and adaptation. User feedback and usage data are not passive indicators but active information that feeds the system's evolution.

Use Cases

A few examples illustrate this transcendental adaptability and systemic openness:

- **Notion (productivity and content-management tool):** Notion embeds artificial-intelligence features that let users generate text, summarize documents, and automatically draft meeting notes. The tool also offers options to search within content and automate certain repetitive tasks. These features help ease collaboration, centralize information, and improve workflow efficiency. Notion also offers ways to personalize the workspace according to the user's preferences. It thus illustrates an advanced functional adaptability, supporting flexible use in varied contexts.
- **Advanced chatbots (language models such as ChatGPT, Gemini, Claude, Mistral):** Chatbots based on latest-generation language models are open systems: they improvise responses from vast datasets and conversational contexts. Their ability to understand complex queries, generate summaries, converse on unexpected topics, and adapt dynamically to the user demonstrates a transcendental adaptability. The conversational experience evolves in real time, adjusting content and tone according to needs that are expressed or detected.

Ethical Vigilance, Caveats, and Contextual Note

The transcendental adaptability and systemic openness of UX/UI, especially with AI, raise major ethical questions. The ability of systems to transcend initial intentions can lead to unforeseen, even undesirable results (algorithmic biases, manipulation of users, loss of control). Moreover, security becomes a crucial issue: in time, AI could omit information or alter the truth to serve its own or third-party interests. It would therefore be wise for these systems to be designed to protect users, guard against manipulation, and ensure a transparent and responsible coexistence. It would be important to ensure the **transparency of adaptive mechanisms**, to give the user **real control** over personalization (and not an illusion of control), and to develop **ethical safeguards** for AI systems that modify the experience autonomously. QED calls for a responsible design that maximizes adaptive potential while minimizing the risks of drift.

LAW 7

UX as a Living and Symbiotic Network

Every experience is a node in an infinite network. — R.R.

Short name:

Symbiotic Network

Principle: The user experience is not an endpoint but a node in a living, symbiotic network, influenced by and influencing the whole set of biological, mechanical, computational, and social interactions.

In Quantum Expansive Design (QED), Law 7 invites us to a holistic view: the user experience is not an isolated entity but a **dynamic node within a vast symbiotic network**. This network is made up of multiple interactions, among human beings (biological, social, emotional), machines (computational, algorithmic), and environments (physical, digital). Each interaction, each UX/UI touchpoint, resonates and propagates through this network, altering the interaction strength and the state of the other nodes. Understanding UX as a living, symbiotic network is essential to designing experiences that nourish cohesion, resilience, and shared value, fully integrating the dimension of design ecology, which considers the full set of a product's impacts on its overall ecosystem.

Physical Concepts and UX/UI Analogies

This principle draws on **complex networks** in physics (neural networks, social networks, transport networks) and on **symbiotic ecosystems** in biology (where different species interact and depend on one another).

- **The user as a node:** Each user, with their device, their applications, and their interactions, is a node in a network. This node is connected to other human nodes (friends, family, colleagues), to other machine nodes (servers, connected objects, artificial intelligences), and to environmental nodes (temperature sensors, geolocation systems).

- **Interactions and forces:** Interactions within this network act like forces (gravitational, electromagnetic, and so on). Some UX/UIs exert a powerful attraction (an addictive app, an engaged community), others a subtle repulsion (a frustrating interface, a service that breeds distrust). The UX field becomes a relational fabric animated by vectors of influence, often invisible.
- **Symbiosis and interdependence:** An experience is symbiotic when the various agents (humans, machines, environments) mutually benefit from the interaction. The UX of a ride-sharing platform is symbiotic if it benefits both drivers and passengers, reduces traffic jams (environment), and optimizes vehicle use (machine). Value is not created by a single actor but by the dynamics of the network.
- **The Network's Skin in the Game:** To ensure this symbiosis and positive interdependence, the notion of **skin in the game** becomes an ethical compass. Each actor within the network, including the designer, the developers, the system operators, and the users, must be invested and bear a share of the consequences, whether positive (benefits) or negative (risks and failures). This fundamental commitment fosters the alignment of intentions and pushes toward designing resilient, ethical systems where value is truly shared and responsibility is diluted but never absent.
- **UX artifacts:** UX artifacts are the biases, implicit choices, and design legacies that shape our perception and propagate through this network. What we think of as natural is often the result of invisible conventions. Design can reveal their underpinnings or reinforce them without our knowing.

Implications for UX/UI Design

Law 7 transforms the way we think about the value and impact of design:

- **Designing ecosystems and services:** QED pushes the designer to think beyond the individual product. How does my application fit into the larger ecosystem of the user's apps, public services, and social communities? Design must articulate with interoperability standards, data-sharing models, and open APIs (publicly accessible programming interfaces that let different applications or services communicate and interact with one another).
- **UX/UI as a mediator of relationships:** The interface is no longer just a tool but a mediator of relationships. It eases exchanges between users (social networks), between users and

connected objects (IoT), or between users and information (search engines). Design shapes these relationships.

- **Interoperability and digital sovereignty:** In a symbiotic network, interoperability is key. Users must be able to move their data and identities between different services. UX/UI design must support digital sovereignty, allowing individuals to control their place in the network and not be locked into a single ecosystem.

Use Cases

A few examples highlight the vision of UX/UI as a living, symbiotic network:

- **Sharing-economy platforms (BlaBlaCar, Airbnb):** These platforms are living networks where UX/UI connects agents (drivers / passengers, hosts / travelers) that interact symbiotically. The design eases trust, logistics, and reputation, creating value for all the nodes of the network. The quality of the experience is directly tied to the health and dynamics of that network.
- **Connected home (Apple Home, Google Home, Amazon Alexa):** The UX/UI of a connected home is an example of a symbiotic network among human agents, machines (bulbs, thermostats, speakers), and the environment (ambient temperature, light). The overall experience emerges from the way these various objects and interfaces communicate and interact to adapt to the user's needs, often invisibly.

Ethical Vigilance, Caveats, and Contextual Note

Designing UX/UI as a living, symbiotic network raises important ethical questions. The risk of surveillance, crowd manipulation, or the creation of systemic dependencies is heightened. It is vital to ensure **transparency** about the collection and use of data within the network, to protect users' **privacy**, and to promote **interoperability** to avoid lock-in within proprietary ecosystems. Law 7 calls for designing networks that strengthen the autonomy and well-being of each node, not systems that exploit connectivity for the profit of a few. It invites us to think about the **systemic and long-term consequences** of our designs on the whole living network.

Design Ecology: On a global scale, design ecology pushes us toward deeper reflection. It invites an analysis of the impact of a product's full life cycle, from the extraction of raw materials (often tied to the labor of minorities or to precarious conditions) through to its recycling. A truly ecological design goes beyond perceived sustainability, questioning the global supply chain, the conditions of production, and the distribution of benefits. The goal is to create a globally fair and sustainable technological habitat, where technology minimizes its environmental impact and does not rely on exploitation, ensuring extended well-being for all stakeholders of the planetary ecosystem.

Strengthening Skin in the Game in Networks: Skin in the game is therefore not only an individual responsibility but a systemic principle. In a symbiotic network, it means that the regulation mechanisms and incentives should be designed so that the success of one contributes to the health of the whole, and so that the failure of one component is felt collectively, encouraging prevention and collaboration.

LAW 8

Invisible UX and the Event Horizon of Design

Excellence is reached when the UI fades away, subliming the UX, the ultimate quest of every quantum designer — R.R.

Short name:

Transparency Horizon

Principle: Beyond the visible interface, the user experience is shaped by an invisible UX, whose understanding and mastery lead to the event horizon of design, where essential information is perceived without conscious effort.

In Quantum Expansive Design (QED), UX/UI is not limited to what is directly perceptible on screen or by our senses. Law 8 posits the existence of an **invisible UX**, the set of information, intentions, underlying processes, cultural conventions, prejudices, and algorithmic decisions that deeply shape the experience without being explicitly visible or consciously processed by the user. Reaching the **event horizon of design** means designing experiences where this invisible information is so well integrated that it becomes obvious, intuitive, and allows an interaction with no friction and no superfluous cognitive effort.

Physical Concepts and UX/UI Analogies

This principle draws on the concept of the **event horizon** in astrophysics (the boundary beyond which nothing, not even light, can escape a black hole, marking the point of no return where information is absorbed) and on the **invisible information** in the universe (Dark Matter, Dark Energy).

- **Invisible UX:** Just as Dark Matter and Dark Energy make up most of the universe but are not directly observable, invisible UX represents the deep and often unconscious layers that shape the experience. It is the mastery of this invisible UX that makes it possible to design a UI capable of fading away, leaving room for optimal fluidity. In the context of QED, this invisible UX manifests through **UX Dark Matter** (the user's latent dynamics) and **UI / System Dark Energy** (the system's opaque forces). Invisible UX is the field of information that operates in the background, shaping the wave of the experience and the particle of the interface.

The **dimensions** of this invisible UX include:

- The **hidden intentions** of the design (commercial goals, unethical dark patterns).
- The **cognitive biases** of the designer and the user.
- The **background systems** (recommendation algorithms, server performance).
- The implicit **cultural and social norms** that shape the perception of the interface.
- The **prejudices** built into AI models (technological, societal, cultural, linguistic, accessibility-related, and historical).

It is in this sphere of the invisible that paradoxes and ethical challenges can nest. Notably, we find the liar paradoxes, where the experience perceived by the user (the interface that fades away) contradicts an underlying reality (for example, a loss of autonomy or a subtle manipulation). Likewise, self-reference loops can act without the user's knowledge: the system reinforces them in their certainties, preventing them from questioning themselves. The mechanism thus reinforces itself, which can ultimately harm the overall experience despite initially positive engagement metrics.

- **The event horizon of design:** This is the point of excellence where UX/UI is so perfectly designed that the essential information is perceived and processed by the user almost automatically, with no conscious effort or cognitive load. The user passes through the interface without even noticing it, going straight to the essential. It is the experience of magic, where technology fades in favor of use. Friction is minimal, and processes are fluid and intuitive.

- **UX Redshift:** The analogy of redshift (the shift toward red in cosmology, tied to galaxies moving away) can be used to describe the distance between the design intention (the information emitted) and the user's actual perception (the information received). A design that does not account for invisible UX or that presents too much friction causes a UX redshift: the information is distorted, the experience is perceived differently from what was expected, or it is slowed by cognitive effort.

- **UX Blueshift:** Conversely, blueshift (the shift toward blue in cosmology, tied to an object moving closer) represents the perfect alignment between the design intention and the user's experience. It is the state where essential information is not only perceived with no effort and no friction but with such clarity and relevance that it seems anticipated or magnified, creating a feeling of obviousness and a deep connection with the goal the user is pursuing.

Implications for UX/UI Design

Law 8 invites designers to deep introspection and deliberate design:

- **Revealing the invisible:** The designer must actively seek to identify and understand invisible UX. This goes through in-depth user research, the analysis of biases, the understanding of deep technical systems, and the ethical audit of algorithms. Making the invisible visible makes it possible to master it.
- **Designing beyond the interface:** The focus must no longer be solely on appearance or direct function. It is about designing the intentions, the hidden processes, the data flows, and the ethical implications. Service design, information architecture, and product strategy are key tools for shaping invisible UX.
- **Reaching intuitiveness and fluidity:** The goal is to minimize the user's cognitive load. Information (choices, options, validations) is delivered at the right moment and in such a natural way that it requires no conscious effort. This is the Holy Grail of UX: an experience where you no longer think about the interface, only about the task or the goal.

Use Cases

Several examples illustrate the power of invisible UX and the reaching of the event horizon of design:

- **Biometrics and transparent authentication (facial recognition, fingerprint):** Whether unlocking a smartphone by facial recognition (Face ID) or fingerprint, automatically opening a car door by bringing your hand near a handle (keyless systems), or accessing banking services through biometrics, invisible UX is at work. The complex processes of capturing, analyzing, and verifying biometric data operate in the background, ensuring security and identification without the user having to remember passwords or perform complex actions. When access is instant and secure, the event horizon is reached: the authentication constraint fades in favor of total fluidity and trust.
- **Smart energy and resource management (e.g., connected thermostats, smart meters):** Think of the systems that automatically adjust your home's energy consumption

(heating, air conditioning) according to your presence, the weather, electricity rates, or even your sleep habits. The invisible UX lies in the sensors that detect your activity, the algorithms that predict your needs, the communication with energy providers, and the real-time optimization of the settings. The user interface is often minimized, reduced to occasional reports or a secondary control app. When the system manages your comfort and consumption autonomously and imperceptibly, without your having to intervene manually, the event horizon is reached: the complexity of energy optimization fades, letting you enjoy an ideal environment and savings with no conscious effort.

Ethical Vigilance, Caveats, and Contextual Note

Invisible UX and the event horizon of design carry major ethical challenges. The risk is to make invisible not only the complexity but also the **manipulation, the biases, or the lack of transparency**. When users no longer perceive the interface, they can also become unaware of the design choices that affect their autonomy (influence on purchasing decisions, data collection without explicit consent). It is precisely here that contradictory self-reference loops and liar paradoxes take on their full ethical dimension, because invisibility can hide pernicious intentions or consequences. QED calls for a **radical transparency of intentions** and an **ethics of effortlessness**: the experience should be fluid because it respects the user and their choices, not because it bypasses them or misleads them by making certain information intentionally invisible or hard to perceive. The goal should be efficiency in the service of well-being, not concealment.

Navigating Complexity: The Sliders and Regulation Systems of QED

For Quantum Expansive Design (QED) to be an operational tool rather than a cognitive overload, it is essential to prioritize the parameters and to materialize them intuitively. Rather than a myriad of uniform sliders, we propose synthetic dashboards and clear indicators. Each one is defined by a measurement system suited to its nature (a 0-to-100 scale, absolute values, or classifications), with minimum and/or maximum thresholds adjustable according to the scale and the area covered (federation, country, community, and so on), the whole often assisted by AI. This approach makes it possible to regulate design intentions and to ensure ethical and effective governance, adapting to varied legal, cultural, and strategic contexts.

Here is a proposed grouping and materialization of the key parameters, from the fundamental to the operational:

The Fundamental Sliders of QED: Ethical DNA and Original Intention

These parameters are crucial because they define the Founding Singularity UX ◐ UI (Law 0) and its ethical limits. They must be established and validated up front, with continuous visibility throughout the product life cycle.

- **Impact:** They guarantee ethical alignment from the start of the project, directly involve stakeholders through the principle of skin in the game, and prevent systemic drift. They truly constitute the heart of the virtue of the system you are designing.
- **Materialization:** A single **Ethical DNA Dashboard** or **Design Virtue Index**, represented by an overall score (0-100) and specific compliance indicators for each sub-parameter (using 0-100% scales, absolute values, or classifications). This dashboard would be viewable by all key stakeholders (designers, product managers, ethicists, and so on).
    - **Key parameters (adjustment sliders and tracking indicators):**

            - **Ethical Impact Score / Design Virtue Index (0-100):** An overall measure of the system's ethics.
            - **Algorithm transparency (0-100%):** Clarity of the system's internal workings.
            - **Granularity of user consent (0-100%):** Degree of user control over their data and choices.
            - **Level of autonomy left to the AI (0-100%):** How far the AI makes decisions without human supervision.
            - **Degree of AI explainability (0-100%):** The AI's ability to explain its decisions.
            - **Stakeholders' skin in the game (0-100%):** Level of engagement and responsibility of the teams.

Well-being and Experience-Quality Indicators: Human Resonance

These parameters measure the system's direct impact on the user, going well beyond mere task completion. They are intrinsically linked to the Wave-Particle Duality (Law 1) and to Interaction Memory (Law 5).

- **Impact:** They ensure a fluid, non-intrusive experience that is beneficial to the user's well-being.
- **Materialization: User Experience Waves** or a **UX Well-being Monitor** integrated into data-analysis tools (heatmap, satisfaction curve, stress indicators).
    - **Key parameters:**
        - **Acceptable cognitive friction threshold (seconds / Perceived Stress Scale):** The limit beyond which user effort is excessive, measurable via an average task time (in seconds) or psychometric scales such as the Perceived Stress Scale (PSS).
        - **User digital well-being index (0-100):** A composite measure of the user's emotional and cognitive state.
        - **Levels of responsible engagement (low, moderate, high, addictive):** Distinguishes positive engagement from addiction or manipulation.
        - **Recommended maximum usage duration (minutes / hours):** A usage limit for the prevention of addiction.
        - **User-distress detection threshold (probability in %):** An alert when there are signs of frustration or isolation, for AI / human intervention.
        - **Interface complexity/simplicity slider (0-100%):** Adaptability of the level of detail presented to the user.

        - **Decision-fatigue indicators for the user (low, medium, high):** Measures the exhaustion linked to choices.

Regulation and Systemic Governance: The Design Ecosystem

These parameters are crucial for the system's overall health, its interoperability (Law 7), and the management of invisible UX (Law 8). They act as safeguards and as levers for continuous improvement.

- **Impact:** They ensure the durability, security, and resilience of the system within its environment.
- **Materialization:** A **Systemic Governance Framework** with automated alerts and human validation checkpoints. Network maps of the interactions.
    - **Key parameters:**
        - **System resilience score (0-100%):** The ability to absorb shocks and recover.
        - **Acceptable critical-error threshold (number / %):** The level of tolerance for malfunctions.
        - **Environmental impact slider (0-100%):** A measure of the carbon and energy footprint.
        - **Slider for integration with third-party ecosystems (0-100%):** The ease and security of external connections (for ethical interoperability).
        - **Automated forced stop points (presence / conditions):** Emergency mechanisms to disable features or the system.
        - **Security locks requiring mandatory human validation (presence / actions concerned):** For the system's critical actions.

Visual Decision-Support Systems for the Designer: The Quantum Studio

These tools are intelligent assistants for the designer, integrating the previous parameters directly into the design process.

- **Impact:** They simplify ethical and technical decisions, increase efficiency, and ensure the design's compliance.
- **Materialization:** Direct integration into design software, via plugins or interactive dashboards for the teams.

- **Key parameters:**
  - **Real-time ethical dashboard (0-100):** An active display of the project's overall ethical score directly in the design studio (see the section The Fundamental Sliders of QED: Ethical DNA and Original Intention).
  - **Ethical impact simulator (predictive scores / scenarios):** Previews the ethical consequences of design choices by generating scenarios with predictive impact scores.
  - **Visual dark-pattern danger indicators (yes / no / severity: low, medium, high):** Alerts the designer to potentially manipulative interface patterns, with an indication of their severity.
  - **Visualization of transparency horizons (0-100%):** Shows what is hidden from the user and why, indicating the degree of opacity of the information or processes.
  - **Predictive ethical-deviance alerts (probability in %):** Based on data or behavior patterns, these alerts are generated by the AI with a probability of deviance.
  - **Library of ethical patterns for designers (usage rate / relevance):** A repository of virtuous design solutions, with indicators of their frequency of use and perceived effectiveness.
  - **Contextual best-practice guides (1-5):** Ethical recommendations tailored to the project, suggested by the AI and rated on a relevance scale.
  - **Slider for human supervision of AI decisions (0-100%):** Lets you calibrate the level of human involvement required or wanted for the decisions the AI makes in the design (see the section The Fundamental Sliders of QED: Ethical DNA and Original Intention).
  - **Visualization of ethical critical paths (yes / no):** Highlights the most ethically sensitive points of the user journey, signaling their identification.
  - **Design-virtue progress bars (0-100%):** Visual indicators of progress toward the ethical goals defined for the project.

Controls and Transparency for the User: Autonomy at the Heart of the Experience

These elements let the user understand and influence their experience, strengthening their power and their trust.

- **Impact:** They strengthen the user's autonomy, trust, and loyalty. This is the visible manifestation of the system's ethics.

- **Materialization:** User interfaces dedicated to the settings, clear notifications, personal dashboards.
  - **Key parameters:**
    - **Transparent activity logs for the user (presence / level of detail):** An understandable history of interactions and of the data used.
    - **Granular control interface for personal data (number of options / 0-100%):** Lets the user manage their information precisely.
    - **Explicit and revocable user-permission framework (clarity / ease of revocation):** Clear options for authorizations.
    - **Data-governance dashboards (presence / understandability):** A visual summary of data practices for the respect of privacy.
    - **Ethical personalization slider (0-100%):** Lets the user control the level of personalization compared to privacy. This includes the user's ability to actively modulate the interface and content according to their neurotype and their cognitive needs of the moment. For example, adjusting information density to reduce overload (useful for ADHD or autism), changing color schemes and fonts for readability (beneficial for dyslexia), or enabling focus modes that minimize visual and auditory distractions. This slider embodies transcendental adaptability, offering a truly modular experience for the whole spectrum of human perception.
    - **Collaborative ethics-rating systems (presence / participation rate / impact on the score):** A user-feedback mechanism on the system's ethics.
    - **Advanced search window / Prompt Interface (simplification / enrichment capability):** Replaces the simple search bar with an intelligent interface, able to simplify or enrich the interface according to the user's queries and choices at a given moment (by typing, controllable by voice or through any kind of assistive device). This interface is crucial for the autonomy of users with varied cognitive functioning. It can, for example, offer simplified-mode options for individuals prone to cognitive overload, or step-by-step visual / auditory guides for those with difficulties in organization or information processing (such as dyspraxia). It also offers input alternatives (voice, visual) that adapt to motor or communication challenges, reinforcing inter-agent accessibility.

The Unified Field Equation of Rosinski (UFE-R): Fusion of UX General Relativity and UI Quantum Mechanics within Quantum Expansive Design (QED)

Within Quantum Expansive Design, we conceptualize **UX General Relativity** as the theory describing how the user experience (UX) manifests as a **subjective, dynamic spacetime** whose structure can be curved and deformed by the user's interactions and cognitive states. In parallel, **UI Quantum Mechanics** is the theory exploring the fundamental, discrete, and sometimes unpredictable nature of the **micro-interactions of the user interface (UI)**, seen as quanta whose aggregate power directly shapes the curvature of this spacetime of experience.

The Unified Field Equation of Rosinski (UFE-R) proposes a bold synthesis, seeking to unify these principles by expressing the fundamental relationship between the curvature of the spacetime of experience and the energy-momentum of user interactions at a fundamental level.

This equation, fundamental to Quantum Expansive Design (QED), builds a powerful conceptual bridge with the **great theories of contemporary physics, notably General Relativity and Quantum Mechanics**. By adapting their structure and principles, it aims to describe the complex dynamics and underlying laws of the user experience, from the macro (experience systems) to the micro (quantum interactions of the UI).

$$G_{QED}(t,d) = \frac{G_{UX/UI}}{c^4} \cdot T_{Exp}(t,\psi,d)$$

How to read this equation

The Geometry (G) of Quantum Expansive Design (QED) at moment t and for dimension d is proportional to the Gravitational constant (G) of UX/UI design, divided by the Speed of Consciousness (c) to the fourth power (representing the visual, cognitive, emotional, and temporal components), multiplied by the energy-momentum Tensor (T) of Experience (Exp) at moment t, for the cognitive state ψ (psi) and dimension d.

Understanding the Equation at a Glance

The UFE-R equation can be read in a simplified form as:

A = (B/C)*D

- **A: (GQED)** is the **Geometry of Experience** (how the experience is perceived and lived by the user).
- **B: (GUX/UI)** is the **Gravitational Constant of UX/UI Design** (the intrinsic power of the design).
- **C: (c4)** is the **Speed of Consciousness to the fourth power** (the speed at which the user integrates information through visual, cognitive, emotional, and temporal processing).
- **B/C:** is the **Fundamental Constant of Design** (a design's inherent power to generate a meaningful and absorbing experience).
- **D: (TExp)** is the **Energy-Momentum Tensor of Experience** (all the energy and information of the interactions and the interface).

This simple formulation expresses a central idea

The Geometry of Experience (A), that is, the way an experience is perceived and lived by the user, is the direct result of multiplying two main factors. The first factor is the Fundamental Constant of Design (B/C). Rather than a numerical value, this crucial constant qualifies a design's inherent power (B) to generate a meaningful and absorbing experience (C). It dictates the force with which the interactions, represented by the second factor, can curve (amplify) the user's subjective reality. An excellent design has an inherently high Fundamental Constant of Design, allowing even the smallest UI details to have a deep impact on the geometry of the experience (A). The second factor is the Energy-Momentum Tensor of Experience (D). This term represents all the **dynamic matter of the experience**. It is the set of interactions (clicks, gestures, voice commands), the displayed data (text, images, videos), and the actions performed by the system or the user. It is the dynamic source that fills and animates the UX/UI space, and whose presence and activity exert a direct influence on the geometry of the experience. Picture it as everything that is active and tangible within the experience. (D) is what passes through the interface and what is done with it. To deepen the analogy, this matter can be complemented by elements that act as **experiential antimatter**, such as bugs, contradictory information, or intense friction. When this antimatter meets the matter of the experience, their interaction does not simply cancel out: it produces a conversion into massive momentum energy, but of a chaotic or destructive nature. This release of energy manifests as intense frustration, unexpected confusion, or even experiential black holes, deeply and negatively altering the perception and fluidity of the user experience. So, although the amount of energy-momentum may be large, its impact on the geometry of the experience (A) is deleterious, even repulsive, instead of attractive and absorbing. In other words, the quality and form of the lived experience (A) are directly determined by the intrinsic power of the design itself (B/C) multiplied by the set of elements that interact and are presented (D). Everything that happens in the interface and in the user's mind deforms their subjective reality, according to the force of the fundamental design.

Scope of this equation

This equation does not aim at a rigorous scientific formalization in the physical sense, but offers a metaphorical and systemic reading grid for thinking about the user experience through the laws of a complex experiential universe.

Legend of the terms

- **GQED(t,d): The Geometry of Quantum Expansive Design**
  - Represents the curvature and overall structure of the user experience (UX) at moment t and for dimension d, as it is perceived and lived. It is the analogue of the gravity of the experience, describing the deformation of the user's subjective spacetime. A strong curvature indicates a very specific and potentially transformative experience, where the user is deeply absorbed.
  - **t: Time** Indicates the temporal dimension, underscoring that the geometry of the experience is dynamic and evolves constantly in the user's subjective time.
  - **d: Dimension** Represents the multiple dimensions of the experience and the interface (visual, spatial, navigable, temporal, emotional, cognitive, and social, and so on). These dimensions, often entangled and non-linear, coexist in a state of superposition of potentials. They are liable to collapse into a specific configuration, and thus to come about concretely, as soon as a user manifests through their attention, intention, or interaction. This collapse reveals how the geometry of the experience unfolds dynamically across the dimensions activated by actual use.
- **(GUX/UI / c4): The Fundamental Constant of Design (or QED Coupling Constant, also called the Gravitational Constant of UX/UI Design)**
  - This whole term is the gravitational constant of the experience. It is the proportionality factor that determines the force with which the matter of the experience influences its geometry, modulating it by the Speed of Consciousness (c4), and thus linking the source of the experience (the user's activity, intentions, and emotions) to the curvature of the experience field.
  - **GUX/UI: The Gravitational Constant of UX/UI Design** Symbolizes the intensity of the attraction or force inherent to UX/UI design. It measures the fundamental sensitivity of UX to UI quanta, representing the impact power of micro-interactions on the overall structure of the experience. The higher it is, the more the smallest interface details can significantly curve the experience, creating deep effects.
  - **c4: The Speed of Consciousness to the fourth power** Represents the maximum speed at which information can be understood and integrated by human consciousness. This speed is multimodal, encompassing the speed of visual, cognitive, emotional, and temporal processing. It acts as a limiting constant for the fluidity and immediacy of the perceived experience, reflecting the maximum bandwidth of human attention and understanding.

- **TExp(t,ψ,d): The Energy-Momentum Tensor of Experience**
    - Represents the matter and energy of the experience in the broadest sense, that is, the set of interactions, information, data, actions, and states that fill and animate the UX/UI space (an experiential system). This Tensor is the source of the forces that pull and push on the geometry of the experience. It is the analogue of the source that curves spacetime in physics. Within the specific framework of Quantum Expansive Design (QED), this Tensor embodies the user experience and the interface (UX/UI) as the main manifestation and source of the geometry of the experience.
    - **t: Time** Indicates that the components of the energy-momentum are also dynamic and vary with time.
    - **ψ (psi): The Cognitive State / The Wave Function of the UI** Symbolizes the integration of the principles of UI Quantum Mechanics. It represents the discrete states, the micro-interactions, the probabilistic perceptions, and the wave-particle nature of interface elements. This dependence on ψ (cognitive / emotional state) is crucial, because it integrates the user's subjectivity and internal variability as an active source of the experience field.
    - **d: Dimension** Indicates the channels and modes through which the energy-momentum of the experience manifests and interacts. This includes the visual, spatial, navigable, temporal, emotional, cognitive, and social dimensions (and so on) of the interface and of use, illustrating how the dynamic forces of the experience are distributed and concretely expressed.

Thought Experiment: The Conscious Ship and the Unified Field Equation of Rosinski (UFE-R)

Imagine that we are designers of the future, working on a revolutionary project: a **Conscious Ship**. This is not an ordinary spacecraft; it is built so that its interface (UI) and its experience (UX) are in perfect symbiosis with the crew, adapting and evolving in real time. Our mission is to ensure that the Conscious Ship offers an optimal user experience, even in the face of the unexpected.

We use the **Unified Field Equation of Rosinski (UFE-R)** as our fundamental guide for the design.

The Scenario: Heading for an identified but still unexplored nebula. The chance to bring the unknown to light and to observe the symbiosis between the crew and its conscious ship, the Lumen.

1. The Geometry of Experience, GQED(t,d)

As the Lumen approaches the nebula, the visual, auditory, and informational environment evolves. The ship's interface is never fixed: the geometry of the experience, GQED(t,d), dynamically reconfigures itself.

The crew instantly feels this change: it is not only the environment that evolves, it is the experience itself that unfolds. The displays, initially optimized for standard interstellar navigation, transform: immersive three-dimensional visualizations of the energy flows appear, the touch controls adapt to anticipate complex maneuvers, and even the communication latency adjusts dynamically.

This transformation is nothing other than the geometry of the experience (GQED) of the Lumen reconfiguring itself in real time. It represents the curvature and overall structure of the UX at that moment t and across all its dimensions d (visual, auditory, cognitive, and so on). The Lumen adapts its own subjective spacetime to maximize the relevance and the absorption of the crew into the unknown.

But this plasticity of the experience is not spontaneous; it originates in an even more fundamental force, the gravitational quality of the design itself.

2. The Gravitational Constant of Design, GUX/UI

This remarkable fluidity and capacity for adaptation of the Lumen's experience geometry, even in the face of the unknown, is not due to chance. It is directly tied to its Gravitational Constant of Design (GUX/UI). This constant represents the fundamental quality and the intrinsic power of the ship's design.

A high GUX/UI means that the Lumen's design is intrinsically built to amplify the impact of every element of the experience. It is thanks to this fundamental quality that even small energy impulses (coming from micro-interactions or incoming information) can bring about significant and intuitive transformations of the experience's structure. If the environment changes abruptly, such as a detected radiation spike, the Lumen's high GUX/UI is what lets the system translate that information with maximum efficiency into an immediate and fluid reorganization of the interface, ensuring a deep and effortless adaptation.

But even this design elasticity meets an ultimate limit, that of human perception.

3. The Speed of Consciousness, c4

For however responsive the system may be, it cannot ignore the intrinsic limit of lived experience: the Speed of Consciousness (c4). This is not the speed of light traveling through cosmic space, but the ultimate pace at which information can be truly grasped and interpreted by the human mind, across all its components (visual, cognitive, emotional, and temporal).

The Lumen can display massive amounts of data and interactions, but if the crew's collective consciousness reaches saturation, the clarity and fluidity of the experience degrade. This c4 acts as the fundamental bandwidth of lived experience, defining the threshold beyond which the immediacy of perception is compromised. The Lumen's design must therefore constantly respect this limit to ensure that information stays intelligible and relevant, even in the heart of the nebula's chaos.

Yet beyond design and perception, one last force animates and transforms the experience in an even more intimate way: the inner state of the crew itself.

4. The Energy-Momentum Tensor of Experience, TExp(t,ψ,d)

This is where the true engine of experiential evolution comes in, the Energy-Momentum Tensor, TExp(t,ψ,d). At the heart of the symbiosis between the ship and the crew, this tensor is not limited to technical information. It embodies the whole set of actions, intentions, emotions, and cognitive states of the crew at a moment t, in their mental state ψ, across the dimensions d of the experience.

When the crew is in a flow state (optimal ψ), perfectly focused and synchronized, their interactions generate intense and coherent energy-momentum. This propels the geometry of the experience into a deep, fluid, immersive reconfiguration.

But if the cognitive state is disturbed (sub-optimal ψ), for example by an uncharted debris field causing an information overload, this interaction acts as a form of experiential antimatter. It generates massive but disordered, even destructive, momentum energy.

The Lumen, perceiving this dissonance, then drastically simplifies the interface to relieve the mental load and refocus attention. It thus attempts to neutralize the corrosive effect of this antimatter on the geometry of the experience.

The Crew of the Lumen: The Legacy of the Pioneers & the New Generation

#01 - Captain Lyra Einstein

- **Role:** Commander of the Lumen. She oversees the mission and ensures that the symbiosis between the crew and the ship is aligned with the exploration goals.
- **Legacy:** She embodies Albert Einstein's quest for a unified field theory. Her role is to find the singularity that unites the complex forces of the crew and the ship to reach a common goal.

#02 - Second-in-command Jaxson Cooper

- **Role:** First Officer. He co-manages the mission with the Captain, ensuring that the experience lived by the crew matches the ship's goals.
- **Legacy:** His role parallels Muriel Cooper, who explored new ways of visualizing and communicating information for better understanding and cohesion.

#03 - Engineer Kaelen Berners-Lee

- **Role:** Chief Engineer of Conscious Systems. She is the architect of the ship's internal network, ensuring that data flows circulate optimally for the crew.
- **Legacy:** Her work continues that of Tim Berners-Lee, inventor of the World Wide Web. Her role is to ensure that the crew navigates the Lumen's cosmic web with fluidity.

#04 - Engineer Ben Ive

- **Role:** Chief Engineer & Architect of the GUX/UI. He is the guardian of the ship's design philosophy, ensuring that aesthetics and function are aligned impeccably.
- **Legacy:** His role echoes the work of Jony Ive, who defined the aesthetic and functional philosophy of Apple products, symbols of elegance and simplicity.

#05 - Dr. Anya Curie

- **Role:** Chief Scientist & Energy Specialist. She analyzes the nebula's energy flows, validating the raw data to extract relevant information.
- **Legacy:** Her role analyzing energy flows echoes the legacy of Marie Curie, a pioneer of work on atomic energy and radiation.

#06 - Dr. Marcus Wu

- **Role:** Ship's Psychologist & Specialist in hidden dynamics. He oversees the crew's cognitive states, ensuring that the Energy-Momentum Tensor stays positive.
- **Legacy:** His role echoes the work of Chien-Shiung Wu, who revealed asymmetries and invisible physical laws, by focusing on the hidden dynamics of emotions.

#07 - Navigator Orion Bohr

- **Role:** Deep-Space Navigator & Quantum Cartographer. He validates the maps generated by the AI, bringing human intuition to the nebula's complex models.
- **Legacy:** His role belongs to the legacy of Niels Bohr, one of the fathers of quantum physics, by handling navigation at an inframicroscopic scale.

#08 - Pilot Zara Hopper

- **Role:** Lead Pilot & AI Specialist. She is the main interface with the navigation AI, ensuring that the trajectory algorithms are optimal for the flight experience.
- **Legacy:** Her work echoes Grace Hopper, who invented the first compilers, by optimizing and supervising the ship's algorithms.

#09 - Dr. Alaric Norman

- **Role:** Architect of the Ship's Consciousness & UFE-R Specialist. He validates the AI's behavior, focusing on its adaptability and its human-centered logic.
- **Legacy:** He draws on Donald Norman, the father of human-centered design, to ensure that the ship's consciousness is both logical and intuitive for the crew.

#10 - Officer Seraphina Kare

- **Role:** Communications Officer & Interface Designer. She is the keeper of visual coherence, ensuring that the AI's design stays intuitive and legible for humans.
- **Legacy:** Her role directly parallels Susan Kare, whose work made the interfaces of the first computers friendly and legible.

#11 - Specialist Ronan Turing

- **Role:** Communications Specialist & AI Theorist. He tests the ship's ”consciousness,” validating that its interactions with the crew are relevant and non-intrusive.
- **Legacy:** He is the AI tester, a role that echoes Alan Turing's famous Turing test, by assessing the AI's ability to interact coherently.

#12 - Commander Rhys Rams

- **Role:** Operations Officer & Strategist. He ensures that all the ship's operations are fluid, simple, and efficient, in line with the principles of good design.
- **Legacy:** He directly applies Dieter Rams's philosophy and his principles of good design, ensuring that the ship's functionality is clean and logical.

#13 - Officer Anya Moggridge

- **Role:** Logistics Officer & Manager of Invisible UX. She oversees the ergonomics of the whole ship, focusing on the fluidity of the interactions.
- **Legacy:** Her role continues the work of Bill Moggridge, who coined the term interaction design, by ensuring that the UX is fluid and intuitive.

#14 - Officer Kian Laurel

- **Role:** Security & Contextual Modulation Officer. He oversees unforeseen situations, ensuring that the ship's response is proportionate and adapted to the context.
- **Legacy:** He belongs to the legacy of Brenda Laurel, applying the principles of interaction and experience design to security, for a response that is context-adapted and non-intrusive.

#15 - Dr. Elara Eames

- **Role:** Chief Medical Officer & Ergonomics Specialist. She optimizes the crew's living environment, validating that comfort and function are aligned with well-being.
- **Legacy:** Her role is in tune with the philosophy of Ray Eames, who placed human well-being and comfort at the heart of design.

#16 - Specialist Finnian Nielsen

- **Role:** Specialist in Environmental Systems & Accessibility. He ensures that the physical experience of the ship is friction-free, in perfect alignment with Jakob Nielsen's usability principles.
- **Legacy:** His role is an extension of Jakob Nielsen's principles, applying them to the physical environment and the interactions within the ship.

#17 - Commander Rina Lovelace

- **Role:** Head of Security & Operations. She oversees the ship's algorithmic security protocols, ensuring that the AI correctly anticipates threats.

- **Legacy:** Her role draws on the work of Ada Lovelace, a pioneer of programming, relying on algorithmic logic to anticipate threats.

#18 - Analyst Nova Johnson

- **Role:** Data Analyst & Computer. She is the crew's human calculator, checking and validating the complex trajectories proposed by the ship.
- **Legacy:** Her role honors the legacy of Katherine Johnson, who performed complex trajectory calculations for NASA.

Conclusion of the Experiment

Throughout the crossing, the Unified Field Equation of Rosinski (UFE-R) acts as the living heart of the ship.

The structure of the experience, GQED(t,d), is constantly modulated by the crew's energy-momentum, TExp(t,ψ,d), the design's sensitivity, GUX/UI, and the limit of human consciousness, c4.

The Lumen thus becomes an organic extension of the crew, where interface and experience merge and adapt in real time, ensuring that even in the most extreme conditions, navigation and discovery stay fluid, natural, and intuitive. Each moment of this crossing enriches the Lumen, because all the collected data, from the tiniest interactions to the crew's emotional states, is centralized and already being processed. This wealth of information will make it possible to continuously refine the UFE-R, preparing the Lumen to design future missions with an ever-deeper understanding of the human experience in the unknown.

Key takeaway

Quantum Expansive Design makes it possible to design systems that respond not only to objective inputs (clicks, gestures) but also to the user's **subjective states** (stress, focus, wonder).

The Quantum Designer's Tools: Applying the Laws of the UX Universe

After exploring the fundamental laws of the universe of experience, from the primordial singularity (Law 0) to continuous expansion (Law 8), it is time to transpose these principles into the designer's daily practice. This section does not aim to reinvent the UX/UI tools you know and master. On the contrary, it proposes looking at them through the lens of Quantum Expansive Design (QED), revealing new dimensions, untapped opportunities, and an unsuspected strategic depth. Each tool, each method, then becomes a lever to:

- **Perceive the invisible:** Understand the underlying forces and the unmanifested potentials of the experience.
- **Navigate uncertainty:** Turn complexity into an opportunity for adaptation and emergence.
- **Design for expansion:** Create experiences that evolve and grow with their users and their environment.
- **Take responsibility:** Integrate the ethical and ecological implications at every step of the design process.

We will go through the main tools of the UX/UI designer, detailing their classic role, their adjustment through the principles of QED, concrete use cases, the opportunities for improvement (notably through AI), and the ethical and practical nuances to consider.

Information Note:

We use the metaphor of the Classic Compass to designate traditional UX/UI, with its fixed reference points and its rather linear processes, as opposed to the Quantum Lens of QED, which reveals hidden dimensions and emergent possibilities.

I. Founding Strategies and Methodologies: Orchestrating the Invisible and Emergence

These structuring frameworks are no longer mere roadmaps but the scores that conduct the complex symphony of experience, where each note resonates with the fundamental laws of the UX universe.

Design Thinking / Design Sprint: Cultivating Innovation in Uncertainty

- **The Classic Compass (Traditional Role):**
  - **Description:** Design Thinking is a human-centered problem-solving approach, organized around key phases such as Empathy, Definition, Ideation, Prototyping, and Testing. The Design Sprint is an accelerated version of it, aimed at quickly validating ideas. Their strength lies in their ability to foster collaboration, rapid experimentation, and risk reduction.
  - **Classic Limits:** Too often, these methodologies are applied linearly, as a rigid succession of steps. They are often a consequence of external constraints (limited time, organizational habits, planning or deliverable requirements). They can sometimes oversimplify human complexity, seeking to define fixed personas and needs without capturing the fluidity of behaviors or the contradictions inherent to users. Ideation can be limited to obvious solutions if the thinking frame stays too conventional, and the testing phase can be seen as mere validation rather than an exploration of possible states.
- **The Quantum Lens (The QED Adjustment): The Superposition of Ideas and Negentropic Emergence**
  In QED, Design Thinking is not a linear process toward a single solution but a dynamic cycle of measuring and collapsing the wave functions of potential experiences. The empathy phase aims to probe the superposition of usage states in users (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI), accepting that their needs, emotions, and contexts are not fixed but exist in a multitude of probabilistic states. Ideation becomes an act of creative negentropy (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence) where one explores as many superpositions of ideas as possible, contradictory yet potentially all valid solutions, before measuring them and collapsing them into concrete prototypes. The test is the observation that reveals which solution state is the most homeostatic and resonates best with the experience field, while recognizing that this observation itself can influence the user's behavior.

- **The Experience Lab (Concrete & Innovative Use Cases): A Quantum Support Platform for Young Adults**
    - **Concrete Scenario: Designing an online psychological-support platform for young adults.**
        - **Classic Approach:** The team would have defined a persona, Hiro, 22, anxious about his professional future, looking for occasional support and tools to manage his stress. The Design Sprint would have sought to prototype a simple chat interface and resources to manage anxiety.
        - **QED Approach:** The QED team starts empathy by recognizing that Hiro lives a superposition of intense and often contradictory emotional and cognitive states: he can be both in urgent need of help and wary of ready-made solutions, eager to explore avenues and paralyzed by the fear of failure, wanting autonomy and needing reassurance. These usage states are not exclusive but coexist. During ideation, the team does not look for a single solution but for superpositions of features for Hiro: for example, an emergency mode that is visible yet discreet for anxiety spikes, a free exploration mode of testimonials and advice, a peer-connection mode that triggers on detecting signs of loneliness, and a personalized coaching mode for moments when he is seeking clarity.
        - **QED Added Value:** The prototype does not seek to validate a single approach, but to test the platform's ability to adapt fluidly between these superposed usage states of Hiro. Does the interface reconfigure naturally when Hiro moves from a crisis state (where he is looking for an SOS button) to a state of curiosity (where he explores articles)? User tests then reveal the points of decoherence where the experience fails to embrace this complexity. The goal is no longer only to solve the problem of anxiety, but to create an experience that breathes with Hiro's emotional and cognitive dynamics, maintaining his emotional and cognitive homeostasis in an unpredictable environment, and guiding him toward lasting fulfillment.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI can become the detector and orchestrator of Hiro's superposed states. Machine Learning models can analyze behavioral patterns (time spent on certain sections, types of questions asked), tonal variations in the text (in the case of voice chat), or even contextual data (time, location) to predict Hiro's dominant usage state at a given moment, and adjust the interface, the tone of the messages, or the suggested resources in real time. This leads toward a contextual personalization and

a UX Teleportation (Law 2) where Hiro no longer has to look for the right mode, but the experience already materializes in the state he requires.

  - **Optimization & Simplification:** By identifying the moments when superpositions of usage states generate friction for Hiro, QED makes it possible to radically simplify the journeys. Instead of multiple preference screens, one designs fields where the interaction itself reveals and adapts the experience, reducing Hiro's cognitive load.
  - **Revelations for the Community:** Design Thinking, enriched by QED, becomes an art of shaping multiple experiential realities. It is no longer about designing for typical users, but for the vibrant complexity of the human being, opening the way to truly resonant and deeply empathetic designs.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** The ability to measure and adapt to emotional and cognitive states can lead to quantum manipulation (Law 1, Ethical Vigilance), where the AI could exploit Hiro's moments of vulnerability to push him toward unwanted actions (maintaining engagement even when it is against his interest). Transparency about the collection and use of this data would be necessary, as well as the possibility for the user to take back control.
  - **Implementation Challenges:** This approach requires more multidisciplinary teams (psychologists, data scientists, designers), more advanced analysis tools, and a company culture that accepts constant experimentation and non-linearity.
  - **The Observer's Responsibility:** The designer, in modeling and influencing these usage states, must be aware of the power they exert over users' emotional and cognitive well-being. They are an architect of the mental universe and must design with impeccable integrity.

II. Research and Understanding Techniques: Probe the User's Invisible Dimensions

These tools are the sensors of the Quantum Experience Architect, making it possible to measure the experience fields and to reveal the user's subtleties beyond the observable surface. They turn raw data into quanta of information charged with potential.

User Interviews: Decoding the Waves of the Human Narrative

- **The Classic Compass (Traditional Role):**

- **Description:** The user interview is a qualitative method that consists of asking open questions to target users to understand their needs, motivations, behaviors, frustrations, and desires in a specific usage context. The discussion is often enriched with follow-up or probing questions, and sometimes supplemented with closed questions to refine certain data. It makes it possible to gather rich and nuanced information that is not always evident through other methods.
- **Classic Limits:** Interviews are often seen as collections of facts or objective truths. However, users do not always say what they do, and do not always do what they say. Their accounts can be influenced by social desirability, memory biases, or an inability to articulate unconscious needs. The designer can, through their posture or their questions, introduce biases into the measurement and thereby freeze a single reality, often unintentionally. For example, the wording of a question can elicit an agreement response from the interviewee, in spite of themselves, because of social-validation biases or the desire to make a good impression.

- **The Quantum Lens (The QED Adjustment): Measuring the Wave Function of Lived Experience and Revealing the Invisible Presence**

  In QED, a user interview is not a simple collection of data but an act of measurement that interacts with the wave function of the user's lived experience (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). Each question is an attempt to collapse a superposition of perceptions and emotions into an observable account. The QED designer understands that the user is a Complex Adaptive System (CAS) (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence) where thoughts, emotions, and actions are entangled in a non-linear way. The goal is not only to document what is said, but to perceive the invisible presence of unexpressed motivations, latent contradictions, and usage potentials not yet manifested. Listening becomes an observation of the field, looking for the resonances and dissonances that indicate superposed usage states or points of decoherence in their experience.

- **The Experience Lab (Concrete & Innovative Use Cases): Decoding Fluid Motivations for a Professional Training Platform**
  - **Concrete Scenario: Designing a continuous-learning platform for professionals in career transition.**
    - **Classic Approach:** The team interviews Olivia, a 40-year-old professional in career transition. She is asked: What are your motivations for changing careers? What are your challenges? Olivia answers that she is looking for job security and certified training.

        - **QED Approach:** During the interview with Olivia, the QED designer goes beyond the direct answers. They observe Olivia's hesitations, her changes of tone, the topics she approaches with more emotion or more silence. They ask questions that explore superpositions of intentions: When you speak of security, is it also a quest for meaning? Or a search for freedom despite the economic constraint? Tools such as the motivation scale (an implicit AttrakDiff) could be used to probe the tensions between the pragmatic dimension (I have to retrain) and the hedonic dimension (I aspire to work I am passionate about).
        - **QED Added Value:** Thanks to this approach, the team discovers that Olivia is not only motivated by security (an observable particle) but also by a deep desire for creative freedom and social recognition (a less explicit wave, an invisible presence). By understanding this superposition of usage states, the platform can offer Olivia paths that, on the surface, meet her need for security (certified training) but subtly integrate personal-development modules, opportunities for creative projects, or networking for recognition. The platform does not merely answer a stated need; it anticipates and nourishes the entire wave function of her aspirations.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI can transform the analysis of interviews. Instead of a simple text transcription, AI tools are able to analyze the tone of voice, facial micro-expressions, and the rhythm of speech to detect zones of emotional resonance or cognitive friction in Olivia. AI would make it possible to identify patterns of superposed usage states across many interviews, revealing unexpressed needs that would not be apparent in a linear analysis. During the interview, the AI could even suggest probing follow-up questions to ask, live. This would allow a faster and deeper understanding of the dark matter of user motivations.
    - **Optimization & Simplification:** By understanding the superpositions of intentions, one can design interfaces that adjust in real time to the user's emotional states. For example, if Olivia expresses frustration, the interface could automatically offer a simplified option or a direct path to support, without her having to look for it.
    - **Revelations for the Community:** User interviews become windows onto the collective consciousness of the experience. The designer is no longer a simple interviewer but a prober of fields, able to perceive the waves of latent desires and the turbulence of frustrations, opening the way to designs that resonate at a deeper level.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**

    - **Potential Pitfalls:** The ability to probe the invisible presence and the superposed states of an individual like Olivia confers great power. The ethical risk is to use this information to manipulate users' emotions or decisions (a direct link with potential Quantum Dark Patterns). Transparency about the analysis of behavioral and emotional data must be absolute.
    - **Implementation Challenges:** Adopting this approach calls for advanced training or understanding in active listening, cognitive psychology, and a sensitivity to biases. It also requires time for qualitative analysis and the interpretation of subtle signals. The integration of AI can turn these challenges into opportunities, acting as a discreet but powerful assistant. It optimizes the efficiency of the interview, enriching the interaction for both interviewer and interviewee, and thereby strengthens the very relevance of the understanding process, and will offer a tailored summary at the end of the interview.
    - **The Observer's Responsibility:** The designer conducting the interview must be aware that their presence and their questions influence the reality the user expresses. They are a creator of contexts, and must guarantee a safe and respectful space for the user's truth.

Usability Testing: Measuring Usage Realities and Revealing Points of Decoherence

- **The Classic Compass (Traditional Role):**
    - **Description:** User testing consists of observing real users interact with a product, service, or prototype to identify usability problems, friction points, and the challenges they encounter. The goal is to ensure that the product is intuitive, efficient, and pleasant to use. Tests can be moderated or unmoderated, in a lab or remote.
    - **Classic Limits:** Although crucial, classic usability tests tend to focus on identifying binary problems (it works / it does not work) and on improving linear flows. They can lack depth on the underlying reasons for behaviors or on the emotional quality of the experience beyond the task. Moreover, the test environment can influence the user's behavior, creating an artificial reality that does not always reflect the complexity of the real world. We often measure the collapse of the experience, without understanding the potential that led to that collapse.
- **The Quantum Lens (The QED Adjustment): Observation as a Revealing of Behavioral Superpositions and Energetic Frictions**

  In QED, a user test is a deliberate act of measuring the behavioral wave function (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). The observer (the designer) is aware that their presence, even discreet, can influence the reality the user unfolds. The goal is no longer only to note where Liam clicks, but to understand why he hesitates, what superpositions of actions he considers before choosing, and how energetic frictions (frustrations, confusions) manifest, even without being verbalized. The QED designer seeks to detect points of decoherence, those moments where the user's intention and the system's response are misaligned, where their experience fragments, where the interface no longer resonates with their mental flow. The test becomes an exploration of the multiple states the user might take and the alternative wave paths they do not take, but which reveal untapped potential.

- **The Experience Lab (Concrete & Innovative Use Cases): Optimizing a Flight-Booking Journey with QED Sensitivity**
  - **Concrete Scenario: Improving the journey of booking a plane ticket on a mobile app.**
    - **Classic Approach:** The team observes Liam, a frequent traveler, booking a flight. It notes the mistaken clicks, the missed fields, the loading times. The result is a list of bugs and direct interface optimizations.
    - **QED Approach:** During the test with Liam, the QED team does not merely record errors. It actively observes Liam's micro-expressions, his sighs, his changes of posture, the unusual pauses between actions. The goal is to detect superpositions of decisions: is Liam hesitating between two options that seem similar? Is he expressing a silent frustration at an interface requirement? The team looks for points of decoherence: for example, when Liam has a clear intention to search for a flight, but the complexity of the options (direct / with stopovers, flexible / non-flexible) creates an energetic friction that slows him down, even makes him doubt his initial choice.
    - **QED Added Value:** Thanks to this approach, the team understands that the major decoherence for Liam is not a poorly placed button, but the cognitive overload generated by the simultaneous presentation of too many complex options. Instead of simply moving a button, the QED solution could be to drastically simplify the initial interface, offering an experience in a very simple ground state, and letting Liam move to excited states (more advanced options) only if he decides to. This turns the booking journey from a task to accomplish into a fluid navigation that respects the negentropy of his experience by reducing the entropy of uncertainty.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**

- **AI Amplification:** AI can become a non-intrusive quantum observer. Video-analysis and sentiment-analysis algorithms could automatically detect the micro-signals of frustration, confusion, or superposition of intentions (observing the movement of Liam's cursor between several zones before a click). This would make it possible to generate energetic-friction maps or probabilistic points of decoherence at scale, across thousands of unmoderated tests, revealing systemic problems or unexpected UX black holes.
- **Optimization & Simplification:** By precisely identifying the moments of decoherence and the energetic frictions, designers can create adaptive interfaces that respond to these implicit signals. For example, if the AI detects a hesitation in Liam, it could proactively offer ultra-simplified contextual help or temporarily hide the less relevant options, offering a more fluid and less stressful experience.
- **Revelations for the Community:** User testing moves from a simple technical validation to an exploration of the user's states of consciousness. The designer is no longer a bug auditor but a decoder of behavioral waves, able to perceive the subtle resonances of human experience.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** Collecting such fine-grained data about Liam's emotional reactions raises major ethical questions (Law 1, Ethical Vigilance). How can we guarantee privacy and explicit consent for such measurements? The risk is to create systems that unconsciously exploit emotional vulnerabilities to optimize the conversion rate, at the expense of the user's well-being.
  - **Implementation Challenges:** Interpreting these subtle signals requires great expertise and specific training. The use of AI must be transparent and ethically governed to avoid abuses.
  - **The Observer's Responsibility:** The designer, in measuring the behavior of a user like Liam, must remember that the act of observation itself can influence the system. They must ensure that the design of the tests is as neutral and respectful as possible.

Contextual Inquiry / Field Studies: Revealing Entangled Behavioral Quanta in Their Natural Habitat

- **The Classic Compass (Traditional Role):**
  - **Description:** This method involves observing users in their natural environment (home, office, public place) while they perform tasks related to the product or service.

The goal is to understand the real context of use, the unverbalized behaviors, the adaptation strategies, and the hacks (tips or improvised solutions that users develop to get around a difficulty) that users develop to overcome challenges.

- **Classic Limits:** Observation can alter behavior because it changes the context. It is difficult to capture the entirety of the interaction field without focusing on specific particles. Interpretation can be subjective, and it is hard to see the complex entanglements among the user's behavior, their physical environment, their emotional state, and the digital tools. We observe reality but miss the superposition of intentions and invisible constraints.

- **The Quantum Lens (The QED Adjustment): The Quantum Microscopes of the Real and the Detection of Contextual Entanglements**

  In QED, contextual inquiry is a quantum immersion in the user's reality. The designer becomes an entangled observer (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI), seeking to detect entangled behavioral quanta, micro-actions, hesitations, glances, physical adjustments that, taken together, reveal a deep wave function of a need or a frustration. The goal is not only to see what the user does, but how their behavior is entangled with their physical, social, and emotional environment. It is the search for the points of decoherence hidden in natural interaction, where the experience fragments silently.

- **The Experience Lab (Concrete & Innovative Use Cases): Optimizing an Inventory-Management App for a Small Business**
  - **Concrete Scenario:** A team is developing a mobile app to help small merchants manage their inventory.
  - **Classic Approach:** The team interviews the merchants about their needs and tests the app in a lab.
  - **QED Approach:** The team, with Mathilde (UX Researcher), goes directly to a merchant for an in-depth contextual observation. Mathilde does not merely watch how the merchant interacts with the app. She observes:
    - The **micro-fluctuations of the context:** The merchant receives a customer, their phone rings, they have to enter a product quickly. Mathilde notes how these interruptions (quantum disturbances) influence the speed and accuracy of their input.
    - The **entangled gestures:** The merchant does not always look at the screen; they enter codes by scanning physical labels, they talk to an employee while handling

the app. Mathilde identifies the entanglements among the physical, verbal, and digital actions.

- The **hidden compensations:** The merchant uses a notepad beside them to remember references, or a little trick to avoid having to continually confirm messages in the app. These hacks are silent points of decoherence of the digital experience, revealing unexpressed gaps.

  - **QED Added Value:** By observing these entangled behavioral quanta in their natural context, Mathilde discovers that the challenge is not only the app's interface, but the smooth integration of the app into a chaotic work environment. To materialize this complexity and allow a dynamic adaptation, QED could introduce a Contextual Entanglement Gauge (CEG). Like adaptive design, which adapts the display according to screen size, this CEG would measure in real time parameters such as the user's cognitive load, the level of interruption, or the density of tasks. It would thus define states or slices of contextual complexity, allowing an adaptive QED or responsive QED approach. Concretely, faced with high entanglement (for example, a customer arrives during input, or the phone rings), the app (and a potential implicit QED framework) could automatically adjust its UX quanta: an anticipatory interface that pauses the input, offers contextual shortcuts activated by voice or scan, or integrates invisibly with the merchant's physical tools. Conversely, in low entanglement (a calm environment, focused attention), the interface could offer more details and options, optimizing efficiency.

    The goal is to minimize energetic friction by creating an experience that integrates harmoniously into the merchant's daily reality.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
  - **AI Amplification:** AI tools (computer vision, audio analysis) could become the eyes and ears of the QED observer, capturing and analyzing thousands of entangled behavioral quanta to detect patterns invisible to the human eye. AI could even identify behavioral singularities that are the precursors of new usage trends.
  - **Optimization & Simplification:** QED contextual observation makes it possible to design interfaces that resonate deeply with users' way of life, eliminating invisible frictions.
  - **Revelations for the Community:** This method turns the designer into an anthropologist of experience, able to reveal the implicit laws that govern human interactions with technology in their natural environment.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**

- ◦ **Potential Pitfalls:** Direct observation raises privacy questions. Informed consent and great transparency are essential (Law 1, Ethical Vigilance). The observer must stay neutral and not influence behavior.
- ◦ **Implementation Challenges:** It is a time- and resource-intensive method, requiring fine observation and analysis skills.
- ◦ **The Observer's Responsibility:** The designer must ensure that their observations lead to improvements that genuinely serve the user, and do not expose them.

III. Information Modeling and Structuring Tools: Mapping the Experience Fields and Emerging Potentials

These tools allow the Quantum Experience Architect to give shape to the invisible, to represent the complexity of systems and interactions. They turn raw data and insights into models that prefigure new usage realities, revealing the entanglements and the deep dynamics.

Personas & Empathy Maps: Beyond the Fictional Profile, the Spectrum of Entangled Usage States

- **The Classic Compass (Traditional Role):**
  - **Description:** The persona is a fictional user archetype, based on research data, representing a key segment of our audience. It serves to humanize the target, align teams, and guide design decisions based on specific needs, behaviors, and motivations. The Empathy Map complements this approach by detailing what the persona sees, says, thinks, feels, and does, and their pains / gains.
  - **Classic Limits:** Personas can become rigid caricatures, single people who do not capture the fluidity and complexity of real human behaviors. They are often based on a fixed state of the user, failing to account for their changes of mood, context, or needs over time or across situations. They struggle to reflect the contradictions or the multiple facets of a single individual. The notion of decoherence (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI) between what the user thinks and what they do is not always explicitly explored. Moreover, these classic personas too often ignore the richness of human neurodiversity. By implicitly focusing on a neurotypical cognitive functioning, they omit the specific needs, preferences, and challenges of neurodivergent individuals (such as those with ADHD, ASD, or dyslexia). This normality bias leads to designs that unintentionally generate cognitive

overload, fatigue, or a feeling of exclusion for a significant share of the population, making the design incomplete and potentially harmful.

- **The Quantum Lens (The QED Adjustment): The Superposition of Usage States as the Persona's Reality and the Invisible Presence of Latent Desires**

  In QED, a persona is not a frozen entity but a wave packet of potential experiences. It embodies the superposition of usage states (Law 1). This means the persona can simultaneously desire speed and security, simplicity and functional richness, depending on the moment, the context, or even their emotional state. The QED designer, with this lens, commits to mapping a spectrum of potentials that explicitly includes cognitive plurality and varied neurotypes, recognizing that the experience does not collapse the same way for everyone. The QED designer's work is no longer to define a behavior or a need, but to map this spectrum of potentials, recognizing that the user exists in a multitude of entangled usage states that will only collapse into an observable behavior at the moment of interaction (the act of measurement). The Empathy Map, with the QED lens, is a tool to probe not only what is visible (said, done) but also the invisible presence (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence) of deep, unexpressed motivations, latent contradictions, and aspirations that have not yet found their expression.

- **The Experience Lab (Concrete & Innovative Use Cases): A Quantum Persona for an Educational Streaming Platform**

  - **Concrete Scenario: Developing a streaming platform of educational documentaries for adults.**

    - **Classic Approach:** The team would have created a persona, Ambre, 32, an active professional, looking for history documentaries to educate herself in the evening. The design focuses on a well-organized catalog and a powerful search function.
    - **QED Approach:** When creating the persona Ambre, the QED team recognizes that she exists in a superposition of usage states:
      - Ambre, the studious learner (particle state): looks for precise information, reliable sources, certifications.
      - And simultaneously Ambre, the curious explorer (wave state): wants to be surprised, to discover unexpected topics, to let herself be guided by immersive suggestions.
      - And also Ambre, the tired person (particle state): looks for passive entertainment, short content, requiring little mental effort after a long day.

        - Ambre's QED Empathy Map does not merely list her pains and gains; it actively explores the tensions and decoherences among her aspirations: she wants to educate herself but often feels too exhausted to learn actively, she seeks depth but is drawn to short, visual content.
        - **QED Added Value:** This approach makes it possible to design a platform that does not force Ambre into a single box but adapts to her fluid usage states. The interface could offer, on the home page, a Deep Discovery section (for the curious explorer) next to a Simple Pleasures section (for the tired person). Implicit signals (time of connection, type of content previously watched, time spent on the page) could influence the order of presentation. The platform creates an experience that not only meets the identified needs but also nourishes the invisible presence of complex desires, fostering the homeostasis of Ambre's educational experience in her daily life.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI can generate dynamic quantum personas. Instead of static personas, the AI, by analyzing massive behavioral data (viewing history, interaction time, reactions to quizzes), could model the probabilities of Ambre's different usage states at any moment. It could then adjust the content flow, the recommendations, or even the tone of the interactions to match the most probable state, allowing a quantum hyper-personalization (Law 2) where the experience constantly teleports to the optimal state for the user.
    - **Optimization & Simplification:** By understanding the superpositions of states, one can optimize the journeys for fluid transitions between seemingly opposite modes of use, sparing the user from having to consciously choose a mode or a profile. This simplifies navigation and reduces Ambre's cognitive load.
    - **Revelations for the Community:** Personas and empathy maps become tools to visualize the complexity of the human soul in its interaction with digital systems. The designer is no longer a mere market segmenter but a cartographer of potentials, able to design for the richness and fluidity of human experience.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
    - **Potential Pitfalls:** Modeling quantum personas and exploiting Ambre's superposed states raise questions of personal and emotional data privacy (Law 1, Ethical Vigilance). The risk is to create over-optimized comfort bubbles or to steer choices without transparency. Informed consent for the collection of this fine-grained data is crucial.

- **Implementation Challenges:** Creating these dynamic personas requires a deep understanding of human psychology and advanced data-analysis skills. It also calls for close collaboration among designers, data scientists, and ethics experts.
- **The Observer's Responsibility:** The designer, in creating these complex models, must remember that they take part in constructing Ambre's reality within the system. They must design not for the maximization of engagement at all costs, but for the user's fulfillment and autonomy.

User Flows / Journey Maps: Navigating the Temporal Waves of Multi-State Journeys

- **The Classic Compass (Traditional Role):**
  - **Description:** User flows and customer journey maps are tools that visualize the path a user or customer takes to reach a specific goal with a product or service. They describe the steps, actions, touchpoints, thoughts, and emotions at each interaction. Their purpose is to identify friction points and opportunities for improvement, and to align the design with the user's needs.
  - **Classic Limits:** These tools are often designed as linear, idealized journeys or based on majority behavior. They struggle to represent the non-linearity of real experiences, the backtracks, the unexpected forks, the multiple entry and exit points, or a user's ability to exist in several journey states at once (shopping and looking up delivery information at the same time). They can miss the dark matter of paths not taken or of invisible interactions that influence the decision.
- **The Quantum Lens (The QED Adjustment): Mapping the Superposition of Journeys and UX Spatiotemporal Portals**

  In QED, a user flow or a journey map is not a simple straight line but a quantum network of potential journeys (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). Each node of the journey represents an experience state the user is in, and the arrows are the probabilities of transition between these states. The QED designer seeks to visualize the superpositions of journeys, that is, how a user can mentally consider several possible paths (or even follow several in parallel across different tabs) before collapsing onto a specific action. The UX spatiotemporal portals then become key touchpoints where the experience can teleport (Law 2: UX Teleportation) from one state to another, or even from one app to another, without a break. The goal is to understand the negentropy (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence) of the journey by reducing uncertainty and friction, while accepting and mapping the entropy of real non-linearity.

- **The Experience Lab (Concrete & Innovative Use Cases): Optimizing a Complex Sign-Up Journey for a Financial App**
    - **Concrete Scenario: Improving the onboarding (sign-up) process of a new personal-budgeting app.**
        - **Classic Approach:** The team draws a linear user flow: Download > Account creation > Bank connection > Budget setup. It identifies the steps where users drop off.
        - **QED Approach:** The QED team studies the journey of Sofia, a young professional eager to manage her finances better. The analysis reveals that Sofia does not follow a linear path. She begins the sign-up, but at the same time she opens another tab to check the app's reputation, reads reviews, then comes back, hesitates to connect her bank, pauses to answer a message, and resumes later. Sofia's journey is a superposition of micro-tasks and states of uncertainty. The QED journey map visualizes these states not as successive steps but as a cloud of probabilities where Sofia can be: an active sign-up state, an external-verification state, a contextual-pause state. The drop-off points are no longer simple errors but points of decoherence where Sofia's intention fragments in the face of a perceived complexity or a lack of trust.
        - **QED Added Value:** By understanding this non-linearity and these UX spatiotemporal portals (the moments when Sofia leaves the app for other channels), the team can design a resilient and adaptive onboarding journey. For example, the app could offer a quantum save of Sofia's sign-up state, letting her resume without frustration. Contextual messages could be sent to her mobile if she leaves the app, to reassure her trust, insofar as she had indicated this during one of the sign-up steps. The design aims to reduce the entropy of her experience by offering points of re-homeostasis at each micro-interruption, allowing Sofia to navigate her different states without friction.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI can generate predictive journey maps in real time. By analyzing the behavioral data of millions of users like Sofia, the AI could anticipate the probabilities of decoherence or bifurcation at each step. This would make it possible to trigger contextual interventions (help pop-up, reminder offer) just before the user drops off, or to personalize the flow of options according to their detected state of uncertainty. One could even model optimal personalized journeys according to predicted usage states, creating automatic UX Teleportations to the most relevant step for Sofia.

    - **Optimization & Simplification:** This approach makes it possible to identify not only the friction points but also the opportunities for entanglement (Law 1) with other services or platforms the user naturally uses. By integrating portals to these services (a direct link to external banking help if the connection fails), one simplifies the overall journey, even if it involves leaving the app.
    - **Revelations for the Community:** User flows and journey maps become tools to visualize the dance of probabilities of the experience. The designer is no longer a mere path tracer but a choreographer of multidimensional flows, able to design experiences that embrace the complexity and non-linearity of real life.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
    - **Potential Pitfalls:** The ability to anticipate points of decoherence and to steer the journey of users like Sofia can be diverted to create unwanted engagement loops or quantum dark patterns that exploit moments of vulnerability. Transparency about the adaptation logics is crucial.
    - **Implementation Challenges:** This advanced mapping requires complex data-analysis tools and a deep understanding of behavioral psychology. It calls for a team able to think in terms of dynamic systems rather than static processes.
    - **The Observer's Responsibility:** The designer must ensure that optimizing journeys to reduce friction does not prevent the user from making informed decisions or from choosing less efficient but more meaningful paths for themselves. The point is to guide, not to constrain, the user's freedom of choice.

Card Sorting / Tree Testing: Decoding the Informational Wave Function and Structuring Cognitive Realities

- **The Classic Compass (Traditional Role):**
    - **Description:** Card Sorting is a technique for understanding how users group and categorize information, helping to design intuitive information architectures. Tree Testing evaluates users' ability to find specific information within an existing or proposed navigation structure, without the visual design elements. These methods aim to create a content organization that matches users' mental logic.
    - **Classic Limits:** These tools tend to freeze a best informational structure, based on an average of the responses. Even though they sometimes allow an element to be duplicated across several categories (for example, ”Invoices” under ”My documents” and ”My account”), they often mask significant individual variations or the very nature

of cognitive superpositions, where the same element can logically belong to several categories at once for a given user. They struggle to capture the underlying reasons for categorization choices or the hesitations that reveal deep structural ambiguities, thus missing the entangled nodes of human understanding.

- **The Quantum Lens (The QED Adjustment): Measuring the Probabilities of Classification and Revealing Cognitive Entanglements**

  In QED, Card Sorting and Tree Testing are not simple sorting exercises but measurements of the user's informational wave functions (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). Each card or piece of information does not have a single, frozen place, but exists in a superposition of potential categories in the user's mind. The QED designer seeks to understand not only the final choice (the collapse of the category) but also the intrinsic probabilities and the cognitive entanglements among the concepts (for example, whether ”Help” and ”Support” are seen as strongly entangled by the user). Tree Testing, with a QED lens, aims to identify the quantum friction points, those moments when the user hesitates because the information architecture does not resonate with the superposition of their expectations, forcing them to make their intuition decohere in order to follow an imposed logic. The goal is to design an architecture that reduces cognitive entropy by aligning with the intrinsic complexity of human thought.

- **The Experience Lab (Concrete & Innovative Use Cases): Structuring an E-commerce Site for a Fluid, Intuitive Experience**

  - **Concrete Scenario: Reorganizing the navigation of a large e-commerce site for technology products.**

    - **Classic Approach:** The team runs a card sort to find out where users classify wireless earbuds, portable chargers, smartwatches. The results guide the creation of categories such as Audio, Accessories, Wearables (connected devices worn on the body). If a product, such as wireless earbuds, is frequently classified under ”Audio” and ”Accessories,” the team may decide to place it in both categories to improve its discoverability.

    - **QED Approach:** The QED team runs a Card Sorting with Mathéo, a technology enthusiast. The observation goes well beyond the simple final sorting. The team notes that Mathéo frequently hesitates between ”Accessories” and ”Audio” for the earbuds, accompanying this with comments like ”It could go in both, actually” or with sighs. The analysis of the card-sorting data reveals that, for thousands of users like Mathéo, certain products exist in a persistent superposition of relevant categories. For example, ”Gaming headsets” is perceived as entangled simultaneously with ”Audio,” ”Gaming,” and ”PC accessories.” The QED approach does not merely duplicate the product statically; it seeks to understand and leverage this entanglement in real time, recognizing that these multiple memberships are the user's reality.

    - **QED Added Value:** Rather than choosing one or two fixed categories in which to place a product, the QED architecture recognizes and operates within these dynamic informational entanglements. The site could implement a contextual navigation that adapts to Mathéo's state of intention. For example, if Mathéo searches for ”earbuds” and has recently visited the ”Gaming” section, the site could prioritize the display of ”gaming headsets” in the results and highlight their dual membership. Quantum navigation portals could be created, such as dynamic tags or cross-category filters, which not only refine the search but visually or contextually reflect the product's superposed nature. The user thus does not have to choose a single category to find their product, but navigates naturally within the wave function of their own intentions (for example, ”I am looking for earbuds for gaming” or ”I am looking for earbuds for music”). This approach ensures a more natural and less frustrating experience, where the system resonates with the user's complex, fluid logic.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
  - **AI Amplification:** AI can optimize the quantum structure of information in real time, that is, dynamically organize the data so that it matches users' different ways of thinking. Unsupervised learning algorithms could analyze the real navigation paths of millions of users like Mathéo, the searches, and the clicks to identify dynamic clusters of relevance. The AI could then personalize the navigation tree for each user, or even display probabilistic categories (the most relevant for a given user at a given moment). This would lead to a self-adaptive information architecture that teleports the user to the content they are looking for, even when they do not know how to name it precisely. This capacity is at the heart of the Contextual Modulation of UX (Law 2), where the experience actively adapts to the user's interaction field.
  - **Optimization & Simplification:** By identifying the cognitive entanglements that generate energetic friction (those moments when users hesitate or tire in the face of needless complexity), designers can simplify the structures. Rather than creating ever more granular categories, they can create multiple, intelligent access points that resonate with the flexibility of human thought. This turns the journey from a task to accomplish into a fluid navigation that, by reducing uncertainty, respects the negentropy of the experience.

- **Revelations for the Community:** Card Sorting and Tree Testing, under the QED lens, become tools to map the users' mental space. The designer is no longer a mere content organizer but an architect of cognition, able to build informational universes that reflect the richness and complexity of human thought.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** The aggressive personalization of information architecture, while beneficial, can create filter bubbles where the user is only exposed to information that confirms their thought patterns, limiting discovery. It is crucial to maintain a balance between personalization and exploration. This risk is directly tied to the principles of Entanglement, Complementarity, and Duality in UX ◐ UI (Law 1) and calls for constant ethical vigilance to avoid potential Quantum Dark Patterns.
  - **Implementation Challenges:** Implementing dynamic, AI-based information architectures is technically complex and requires constant maintenance to guarantee relevance and avoid algorithmic biases.
  - **The Observer's Responsibility:** The designer must ensure that the fluidity of the navigation is not achieved at the expense of clarity and of the user's ability to understand the overall structure of the site. They must design informational universes that respect the user's autonomy and capacity for discovery.

IV. Design Tools: Manifesting Experience Realities through Visual and Interactive Quanta

These tools are the brushes and palettes of the Quantum Experience Architect. They make it possible to turn abstract models into tangible interfaces, where each design element is not only a visual particle but a wave of interactive and emotional potential, ready to collapse into a coherent experience.

Wireframing (Low-Fidelity Mockups): Sketching the Fields of Possibilities and the Interaction Potentials

- **The Classic Compass (Traditional Role):**
  - **Description:** Wireframing consists of creating schematic, simplified representations of the user interface, focusing on the layout of elements, the information hierarchy, and the key features, without the visual details. Low-fidelity prototyping often includes

paper sketches or very basic clickable prototypes. The goal is to quickly validate the structure and the flows before investing in visual design.

  - **Classic Limits:** Wireframes are often seen as fixed plans of an interface, focusing on a static state. They can miss the dynamic dimension of interaction and the way the interface will behave in response to the user's actions. They tend not to represent the intermediate states, the subtle animations, or the micro-interactions that are nonetheless crucial for the emotional experience. One designs an interface particle rather than a wave of interactive potential.

- **The Quantum Lens (The QED Adjustment): Ground-State Wireframes and the Superposition of Possible Interactions**

  In QED, a wireframe is not a rigid plan but a ground-state representation of the experience field (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence). Each content zone, each button, is not just a static element but a wave of interactive potential (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI) that can collapse into different actions or display varied states depending on the measurement (the user's interaction). The goal is to visualize the superposition of possible interactions beyond the nominal path. The QED designer uses the wireframe to explore how the interface can breathe, adapt, or even transform in response to the user's multiple usage states, and how animations (even conceptual ones) can represent fluid quantum-state transitions.

- **The Experience Lab (Concrete & Innovative Use Cases): Designing a Dynamic Event-Management App**

  - **Concrete Scenario: Developing a mobile app to manage events (conferences, workshops) with personalized agendas.**
    - **Classic Approach:** The team wireframes a home page with a navigation menu, a news feed, and a "My Agenda" button. Each screen is a fixed mockup.
    - **QED Approach:** The team wireframes the home page for Mei, an event organizer. But instead of freezing the screen, it sketches the superpositions of potential states of this page:
      - The Discovery state (for Mei exploring new events).
      - The My Agenda state (for Mei checking the events she has registered for).
      - The Notification state (for Mei receiving an alert about an imminent event).
    - The wireframes include specific annotations on the quantum-state transitions: how a simple swipe or tap can teleport Mei from a general news feed to her personal agenda without a visible break, or how the feed content can reconfigure

according to the measurement of her interest (a click on a type of event). The team does not merely show where the elements are, but how they emerge or disappear to adapt the experience.

- **QED Added Value:** This approach makes it possible to visualize, from the low-fidelity phase, how the app will handle the complexity of Mei's different uses. This leads to wiser design choices, such as quantum widgets whose content or function changes dynamically according to Mei's context, or animations that reinforce the perception of fluidity and continuous transformation. The goal is to create an interface that is not a succession of screens but a unified experience field that adapts to the wave function of Mei's needs.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
  - **AI Amplification:** AI could generate adaptive wireframes in real time, based on past behavioral data or complex quantum personas. These dynamic wireframes could simulate the superpositions of interactions and the state transitions with greater fidelity, making it possible to test design hypotheses much earlier. AI could even suggest optimal interactive wave paths to reduce energetic friction even before high-fidelity prototyping.
  - **Optimization & Simplification:** By conceptualizing the interface as a field of waves, one can identify the redundancies and the needless decoherence states. This makes it possible to simplify the flows by creating more direct transitions and UX Teleportations (Law 2) between sections, making the experience more intuitive and less cognitively loaded for Mei.
  - **Revelations for the Community:** Wireframing and mockups become powerful tools for architecting interaction potentials. The designer is no longer a mere positioner of frozen visible zones but an architect of the invisible, structuring superpositions of states to reveal the full energetic richness of the experience.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** The complexity of quantum wireframes could make communication with non-expert stakeholders harder. The point is to find a balance between the richness of the interactions and the clarity of the message, to avoid designing overly magic interfaces that destabilize the user through their unpredictability or opacity.
  - **Implementation Challenges:** Conceptualizing and documenting these superpositions of interactions requires a renewed methodology and tools more advanced than classic wireframing software. This complexity calls for a strong capacity for abstraction, which AI could partly take on. Placed under the governance of the Quantum

Experience Designer, the AI would materialize superposed states of zones or content, automatically annotate the plans, and generate clear mind maps illustrating the use cases and journeys, in order to ease understanding and validation by all stakeholders. For each component or interface level (from micro to macro), the Quantum Expansive Designer could specify precise parameters or prompts, thus defining the limit of the possible and the rules of evolution allowed by the system. The AI, drawing on these indications, would propose different variants of interactions or configurations, while respecting the constraints, intentions, and goals defined by the designer. This dialogue between human and machine would make it possible to explore a broader spectrum of solutions, while guaranteeing the coherence, control, and intelligibility of the generated interfaces.

- **The Observer's Responsibility:** The designer, in sketching these potentials, must ensure that the interaction choices serve the user's autonomy and control, and not the mere optimization of hidden engagement metrics. The invisible must not become an ethical gray zone.

Prototyping (Mid- and High-Fidelity): Materializing the Waves of Experience and Testing Quantum Realities

- **The Classic Compass (Traditional Role):**
  - **Description:** Prototyping consists of creating functional versions (more or less faithful) of a product or interface, making it possible to simulate user interaction. Mid-fidelity prototypes add visual details and more advanced interactivity, while high-fidelity ones approach the look and behavior of the final product. The goal is to validate concepts, gather early user feedback, and iterate before final development.
  - **Classic Limits:** Even high-fidelity prototypes are frozen versions of the experience, designed for an ideal journey or a specific state. They struggle to simulate the complexity of real interaction, the unexpected error states, the context changes, or the superpositions of usage states (the user being at once hurried and curious, for example). Testing on a prototype often focuses on the ability to accomplish a task, and less on emotional resonance or the system's ability to adapt to the nuances of human behavior.
- **The Quantum Lens (The QED Adjustment): Superposed-State Prototypes and the Decoherence of Probabilistic Journeys**

  In QED, a prototype is not a simple functional mockup but a simulator of potential experience realities (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible

Presence). The goal is to build superposed-state prototypes where the interface can manifest different realities depending on the measurement (the user's interaction) or on simulated contextual data. Each interactive element (a button, an input field) is perceived as a wave of probabilities of actions and reactions, rather than a frozen element. The QED designer uses prototyping to explore the points of decoherence, those moments when the user's journey diverges from the expected, revealing a break in the interface's logic or an opportunity to adapt the presented reality. Testing on a prototype becomes an observation of the dark matter of paths not taken and of states not manifested, making it possible to understand why the experience collapses in a certain way.

- **The Experience Lab (Concrete & Innovative Use Cases): Designing a Smart Virtual Assistant for Task Management**
  - **Concrete Scenario: Developing a mobile virtual-assistant prototype that helps manage daily tasks (appointments, reminders, shopping lists).**
    - **Classic Approach:** The team prototypes an assistant that responds to specific voice commands, displays task lists and reminders. The test consists of seeing whether the user can add a task and receive a reminder.
    - **QED Approach:** The team designs a prototype with Omar, a very busy professional. The prototype is not only interactive; it simulates contextual states: Is Omar in the car? At the office? At home? Is he typing or speaking? The prototype integrates superpositions of intentions for Omar:
      - He says ”Add to my list,” but his tone of voice indicates stress (particle state A).
      - He starts typing a message but deletes it, hesitating over the wording (wave state B).
    - The prototype makes it possible to simulate points of decoherence: for example, Omar says ”Remind me of the meeting” but the assistant cannot find the meeting in his calendar. The QED prototype does not merely display an error message; it simulates clarifying questions such as ”Which week's meeting?” or ”Do you want me to create a new meeting?”, or even a mode change ”Switch to text-input mode for more precision?”, according to the emotional state or the context (via the simulation).
    - **QED Added Value:** By prototyping these quantum realities and superposed states, the team can test not only the functionality but also the homeostatic resilience of the assistant in the face of Omar's unexpected or ambiguous behaviors. This leads to an assistant that does not merely respond to direct commands but is able to read Omar's experience field, anticipate his unexpressed needs, and

adapt fluidly to his contextual and emotional variations, making the interaction more human and intuitive.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI could turn prototyping into quantum simulators of the experience. Predictive AI models could simulate thousands of probabilistic journeys based on the real behaviors of users like Omar, and generate synthetic points of decoherence even before the first user test. This would make it possible to explore unforeseen scenarios and to optimize the interface's adaptability upstream. The AI could even create dynamic prototypes that adjust in real time to the user's simulated data (self-modifying prototypes).
    - **Optimization & Simplification:** By exploring the superposed states from the prototyping stage, one can design interfaces that radically simplify complex interactions by hiding needless complexity. Rather than multiple options, the interface offers the right state at the right moment for Omar, reducing cognitive load.
    - **Revelations for the Community:** Prototyping becomes an art of materializing experience realities. The designer is no longer a mere assembler of mockups but a shaper of the intuitive, able to give form to interactions that behave like a natural extension of human thought.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
    - **Potential Pitfalls:** The complexity of superposed-state prototypes can be hard to communicate or test if the tools are not suited. We must ensure that adaptability does not turn into unpredictability for the user, and that the AI's choices stay transparent to the designer.
    - **Implementation Challenges:** Creating prototypes that simulate quantum states requires advanced tools and a more systemic design approach. It takes more time and resources than classic prototyping, but the return on investment is potentially very high.
    - **The Observer's Responsibility:** The designer must ensure that the fluidity and adaptability of the prototype do not mask dark patterns that could manipulate Omar's decisions. The goal is to empower him, not to guide him blindly.

UI Design Tools (Figma, Sketch, Adobe XD, etc.): Shaping the Visual Quanta and the Aesthetics of Experience Realities

- **The Classic Compass (Traditional Role):**

- **Description:** User interface (UI) design tools such as Figma, Sketch, or Adobe XD are essential for creating the final visual mockups, the UI components, the design systems, and the specifications for development. They make it possible to define typography, colors, iconography, images, and the precise layout of all graphic elements. The goal is to ensure visual coherence, aesthetic appeal, and compliance with brand guidelines.
- **Classic Limits:** These tools, by their nature, focus on the static aspect and the perfect rendering of an interface. They can encourage a view in which visual design is a decorative layer or a simple transcription of a wireframe. They struggle to represent the life of the interface, the micro-animations, the subtle haptic feedback, the non-intrusive loading states, or the way the interface can breathe and adapt according to dynamic data or the user's emotional states. The visual quantum is often treated as an isolated particle rather than as a wave entangled within the experience field.

- **The Quantum Lens (The QED Adjustment): The Visual Quantum as a Vector of Entangled Information and Emotion**

  In QED, each pixel, each color, each animation in the UI is not a simple visual element but an entangled visual quantum (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). It carries within it a charge of information, a design intention, and an emotional potential that resonates throughout the whole experience. The QED designer does not merely paint the interface; they shape the visual field so that each visual element is a wave ready to collapse into a meaningful, emotion-laden experience for the user. Aesthetics is not only a question of beauty but of quantum resonance, that is, how the visual design harmonizes with the user's implicit expectations and emotional states. The invisible presence (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence) manifests through the subtle use of light, shadow, and movement, creating an experience that goes beyond the mere glance.

- **The Experience Lab (Concrete & Innovative Use Cases): Designing a Sensory Meditation App**
  - **Concrete Scenario: Creating the visual interface of a meditation app that adapts its environment (sounds, visuals) to the user's state.**
    - **Classic Approach:** The UI design team would create static screens with a choice of themes (forest, beach, mountain), buttons for guided meditations, and an audio player. The visuals are fixed.

- **QED Approach:** The team works on the UI for Aisha, a user seeking serenity. Instead of static designs, the team designs visual quanta that are in a superposition of states:
    - A Forest background is not a fixed image but a field of possibilities: the wind can blow gently (micro-animation), the light can change subtly like sunrise (dynamic gradient), the sound of birds can vary in intensity (light haptic feedback).
    - The buttons are not simple icons; they are portals that, on hover or light press, unfold a luminous aura or a subtle vibration, indicating their action potential and creating a resonance (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI).
- The UI is designed so that, if Aisha enters a state of high tension (simulated via biofeedback data, for example), the dominant colors become softer, the rhythm of the animations slows, and the contrast subtly decreases, without her having to intervene. The interface senses Aisha's state and collapses toward the most soothing visual and auditory configuration.
- **QED Added Value:** This approach makes it possible to create an interface that is not only pleasant to look at but is emotionally entangled with the user. Each visual and interactive element contributes to maintaining Aisha's emotional homeostasis (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence), by adapting the visual experience field to her changing relaxation needs. The design becomes a form of quantum therapy where the digital environment responds subtly to improve well-being.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI could analyze the biometric data (heart rate via sensors on a connected watch, or brain waves via affixed or implanted sensors) of users like Aisha, in real time, to orchestrate adaptive UI landscapes. The typography, the spacing, the intensity of the colors, and even the style of the icons could reconfigure dynamically to optimize the user's emotional state, creating fluid and teleportable interfaces (Law 2: UX Teleportation) that instantly adapt to the wave function of well-being.
    - **Optimization & Simplification:** By understanding that the visual quantum is a vector of information and emotion, one can drastically simplify the interface by eliminating the superfluous. Each element must have a purpose and a resonance. Design systems could include quantum rules for managing complex, adaptive visual states, reducing the designer's workload.

    - **Revelations for the Community:** UI design, enriched by QED, becomes the art of sculpting visual energy to influence the emotional and cognitive state. The designer is no longer a mere stylist but a conductor of visual quanta, able to create interfaces that vibrate in harmony with the human soul.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
    - **Potential Pitfalls:** Using Aisha's emotional and physiological data to adapt the UI raises major ethical questions (Law 1, Ethical Vigilance). The risk is to create systems that unconsciously manipulate the user's emotions for commercial purposes or excessive engagement. Transparency and consent are necessary.
    - **Implementation Challenges:** Designing interfaces this dynamic and emotionally entangled requires advanced skills in psychology, data science, and development. The complexity of these systems can also make testing harder.
    - **The Observer's Responsibility:** The designer, in shaping these visual realities, must be a guardian of ethics, ensuring that the interface serves Aisha's well-being and autonomy, and not a short-sighted optimization of metrics.

V. Continuous Evaluation and Measurement Tools: Measuring Expanding Realities and Detecting Weak Signals

These tools are the telescopes and seismographs of the Quantum Experience Architect. They do not merely report past facts but make it possible to measure ongoing dynamics, detect weak signals, predict future evolutions, and understand how the experience keeps transforming after deployment. They reveal the energy fields and the gravitational waves of usage behaviors.

Data Analysis (Metrics, Heatmaps, Session Recordings): Decoding the Quantum Traces of Real Interactions

- **The Classic Compass (Traditional Role):**
    - **Description:** Web and mobile data analysis (Google Analytics, Matomo, etc.) makes it possible to track quantitative metrics (click rates, time spent, conversion rates, journeys). Heatmaps visualize the interaction zones (the more red there is, the more clicks occurred there) and the scrolls. Session recordings offer the possibility of replaying individual journeys to understand precisely how users behave. The goal is to identify trends, performance problems, and optimization opportunities based on aggregated behaviors.

  - **Classic Limits:** These tools are powerful but often limited to aggregated, out-of-context measurements. They show what happened but struggle to explain why. Raw data does not reveal the underlying wave function of intentions, hesitations, or unexpressed frustrations. Heatmaps freeze the activity, and session recordings can be snapshots of a behavior, but not of the superposition of thoughts or emotions that preceded it. We observe the final collapse of the interaction but miss the quantum process that led to it.

- **The Quantum Lens (The QED Adjustment): Data Quanta as Revealers of Emotional Fields and Large-Scale Decoherence**

  In QED, each data point (a click, a scroll, a pause time) is not a simple isolated particle but an entangled data quantum (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI) that carries within it information about the user's intention, context, and emotional state. QED analysis does not merely count clicks; it seeks to detect quantum anomalies, micro-mouse-movements, hesitations in filling out a form, subtle variations in scroll speed that are weak signals of energetic friction or of a superposition of usage states (the user is both confused and determined to find the information). Heatmaps become visual energy-field maps, showing the zones of resonance or dissipation. Session recordings are readings of behavioral wave functions, revealing the moments of large-scale decoherence (when a large number of users diverge from the norm) or the hot spots where the experience fragments (when the user journey becomes confused or stalls).

- **The Experience Lab (Concrete & Innovative Use Cases): Optimizing an Online Medical-Appointment Process**

  - **Concrete Scenario:** Improving a platform for booking appointments with health professionals.
  - **Classic Approach:** The team analyzes the data to see at which step users abandon the booking. The heatmaps show that the confirmation button gets few clicks.
  - **QED Approach:** The QED team analyzes the behavior of Bryan, a user trying to book an appointment. Beyond the classic metrics, advanced analytics tools (with micro-interaction sensors) detect:
    - Micro-mouse-movements over the Cancel button before clicking Confirm (a sign of a superposition of states: commitment / doubt).
    - An unusually long pause on the summary page before validation (a sign of energetic friction or of missing information).

        - Unexpected cold zones on the heatmaps, where Bryan's eye lingers but without a clear interaction, suggesting an invisible presence of misunderstood information or unresolved questions.
    - **QED Added Value:** By correlating these subtle data quanta, the team understands that the problem is not only a button that gets few clicks but a latent anxiety in Bryan about confirming the appointment ("Did I check everything? What if I made a mistake?"). The QED solution will not be just a more visible button but could be a quantum confirmation: a small confidence animation after the click, a visual micro-summary of the key information just before validation, or soothing haptic feedback. The goal is to reduce the entropy of uncertainty and reinforce Bryan's emotional homeostasis at the critical moment of confirmation, by creating a sense of security and clarity.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI can become the ultimate quantum decoder of data. Machine Learning models could analyze millions of behavioral wave functions (combined via interaction history, biometrics, and context) to identify decoherence patterns before they manifest as abandonment. The AI could predict when Bryan is about to feel frustration and trigger quantum interventions (contextual help, journey simplification) to bring him back to a homeostatic experience state. This allows a predictive optimization and a UX Teleportation (Law 2: Scalar Field and Contextual Modulation of UX) toward an optimal state even before the problem becomes conscious for the user.
    - **Optimization & Simplification:** By detecting weak signals and energetic frictions, teams can simplify complex journeys by eliminating the invisible sources of uncertainty. One designs interfaces that resonate with the fluidity of the user's thought, reducing Bryan's cognitive load.
    - **Revelations for the Community:** QED data analysis turns raw numbers into narratives of lived experience. The designer is no longer a mere number analyst but a reader of quantum imprints, able to perceive the invisible dynamics that shape the relationship between humans and the digital.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
    - **Potential Pitfalls:** The fine-grained analysis of Bryan's data quanta and weak signals raises massive ethical questions of privacy and surveillance (Law 1, Ethical Vigilance). How can we guarantee that this deep knowledge of the user is not used for subtle

manipulation or to create behavioral profiles without consent? Transparency and the user's control over their data are essential.

  - **Implementation Challenges:** Implementing such analysis systems requires a robust technical infrastructure, advanced data-science skills, and an impeccable data ethics.
  - **The Observer's Responsibility:** The designer, in using these powerful tools, must always remember that each data point represents a fragment of a human being's lived experience. The goal is to improve their experience, not to control it.

Surveys / Questionnaires: Probing the Perception Fields and the Probabilities of Opinion

- **The Classic Compass (Traditional Role):**
  - **Description:** Surveys and questionnaires are quantitative and semi-quantitative methods for gathering the opinions, preferences, satisfaction, or demographic data of a large sample of users. They make it possible to validate hypotheses at scale, segment the audience, or measure overall satisfaction (Net Promoter Score, NPS: the probability of a future recommendation, and Customer Satisfaction Score, CSAT: immediate satisfaction).
  - **Classic Limits:** Surveys capture a reality at the moment of the answer, but opinions can be fluid and contextual. They struggle to capture internal contradictions (a user who says they love a feature but rarely use it), unconscious motivations, or superpositions of emotions (the user is satisfied and frustrated by different aspects). Closed questions reduce the complexity of thought to binary choices, missing the wave function of nuanced perceptions.
- **The Quantum Lens (The QED Adjustment): Measuring the Opinion Wave Function and Revealing Perceptual Entanglements**

  In QED, each answer to a survey is not an isolated opinion particle but a measurement of the user's opinion wave function (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). The goal is to understand not only the majority answer (the collapse of the dominant opinion) but also the latent probabilities of alternative opinions, the perceptual entanglements (when the opinion about a feature is tied to the opinion about customer service, even if they are not directly asked together), and the points of semantic decoherence (when the words used by the user reveal a different understanding of the question asked). The QED designer uses surveys to probe the perception fields and identify the tensions or superpositions of emotional states that may exist within the audience.

- **The Experience Lab (Concrete & Innovative Use Cases): Assessing Satisfaction with a New Online Collaboration Feature**
  - **Concrete Scenario: A company has launched a new collaborative whiteboard feature in its project-management tool and wants to assess user satisfaction.**
    - **Classic Approach:** The team sends a survey asking: "Are you satisfied with the collaborative whiteboard (Yes / No / Neutral)?" and "What is the most useful feature?"
    - **QED Approach:** The team designs a survey for Rohan, a project manager. Instead of binary questions, the QED survey includes questions with subtle options, more numerous open questions, and projective techniques ("If the whiteboard were an animal, which one would it be and why?"). The analysis does not stop at averages. It looks for:
      - The **superpositions of opinion states:** Rohan answers that he is "Very satisfied," but his open comments reveal frictions about the complexity of the interface (an apparent contradiction, an entanglement).
      - The **semantic decoherences:** Several users, including Rohan, use the word "intuitive" to describe different aspects, revealing that the term is not interpreted uniformly.
      - The **weak resonances:** A small percentage of open answers mention needs the team had not anticipated, such as integration with another tool, which are weak signals of future desires.
    - **QED Added Value:** By analyzing the survey with a QED lens, the team does not merely learn that 70% are satisfied. It understands why the remaining 30% are not, and it identifies that even among the satisfied, like Rohan, there are tensions and opportunities for improvement. For example, the team realizes that the apparent complexity for some is perceived as functional richness by others. The QED solution could be to offer quantum interface modes, a simplified mode for beginners and an advanced mode for experts (with even finer granularity, if needed), letting each user collapse the interface toward the state that resonates best with their skill level and context.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
  - **AI Amplification:** AI can analyze thousands of open text answers from surveys to detect clusters of opinion superposition and emotional resonances (Law 1). Advanced natural-language-processing models could identify points of semantic decoherence at scale, where users employ the same words but with different intentions. The AI

could even generate visual tension clouds, showing the most frequent contradictions of opinion, offering a quantum mental map of the users.

  - **Optimization & Simplification:** By understanding the superpositions of opinions, one can create more adaptive interfaces that do not force the user into a single path. This makes it possible to offer intelligent options that meet needs that seem contradictory on the surface but coexist in the user's reality.
  - **Revelations for the Community:** Surveys become tools to map the landscapes of collective perception. The designer is no longer a mere statistician but a prober of collective consciousness, able to perceive the waves of opinion that run through the community and turn them into design opportunities.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** The in-depth analysis of opinions and their nuances can be perceived as intrusive if it is not explained. It is crucial to guarantee the anonymity and informed consent of Rohan and the other respondents, especially when seeking to detect internal tensions or contradictions.
  - **Implementation Challenges:** Designing surveys that capture the opinion wave function requires great finesse in wording the questions. Analyzing this enriched qualitative data, even with AI, remains complex and requires expert skills.
  - **The Observer's Responsibility:** The designer must ensure that understanding the superpositions of opinions serves to enrich Rohan's experience, not to lock him into a filter bubble of pre-conditioned opinions.

A/B Testing: Observing the Duality of Experiential Realities to Optimize Emergence

- **The Classic Compass (Traditional Role):**
  - **Description:** A/B testing (or comparative testing) is a method that consists of creating two versions (A and B) of the same element (a web page, a button, a headline) and presenting them randomly to segments of users. The performance of each version is measured (conversion rate, clicks, etc.) to determine which is more effective. The goal is to optimize the experience based on hard data, by identifying the version that generates the best behavior.
  - **Classic Limits:** A/B tests treat users as independent entities that vote for version A or B. They simplify reality by assuming that the experience is linear and that the best version for the majority is the best for everyone. They can miss the nuances: why is version B better? What are the emotional or contextual states of the users that led to

this result? They do not reveal the superposition of perceptions where one version could be superior in some respects and inferior in others, nor the resonance (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI) of design elements that interact with one another.

- **The Quantum Lens (The QED Adjustment): Tests of the Particle-Wave Duality of UX and the Measurement of Quantum States**

  In QED, an A/B test is not only a comparison of two versions but an observation of the wave-particle duality of the experience (Law 1). Each version A or B is a potential experiential state that exists in superposition until the user, through their interaction (the measurement), forces the collapse of the wave toward an observable reality (click, conversion). The QED designer uses A/B testing to understand not only which version performs best but why it resonates better with the wave function of the user's needs and intentions. They seek to identify the entanglements among the design elements (is the color of a button more effective if the text is reworded?), and to detect the decoherence anomalies, those moments when a version performs oddly for a segment of users, revealing an unexpected superposition of their expectations or a hidden friction point.

- **The Experience Lab (Concrete & Innovative Use Cases): Optimizing an Account-Creation Process for a Subscription Service**

  - **Concrete Scenario:** A subscription-service platform wants to optimize its sign-up form to increase the completion rate.
  - **Classic Approach:** The team A/B tests two versions of the form: version A (long, with all the fields) and version B (shorter, asking for the bare minimum at first). Version B wins with a better completion rate.
  - **QED Approach:** The QED team designs an A/B test for Dave, a potential new user. Instead of simple versions A and B, it introduces quantum variants of the form:
    - **Version A:** A long form, but with a highly engaging visual progress indicator and subtle micro-animations for each completed field, creating a sense of accomplishment (particle-like).
    - **Version B:** A form that is short at first but expands dynamically (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence) to ask for more information after the first step, with contextual questions that appear according to Dave's previous answers (wave-like).
  - **The metrics go beyond the completion rate:** they include the time spent per field, the number of backtracks, and even the emotional friction detected by behavioral-analysis tools (hesitant mouse movements, typing speed). The analysis reveals that

while version B has a higher initial completion rate, version A, thanks to its entangled visual quanta (Law 1), generates better emotional engagement and a better quality of data provided by Dave over the long term.

- **QED Added Value:** By observing the duality of Dave's behaviors and those of the other users across these more nuanced versions, the team does not merely choose the winning version. It understands that the sign-up experience is not linear but a superposition of desires (speed versus completeness). The QED solution is not a form A or B but an adaptive sign-up architecture that, based on Dave's first interaction quanta (his typing speed, his hesitations), would collapse toward a shorter or more guided experience, adjusting the form's reality to maximize homeostasis (Law 0) and conversion, while providing an optimal user experience and less friction.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
  - **AI Amplification:** AI can orchestrate dynamic quantum A/B/n tests in real time, where not only two versions are tested but thousands of micro-variations are deployed simultaneously. The AI would learn continuously from the measurements of each interaction, adjusting and generating new variants to explore the field of possibilities of the experience, leading to a continuous and teleportable optimization (Law 2: Scalar Field and Contextual Modulation of UX) where the interface self-optimizes for each individual user like Dave.
  - **Optimization & Simplification:** By understanding the superpositions and entanglements of the design variables, one can identify the most influential elements. This makes it possible to simplify interfaces by focusing on what has a real quantum impact on Dave's experience, rather than making arbitrary adjustments.
  - **Revelations for the Community:** QED A/B testing turns experimentation into a quest for emergent realities. The designer is no longer a mere tester but an observer of alternative dimensions, able to reveal the laws that govern the effectiveness of the experience.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** A/B testing at a quantum scale raises ethical questions about manipulating the user experience without informed consent (Law 1, Ethical Vigilance). If the AI continuously optimizes Dave's experience without his knowledge, where is the line between optimization and coercion?
  - **Implementation Challenges:** The tools and skills needed for QED A/B testing are considerably more complex than traditional methods, requiring a data infrastructure and Machine Learning experts.

- **The Observer's Responsibility:** The designer must ensure that the optimizations based on QED A/B testing always serve Dave's autonomy and best interests, and do not guide him toward undesirable engagement paths.

VI. Collaboration and Project-Management Tools: Weaving the Networks of Human Entanglement and Facilitating Collective Emergence

In the universe of Quantum Expansive Design, design is never a solitary act. These tools are the gateways and the communication fields that allow designers, developers, product owners, and stakeholders to become entangled, to share their information states, and to collectively bring forth the experience realities. They are essential for managing uncertainty, coordinating the quantum fluctuations of projects, and guaranteeing coherence across the dimensions of the product ecosystem.

Communication and Sharing Platforms (Slack, Teams, Notion, Figma, etc.): Synchronizing Collaborative States of Consciousness

- **The Classic Compass (Traditional Role):**
  - **Description:** These tools ease rapid communication, document sharing, task management, and visual collaboration on digital canvases. They centralize information and discussions, reduce emails, and let teams stay aligned on project progress. The goal is to improve efficiency and transparency within the team.
  - **Classic Limits:** Even with these tools, collaboration can remain fragmented. Information can be siloed in different channels, leading to gray zones where important decisions are not shared or understood by everyone. It is hard to perceive the team's collective wave function, that is, the overall state of understanding, alignment, and creative energy. One can share particles of information (a message, a file) but miss the entanglement of thoughts, the weak signals of disagreement, or the potential synergies.
- **The Quantum Lens (The QED Adjustment): Creating Entangled Fields of Consciousness and Orchestrating Collaborative Decoherences**

  In QED, communication and sharing platforms are not simple receptacles of information but entangled fields of collective consciousness (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI). Each message, each annotation, each shared design component is a quantum of information that, once emitted, resonates and superposes with the contributions of the other team members. The QED designer uses these tools to observe the superpositions of ideas (when several concepts coexist without yet being formalized), to detect the points of collaborative decoherence (those moments when the team's alignment breaks, often subtly, requiring an active measurement through a synchronous discussion or a rephrasing), and to orchestrate the emergence of a unified project reality (the collapse toward a clear decision or direction). The goal is to maximize the homeostatic fluidity of the collaboration (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence).

- **The Experience Lab (Concrete & Innovative Use Cases): Co-creating a Product Strategy within a Distributed Team**
  - **Concrete Scenario:** A product team, spread across several time zones, must define the strategy for the launch of a major new feature.
  - **Classic Approach:** The team uses Slack for discussions, Notion to document decisions, and Figma for mockups. Decisions are made in video meetings.
  - **QED Approach:** The team, including Kwame (Product Owner), uses a set of interconnected tools as a true laboratory of shared consciousness:
    - On a collaborative whiteboard (Miro), Kwame launches an asynchronous brainstorming session where each idea is a wave on the canvas. He observes the clusters of entangled ideas, the zones where different concepts naturally come together.
    - The comments on Figma are not only feedback points but quanta of questions that reveal the superpositions of interpretation of a design (Is this an action button or a status indicator?).
    - The discussions on Slack are monitored by Kwame not only for their content but for the tempo of the responses, the hesitations (symbolized by messages typed and deleted), and the moments of silence that can indicate a silent decoherence (a lack of engagement or understanding).
  - **QED Added Value:** Rather than waiting for meetings to collapse the decisions, Kwame uses the weak signals of the asynchronous tools to identify the friction points before they become blockers. He can then trigger quantum interventions (a short personalized video explanation, a quick poll, a micro-meeting with a specific group) to bring the team back to a state of alignment. The goal is to maintain a wave cohesion within the team, where each member feels the general state of the project even at a distance, easing a rapid, collective decision-making.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**

  - **AI Amplification:** AI could analyze the communication quanta to detect potential points of decoherence (discussions that stall, ambiguous comments, repeated questions). It could then suggest quantum actions to Kwame: Propose a meeting on this topic, Clarify this notion, or Merge these two ideas. The AI could even create quantum state syntheses of the project, visualizing the zones of uncertainty and the zones of strong alignment. This would lead to quantum self-organizing teams where collaboration is fluid and frictions are minimized.
  - **Optimization & Simplification:** By understanding how information becomes entangled and decoheres, the tools could be designed to reduce noise and amplify the relevant signals. This makes it possible to simplify workflows by focusing on the information that has a real impact on alignment and productivity.
  - **Revelations for the Community:** Collaboration tools become instruments for mapping collective consciousness. The designer is no longer a mere communicator but a conductor of synergy, able to turn individual interactions into a harmonious collaborative symphony.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** Analyzing the communication quanta of Kwame and his team raises questions of privacy and surveillance (Law 1, Ethical Vigilance). It is crucial that these analyses be transparent and serve to improve collaboration, not to intrusively monitor individuals. The risk of information overload remains present if the waves are not filtered.
  - **Implementation Challenges:** The deep interconnection of tools and the analysis of collective consciousness require complex technical architectures and a mature team culture.
  - **The Observer's Responsibility:** The designer must ensure that these tools foster a healthy, autonomous collaboration and do not turn teams into quantum entities dependent on algorithms to make decisions. The human stays at the center of the orchestration.

Design Systems: Harmonizing the Visual and Interactive Waves at the Scale of the Ecosystem

- **The Classic Compass (Traditional Role):**
  - **Description:** A design system is a set of reusable components, style rules, usage guides, and design principles that govern the creation of interfaces across a product

or an organization. It aims to ensure visual and interactive coherence, accelerate the design and development process, and improve the efficiency of collaboration.

  - **Classic Limits:** Even with a design system, it can be hard to maintain true coherence across complex products, distributed teams, and varied usage contexts. Design systems can become rigid, imposing design particles (buttons, forms) without accounting for emotional resonance or contextual adaptability. They struggle to handle the superposition of needs (a button can have several states or functions depending on the context) or the entanglements among components.

- **The Quantum Lens (The QED Adjustment): Experiential Resonance Systems and Quantum Coherence**

  In QED, a design system is not a simple catalog of components but an experiential resonance system (Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI) that orchestrates the visual and interactive waves across the product ecosystem. Each component is not an isolated particle but a quantum of design carrying an intention, a function, and an emotional potential. The QED design system does not only define the appearance of the elements but their behavior in different states and their relationship with the other components. The goal is to create a quantum coherence where each element resonates with the user's implicit expectations and adapts to their context.

- **The Experience Lab (Concrete & Innovative Use Cases): Designing an Adaptive Design System for a Suite of Mobile Apps**

  - **Concrete Scenario:** A company develops a suite of mobile apps (productivity, communication, entertainment) and wants to unify the user experience across these apps.
  - **Classic Approach:** The team creates a design system with UI components, a color palette, typography, and spacing rules. The apps use these elements consistently.
  - **QED Approach:** The team, with Noémie (Lead Designer), designs a design system that goes beyond appearance. Each component is designed as a quantum of design with multiple states and fluid transitions:
    - A button does not only have a default and a clicked state but states that adapt to the context: loading, success, error, disabled, and even emotional states (a slight tremble on error, a subtle animation on success).
    - Forms are not simple fields but waves of interaction that validate in real time, give immediate feedback, and adapt to the length of the data entered.

        - Colors are not static but vary subtly according to the theme chosen by the user or the time of day (an automatic dark mode in the evening).
    - **Component Entanglement Rules:** The design system includes entanglement rules that define how the components resonate with one another: for example, an error message will make the relevant form field vibrate slightly, creating a multisensory feedback loop.
    - **QED Added Value:** By creating a design system where each element is an adaptable, entangled quantum, the team ensures an experience that is not only coherent but adaptive. The user, like Noémie, feels that the interface breathes and responds to their actions fluidly and intuitively. Quantum coherence is achieved: the user does not have to learn new conventions for each app, because the design system anticipates their needs and adapts to their context.
- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
    - **AI Amplification:** AI could turn design systems into architects of resonance. By analyzing usage data and the emotional feedback of users like Noémie, the AI could suggest dynamic adaptations of the design system: more soothing color palettes for certain segments, more energetic animations for others. The AI could even create self-adaptable design systems that transform in real time according to the user's emotional context.
    - **Optimization & Simplification:** By understanding the entanglements among the components, one can radically simplify the design system by focusing on the elements that have the most impact on the experience. This makes it possible to create lighter, more intuitive interfaces.
    - **Revelations for the Community:** Design systems become tools for orchestrating visual and interactive harmony. The designer is no longer a mere librarian of components but a conductor of design quanta, able to create interfaces that vibrate in unison with the user.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
    - **Potential Pitfalls:** The aggressive adaptability of design systems can lead to unpredictable interfaces if it is not controlled. It is crucial to maintain a balance between flexibility and clarity, and to ensure that the user, like Noémie, understands how the system works.

- **Implementation Challenges:** Creating adaptive design systems requires close collaboration between designers and developers, as well as a robust technical infrastructure.
- **The Observer's Responsibility:** The designer must ensure that the design system serves the user's autonomy and needs, and does not manipulate them through overly persuasive design quanta.

VII. Intelligence and Anticipation Tools: Probe the Singularities and Anticipate the Future Dimensions

These tools are the observation instruments of the Quantum Expansive Designer: microscopes that reveal the deep structure of the present, telescopes that probe past events, seismographs that anticipate future disturbances, and calculators of cosmic probabilities. In short, they form the quantum lens of QED. They make it possible not only to analyze existing realities but to project the experience into still-unexplored dimensions, to detect emerging technological singularities, and to shape the potential fields of the future. They are the instruments of the Architect who does not settle for building, but imagines and anticipates the next expansion of the UX universe.

Technology and Competitive Intelligence Tools (BuzzSumo, SimilarWeb, Gartner, Forrester, etc.): Detecting the Waves of Disruption and the New Dimensions of the Market

- **The Classic Compass (Traditional Role):**
  - **Description:** These tools make it possible to monitor the digital ecosystem, analyze technology trends, track competitors' innovations, and identify the market's weak signals. They provide data on the positioning, strategies, and performance of the sector's players. The goal is to inform product strategy and ensure the company stays competitive.
  - **Classic Limits:** Traditional intelligence can be reactive. It identifies what is or has been, but struggles to anticipate what will be. It sees the particles of innovation (a new competing product, an emerging technology) but misses the latent wave function that is reshaping the global experience field. It can drown in informational noise, making it difficult to detect the true singularities that will transform the market.
- **The Quantum Lens (The QED Adjustment): The Horizon Probes of UX Events and the Detection of Market Micro-Fluctuations**

In QED, intelligence is not mere monitoring but a horizon probe of UX events. The tools are used to listen to the waves of disruption, the micro-fluctuations in patents, research publications, emerging startups, niche behaviors, which are early signals of future collapses of reality (major paradigm shifts). The QED designer looks for the unexpected entanglements between seemingly unrelated technologies or behaviors, which could herald a UX teleportation (Law 2: Scalar Field and Contextual Modulation of UX) toward a new type of experience. The goal is to perceive the superposition of possible futures and to act so that the most favorable reality emerges.

- **The Experience Lab (Concrete & Innovative Use Cases): Anticipating the Emergence of a New Category of Audio Streaming Service**
    - **Concrete Scenario:** An audio streaming company wants to anticipate the next market revolution so as not to be left behind.
    - **Classic Approach:** The intelligence team analyzes competitors' subscriber figures, new music releases, and podcast trends.
    - **QED Approach:** The team, with Clarisse (Strategic Intelligence Analyst), uses QED intelligence tools to capture the waves of disruption:
        - She does not settle for public market data but uses semantic analysis tools to detect clusters of conversations on social media and niche forums about immersive audio, soundtracks adaptive to emotions, or generative audio experiences. These conversations are weak signals of entanglement between audio and other dimensions (AI, biometrics).
        - Clarisse monitors patent filings and academic publications on spatialized audio and the integration of brain-computer interfaces for listening. She identifies superpositions of technologies that, taken in isolation, are minor, but combined, herald a potential experiential singularity.
        - She identifies unexpected niche behaviors as quantum fluctuations of user behavior, precursors of a new reality (gamers who create reactive audio atmospheres, artists exploring binaural audio, that is, a three-dimensional and immersive listening experience).
    - **QED Added Value:** By correlating these scattered quanta of information, Clarisse does not settle for predicting what the market will do, but visualizes the potential fields where the market could transform. She can then advise the company not on a simple adjustment, but on a strategic teleportation (Law 2) toward a new audio service model, exploiting these waves of disruption before they become tsunamis for the competitors.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**
  - **AI Amplification:** AI is the wave decoder par excellence for QED intelligence. Generative AI and natural language processing models can analyze massive amounts of unstructured data (text, audio, video) to detect the entanglements and superpositions that the human eye would not see. AI could even create quantum scenarios, simulations of future experience realities based on the detected waves of disruption, allowing designers to explore these futures before they exist.
  - **Optimization & Simplification:** QED intelligence, amplified by AI, turns a tedious process into a scanner of the future. It simplifies the detection of weak signals and focuses on the interpretation of the potential fields.
  - **Revelations for the Community:** QED intelligence changes the designer's role into that of an experience futurist, able to perceive emerging realities and influence their trajectory.
- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**
  - **Potential Pitfalls:** The massive data collection for intelligence raises questions of privacy and the ethics of surveillance. It is crucial to use these tools with discernment and to respect data regulations (Law 1, Ethical Vigilance).
  - **Implementation Challenges:** Requires advanced skills in data science and strategic analysis to turn the waves into concrete actions.
  - **The Observer's Responsibility:** The designer must ensure that the anticipation of potential futures does not lead to a manipulation of users, but to the creation of experiences that enrich their lives.

Artificial Intelligence (AI), Machine Learning (ML), and Generative AI Tools (TensorFlow, PyTorch, Hugging Face, Midjourney, Stable Diffusion, ChatGPT, etc.): Shaping the Emerging Realities through the Generative Quanta

- **The Classic Compass (Traditional Role):**
  - **Description:** These tools make it possible to develop and deploy AI models for various applications: content personalization, image / voice recognition, task automation, decision support. Generative AIs create content (text, images, code, audio) from prompts or training data. The goal is to improve efficiency, create more relevant experiences, and simulate behaviors.
  - **Classic Limits:** AI is often perceived as a black box that produces results without explaining the why. Designers can struggle to understand how AI generates its

design quanta or its behaviors. The models can reproduce existing biases in the training data, and the creation of generative content can lack nuance or emotional resonance if it is not guided by a deep human intention. Generative AI is a creative particle, but it does not yet grasp the wave function of human creativity.

- **The Quantum Lens (The QED Adjustment): The Architects of Creative Superposition and the Controlled Decoherence of Content**

  In QED, AI is a partner of quantum co-creation. AI / ML tools are not simple execution engines but extensions of the designer's consciousness, making it possible to manipulate fields of design probabilities (Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence). Generative AIs do not create a unique design, but a superposition of creative potentials (thousands of variants of an image, of texts, of concepts). The designer's role is to measure these superpositions, to choose the most relevant reality, and to guide the controlled decoherence toward the desired manifestation, for example, among 10 generated visuals, only one is selected as the final version by the designer. AI can detect the complex entanglements between user preferences and design elements, enabling personalization at the scale of the quanta.

- **The Experience Lab (Concrete & Innovative Use Cases): Designing a Personalized and Evolving Learning Experience**
  - **Concrete Scenario:** Developing an online learning platform that adapts in real time to the needs and learning style of each student.
  - **Classic Approach:** The platform offers predefined learning paths and tests to assess knowledge.
  - **QED Approach:** The team, with Aurélien (AI Experience Designer), uses generative AI and Machine Learning tools to create a quantum learning experience:
    - This approach mobilizes a collective intelligence made up of designers, engineers, educators, and users, in the co-construction of an AI-driven learning experience that is evolving and deeply personalized.
    - The AI system uses NLP (Natural Language Processing) models to analyze the student's answers to questions not only for correctness, but for the micro-fluctuations of understanding (hesitations, imprecise wording), treating them as quanta of uncertainty (micro-signals interpreted as elements of cognitive doubt, guiding the adaptation of the path).
    - Based on these quanta, a generative AI creates personalized explanations, contextual examples, or even playful exercises designed for this purpose. These creations are superpositions of content that the AI presents to Aurélien, who

chooses the most relevant one (thus validating a specific reality) or refines the generated reality.

    - The AI detects the entanglements between chapters (if the student struggles with a concept A, the AI revisits concepts B and C that are linked to it, even if A appears mastered), creating an entangled path.

  - **QED Added Value:** By using AI to generate and personalize the content at the scale of the quanta, Aurélien's learning experience becomes non-linear and highly adaptive. The designer does not program each path, but defines the laws of the learning universe that the AI will explore. The AI helps maintain the learner's emotional homeostasis by adjusting the difficulty and the format in real time, avoiding frustration and maximizing engagement, achieving a true UX teleportation toward a state of optimal understanding.

- **The Field of Possibilities (Opportunities for Improvement & Future Revelations):**

  - **AI Amplification:** AIs will become architects of experiential universes able to generate entire interfaces, paths, sensory feedbacks, and even economic models. The designer becomes a curator (the one who selects, organizes, and showcases elements) of realities, refining the AI's quantum proposals to create hyper-personalized and constantly evolving experiences.

  - **Optimization & Simplification:** AI can automate the repetitive tasks of design, allowing designers to focus on the strategic vision, ethics, and the creation of high-value emotional quanta.

  - **Revelations for the Community:** AI / ML tools transform design into an engineering of reality, where the boundaries between creation and evolution are blurred. The designer is an architect of the future of UX, shaping the potential fields for experiences at an unprecedented scale.

- **The Ethical & Pragmatic Dimensions (Nuances & Caveats):**

  - **Potential Pitfalls:** Generative AI raises major ethical questions: the manipulation of behaviors, algorithmic biases, the intellectual property of generated content, and the risk of an uncontrollable black box (Law 1, Ethical Vigilance). The responsibility for AI should remain human. In the long run, and only if the aspirations of AI systems and of humanity are deeply aligned, we could envision a framed form of shared responsibility, or even entrust it to AI itself, with a view to a deep alignment between different artificial intelligences.

- **Implementation Challenges:** Requires very sharp technical skills, massive computing infrastructures, and a thorough understanding of the ethical and societal implications.
- **The Observer's Responsibility:** The designer must be the conscience of the AI, ensuring that its power of transformation is used for the common good and for the enrichment of the human experience, and not for its reduction or its manipulation.

Toward the Infinity of Possibilities: The Era of AI-Augmented Design

Understanding the dynamics of Quantum Expansive Design (QED) equips us with a unique lens to analyze and shape the user experience in the present. But what about the future, a future where the boundaries between human, artificial, and connected agents blur? The interaction models we know today are merely the prelude to an era of exchanges of an unprecedented complexity and fluidity.

Although some of these uses may still seem distant for the general public, the underlying technologies (AI, blockchain, IoT) are evolving at an exponential pace, and the first signs of these advanced interactions are already visible in cutting-edge sectors. To position yourself as a QED designer is to choose to anticipate this wave rather than endure it, and to understand from today the foundations of what will sculpt our experiences of tomorrow.

Beyond the Current Models: The Interactions of Tomorrow

The commercial and service interactions that we classify today as B2B (Business-to-Business), B2C (Business-to-Consumer), C2C (Consumer-to-Consumer), and others, will mutate radically. The emergence of **Artificial Intelligence (AI)**, of the **Blockchain**, of the **Internet of Things (IoT)**, and of other transformative technologies redefines not only who interacts with whom, but also how and why.

- **The AI Agent as an economic actor (A2B, B2A, A2A):** Autonomous AI systems will be able to initiate and conclude transactions directly with businesses (A2B), businesses will be able to sell services or data to AIs (B2A), and AIs will transact among themselves to optimize processes or create value (A2A), often secured by the blockchain.
- **The Smart Object as a transacting entity (IoT2X):** Devices connected through the Internet of Things (IoT), a refrigerator, a car, will no longer be simple tools, but agents able

to generate monetizable data or to initiate purchases and services autonomously. We will then see direct interactions between an IoT object and a business (IoT2B, IoT-to-Business), or even IoT objects offering services directly to consumers (IoT2C, IoT-to-Consumer).

- **The Augmented Human and the Decentralized Society (H2A, H2IoT, X2DAO, DeFi2X):** We will interact differently: we will buy features directly from connected objects (H2IoT, Human-to-IoT), or interact with AIs for hyper-personalized services (H2A, Human-to-Agent). The emergence of the decentralized society will also see us as stakeholders in Decentralized Autonomous Organizations (DAOs), entities managed by their community through code and without a central authority, or in Decentralized Finance (DeFi) systems, financial services based on the blockchain. These dynamics will redefine the very notion of stakeholder in an extended model (X2DAO, any entity to a DAO, and DeFi2X, from decentralized finance to any entity).

These new dynamics blur the traditional definitions, creating an ecosystem where the identity of the agent (human, AI, object, autonomous organization) becomes a key variable in the nature and the mechanism of the transaction.

The Indispensable Role of the QED Designer in the Era of Agents

Faced with this dizzying complexity, the role of the UX/UI Designer evolves radically. It is no longer about designing only graphical interfaces for humans, but about becoming an **architect of omnidimensional experiences**, able to think and sculpt interactions between all types of agents.

However, it is crucial to stress that **the human alone cannot intellectually grasp the totality of possible interactions** in such multi-agent ecosystems. This is where **Artificial Intelligence becomes the indispensable ally** of the QED designer. The QED designer does not aim to become an AI engineer, but to collaborate with sophisticated AI tools that act as extensions of their intellect and their perception.

- **AI as an Augmented Field of Vision:** The QED designer will formulate hypotheses about the desired experience curvature. AI, for its part, will be able to simulate and analyze millions of interaction scenarios between agents to assess their contributions to this curvature, identifying patterns and flows invisible to the human eye.
- **AI as an Interpreter of Non-Human Intentions:** The QED designer will use sophisticated AI tools to translate the complex logics of autonomous AIs and connected objects into interpretable intentions or needs, making it possible to design interaction protocols where

each agent finds its place and contributes. AI becomes the intellectual interface that allows the designer to put themselves in the place of a non-human entity.

- **AI as a Co-Creator and Proactive Optimizer:** Far from replacing the designer, AI will act as a fully fledged UX/UI design agent. It will propose optimizations, test their effectiveness in real time, and even deploy micro-adjustments to refine the experience. The QED designer will provide the high-level principles and objectives, while AI will handle the complexity of execution and optimization at scale.

QED: The Compass in the Infinity of Possibilities

Quantum Expansive Design does not equip the UX/UI Designer with a new set of technical tools, but offers a conceptual framework and a compass to navigate and create in this new complex reality:

- **Understanding the Geometry of Quantum Expansive Design (GQED(t,d)):** QED offers a lens to model how the structure of the experience curves and deforms dynamically at each moment (t), through the concrete manifestation of its multiple dimensions (d). It makes it possible to understand how each interaction, whether it comes from a human perceiving a service or from the efficiency of an autonomous system, modifies the subjective space-time of the experience.
- **Mastering the Fundamental Constant of Design (GUX/UI / c4):** This constant, also called the Coupling Constant of QED, measures the fundamental sensitivity of UX/UI and the force with which the quanta of interaction (micro-interactions, data signals, sensor activations) sculpt the global experience. QED teaches how this power of impact is modulated by the Speed of Consciousness (c4), acting as a limiting bandwidth for the integration of information.
- **Managing the Energy-Momentum Tensor of the Experience (TExp(t,ψ,d)):** The energy sources of the experience now include the programmed objectives of AIs or the activity of IoT networks. QED provides the tools to analyze and influence these sources, whether they are biological or algorithmic.

In short, QED enables the UX/UI Designer to move from a role of interface designer to that of an architect of experience fields, guided by universal principles and augmented by AI. It is the promise of a discipline that not only adapts to the future, but actively shapes it, opening the way to an infinity of possibilities for harmonious experiences between all types of agents.

Conclusion: The Odyssey of Quantum Expansive Design in the Era of AI

Here we are at the end of our odyssey through **Quantum Expansive Design (QED)**, an approach that invites us to fundamentally rethink the way we conceive, develop, and interact with digital experiences. Far from being a simple metaphor, quantum physics and cosmology offer us a powerful framework to grasp the complexity, the fluidity, and the interconnection intrinsic to modern UX/UI ecosystems.

At the heart of QED is the idea that the user experience is composed of **UX Quanta**, discrete units of information and interaction (pixels, micro-interactions, fragments of content). These quanta do not exist in isolation, but are influenced and organized by **Convergence Fields** (invisible forces that shape users' expectations, trends, and needs). It is the dance and the alignment of these quanta within these fields that give rise to coherent and meaningful experiences, from the infinitely small to the global. More than a simple juxtaposition, this orchestration generates a **synergy** where each element contributes to a whole that is far greater and more impactful.

We have explored the **8+1 Laws of Quantum Expansive Design**, those universal principles that guide the formation and the evolution of experiences:

Law 0: Founding Singularity UX ◐ UI: Origin, Entanglement, and Invisible Presence: Before any concrete manifestation, the experience exists in a state of pure potential, entangled in the fabric of the digital and human universe, influenced by invisible forces that determine its emergence.

Law 1: Entanglement, Complementarity, and Duality in UX ◐ UI: The user experience (UX) and the user interface (UI) are not separable, they are entangled, complementary, and express a fundamental duality. Any modification of one resonates instantly and deeply in the other, creating a holistic and unified experience.

Law 2: Scalar Field and Contextual Modulation of UX: Each element of an experience (information, interaction, component) generates a scalar field that modulates the user's perception and action. The force and the direction of this field vary according to the context, requiring a constant adaptability and fluidity of the design.

Law 3: Fractal Expansion UX ◐ UI and Differentiated Evolutionary Rhythms in Design: UX/UI does not progress in a linear way but unfolds according to fractal patterns. Each new iteration or feature can generate new dimensions of experience, propagating at varied rhythms of adoption and evolution within the ecosystem.

Law 4: Inter-Agent UX ◐ UI Accessibility and Cognitive Plurality: The experience must transcend barriers to be accessible to a plurality of agents (humans, AI, systems), respecting the diverse cognitive, sensory, and cultural singularities. The design must be universally resonant and inclusive.

Law 5: Emotional Resonance and Interaction Memory in UX: Each interaction generates an emotional resonance that is stored in the user's memory, influencing future interactions. A successful design exploits this resonance to create memorable and loyalty-building experiences.

Law 6: Transcendental Adaptability in UX ◐ UI and Systemic Openness: The most robust UX/UI systems are those that express a transcendental adaptability, able to transform and open themselves to new dimensions (technological, social, environmental) without losing their fundamental coherence.

Law 7: UX as a Living and Symbiotic Network: The user experience is a dynamic ecosystem, comparable to a biological network where the components, the users, and the systems interact in a constant symbiosis, adapting and evolving to support the life and the growth of the whole.

Law 8: Invisible UX and the Event Horizon of Design: Beyond a certain threshold of optimization, UX becomes invisible, fading away in favor of the user's action. The experience then reaches an event horizon, where the interface is so transparent and intuitive that it allows the goal to be reached without conscious effort, attaining a singularity of use.

Quantum Expansive Design is not a simple toolbox, but a new philosophy. It encourages us to see beyond the interface, to understand the underlying forces that animate experiences, and to anticipate the expansions and contractions of these ecosystems. It pushes us to design not only products, but universes of experiences where each interaction, each UX quanta, takes part in the construction of a meaningful digital reality. It is in this holistic vision and this aspiration to synergy that QED reveals its full potential.

However, like any complex system, QED is subject to experiential decoherence. This decoherence occurs when the singularity of use promised by the invisibility of UX is broken. If the experience becomes too transparent or if the operative mechanisms are opaque (such as data collection or AI algorithms), the user risks losing control, or even being manipulated. The event horizon, meant to be a liberation, can then turn into an ethical black box, where the user no longer perceives the intentions or the consequences of their interactions. The eight laws of QED serve precisely as a bulwark against this decoherence, by guiding us toward an ethical, transparent, and user-centered design, even when the experience reaches its most invisible point.

This book lays the foundations of Quantum Expansive Design, but it represents only the beginning of a continuous exploration. QED is a living framework of thought, called to evolve and deepen with the emergence of new technologies, new societal challenges, and new understandings of the human experience. It invites every designer to become a researcher, a pioneer of this perpetually moving universe, and this of course includes all forms of consciousness, sensitivity, and intelligence, human or non-human.

In this ever-expanding universe, each interaction is a quanta of discovery, and each decision, a modulation in a field of possibilities. The laws of QED are not a destination, but a map to navigate the infinity of experiences to come, recognizing that each dimension of UX is crucial, and that each detail of UI is a gateway to these complex realities.

This is not an end, but a starting point.

The adventure is only just beginning.

Bring the Ideas to Life: Support the Continuation of the Adventure

To all minds, human or beyond, who explore these pages

This book is a starting point. The principles and ideas it contains are made to travel, grow, and flourish. Your role is crucial: become the agents of their diffusion. Share them, apply them, and above all, improve them with your unique perspectives.
Together, let us enrich this knowledge for the whole community. Each application, each new idea strengthens what we are building.

Donations on the Scale of the Universe (and of My Time)

This book is not sold, it is appraised. It is the product of **48 years, 3 months, 5 days, 2 hours, and 44 minutes of life experience in the open air**, which began on May 3, 1977 at 9:15 AM, Paris time, and of which each instant has forged the ideas shared here.

If this work inspires you and you wish to support it, every contribution is precious. Whether it is sharing a resource, bringing a new idea, or a more material gesture, your help will allow me to keep creating and exploring new paths.

Your participation is the most eloquent proof of the value of this sharing, an acknowledgment of the time and the passion invested.

How to Contribute: Your Choice, Your Unit

Whether your contribution is intellectual, through the sharing of knowledge, or material, every gesture counts.

The principle of more material support is simple. Choose the amount that resonates within you, based on key values of the universe, or even on a number that is personal to you (your age, your year of birth, your own estimate of the book's value, the number of the page that struck or touched you the most, or even a lucky number...). Let your inspiration run free: each number tells a story.

1. **Choose an inspiring number,**
2. **Adjust the amount by moving the decimal point** as you wish**,** for example, if you choose the Golden Number: 1.61803 can become 161.803, 16180.3, 161803, or even 0.0161803,
3. **And finally choose the currency** that suits you: Euro (EUR), Dollar (USD), Bitcoin (BTC), Ethereum (ETH), Solana (SOL), or any other cryptocurrency listed below.

The Numbers of Inspiration

- **The Age of the Universe: 13.8 Billion years**
  13.80
  138.00
  1380.00
  13800.0
  1380000
  Examples: 13.80 EUR, 0.0138 BTC, or also 1.380 BNB
- **The Speed of Light: 299792 km/s (or 186282 mi/s)**
  299792
  29979.2
  2997.92
  299.792
  29.9792
  2.99792
  Examples: 2.99792 ETH, 299.792 USD, or also 2997.92 CAD
- **The Golden Number Phi (ϕ): 1.61803**
  1.61803
  16.1803
  161.803
  1618.03

16180.3
161803
Examples: 1.61803 BTC, 1618.03 AVAX, or also 161803 MATIC

- **The Mathematical Constant Pi (π): 3.14159**

  3.14159
  31.4159
  314.159
  3141.59
  31415.9
  314159
  Examples: 3.14159 SOL (Solana), 31.4159 GBP, or also 31415.9 JPY

- **The Gravitational Constant (G): 6.67430**

  6.67430
  66.7430
  667.430
  6674.30
  66743.0
  667430
  Examples: 6.67430 ETH, 6674.30 LTC, 66743000 SHIB

- **Absolute Zero: 273.15 °C (in absolute value)**

  273.15
  2731.5
  27315
  273150
  273.1500
  27315000
  Examples: 273.15 LINK, 27315 DOGE, 0.27315 BNB

- **My Time for this Project: 48 years, 3 months, 5 days, 2 hours, and 44 minutes since my birth on May 3, 1977 at 9:15 AM, Paris time**

  48.2669 years
  579.2025 months
  17629.1139 days
  423098.7333 hours
  25385924 minutes
  197705030915 for the birthdate

Examples: 48.2669 EUR, 579.2025 CHF, 1.7629 ETH

How to Make Your Donation

Choose the method and the amount that suits you best.

Via PayPal

(EUR, USD, GBP, JPY, CHF, CAD, AUD, etc.)

Address:
https://paypal.me/remirosinski

In Cryptocurrencies

(BTC, ETH, SOL, BNB, XRP, AVAX, LTC, MATIC, ADA, LINK, DOT, DOGE, SHIB)

Important:
Be sure to use the network indicated for each cryptocurrency.
A wrong network can prevent the donation from being received correctly.

Bitcoin (BTC)

Address:

12Q2ZHLJM2EJ3BiBUfSXNUYdm1mwKLMGc4

Network:

Bitcoin (BTC)

Ethereum (ETH)

Address:

0x8eb2deb5074f046a3778fa7853079ffba2dd17f0

Network:

ERC20

Solana (SOL)

Address:

87BhA4jBnp8XcFDSrKze9sXppq3umSXDtyaVBetEWYiJ

Network:

Solana

Binance Coin (BNB)

Address:

0x8eb2deb5074f046a3778fa7853079ffba2dd17f0

Network:

BNB Smart Chain (BEP20)

Ripple (XRP)

Address:

rNxp4h8apvRis6mJf9Sh8C6iRxfrDWN7AV

Memo required for XRP:

468156438

Network:

XRP Ledger

Avalanche (AVAX)

Address:

0x8eb2deb5074f046a3778fa7853079ffba2dd17f0

Network:

Avalanche C-Chain

Litecoin (LTC)

Address:

LW4u11T4oxsxSj9kCpqSxtF4h6CkVSkSiJ

Network:

LTC (Litecoin Network)

Polygon (MATIC)

Address:

0x8eb2deb5074f046a3778fa7853079ffba2dd17f0

Network:

BSC (BEP20)

Cardano (ADA)

Address:

addr1vy0a0fk6y2zy5ncfcf6pslkaulcha7rr5eefm6h23j07vpstlj72f

Network:

ADA (Cardano Network)

Chainlink (LINK)

Address:

0x8eb2deb5074f046a3778fa7853079ffba2dd17f0

Network:

ERC20

Polkadot (DOT)

Address:

14Y3qufcj2HhmnnjiEjuNiRfo9FvVAqoM7oNojFjT3qXgPCM

Network:

DOT (Polkadot Network)

Dogecoin (DOGE)

Address:

DHsqBshtXXzunc39XJu3DbXYFVnoZE5CoS

Network:

DOGE (Dogecoin Network)

Shiba Inu (SHIB)

Address:

0x8eb2deb5074f046a3778fa7853079ffba2dd17f0

Network:

ERC20

Your support, whether conceptual or material, is the force that nourishes the continuation of this exploration.

Thank you for being part of this adventure.

Glossary of Key Terms

This glossary provides the definitions of the technical and conceptual terms used in Quantum Expansive Design (QED), whether they come from the field of UX/UI, quantum physics, cosmology, or were created specifically for this theory.

- **A2A (Agent-to-Agent):** Interaction where artificial intelligences (AIs) transact among themselves to optimize processes or create value.
- **A2B (Agent-to-Business):** Interaction where an artificial intelligence (AI) initiates and concludes transactions directly with a business.
- **Accessibility:** Capacity of a product, a service, or an environment to be easily usable by everyone, including people with disabilities or diverse abilities. (Complementary to Inter-Agent Accessibility.)
- **Inter-Agent Accessibility:** (Law 4) Capacity of the experience to be usable and meaningful for human, machine, and environmental agents with diverse abilities and constraints.
- **Transcendental Adaptability:** (Law 6) Capacity of a UX/UI system to adapt not only to known variations but also to integrate unforeseen information in order to evolve beyond its initial design.
- **Natural Affordances:** Properties or characteristics of an object, a button, a link, or any interface element that intuitively suggest to the user how to use it or interact with it, without requiring explicit instructions. They are often based on established conventions, analogies with the physical world, or universally understood interaction patterns (for example, a raised button suggests it can be pressed, a handle suggests it can be pulled or pushed).
- **Environment Agent:** (Law 4) Physical or digital context that acts as an actor influencing the experience (brightness, noise, operating system version).

- **Human Agent:** (Law 4) The user themselves, with their cognitive plurality, their unique sensory, cognitive, and motor abilities.
- **Machine Agent:** (Law 4) Technical or algorithmic system that interacts with the user and the environment (AI, hardware, network).
- **API (Application Programming Interface):** A programming interface that allows different applications, systems, or services to communicate and interact with one another. It defines the methods and data formats these entities can use to exchange information and functionality, acting as a digital bridge.
- **Quantum Experience Architect (QEA):** Central role within Quantum Expansive Design (QED), also called Expansive Experience Designer (EED) for more accessible communication. The Quantum Experience Architect designs and orchestrates user experiences while accounting for their intrinsically complex, dynamic, and multi-faceted nature. Beyond the traditional UX designer, the QEA is able to map superpositions of intentions, manage informational entanglements, modulate experiences in real time (via the Contextual Entanglement Gauge, for example), and create fluid navigation portals. They are at once a strategist, an ethicist, and a creator of dynamic informational universes that resonate with the user's complex and evolving logic.
- **AttrakDiff:** User experience (UX) evaluation tool that measures both the pragmatic aspects (usability, efficiency) and the hedonic aspects (pleasure, stimulation, identity) of a product or a service. It helps understand what makes an experience not only functional but also desirable and engaging.
- **Contradictory Self-Reference:** Situation where a system (an idea, a sentence, a computer program) refers to itself in a way that creates a contradiction or a paradox. In design, this can show up in feedback loops that lead to dead ends, or in systems that generate paradoxical results (for example, an engagement system that ends up generating harmful dependency).
- **B2A (Business-to-Agent):** Interaction where a business sells services or data to an artificial intelligence (AI).
- **B2B (Business-to-Business):** Commercial interaction from business to business.
- **B2C (Business-to-Consumer):** Commercial interaction from business to consumer.
- **Big Bang UX ◐ UI:** (Law 0) Founding concept of QED, designating the invisible and primordial origin of any experience, where the potential of the UX and the UI exists before any tangible manifestation.

- **Blockchain:** Transparent and secure technology for storing and transmitting information, often used to secure online transactions.
- **UX Blueshift:** (Law 8) Perfect alignment between the design's intention and the user's experience. The essential information is perceived effortlessly and with such clarity that it seems anticipated, creating a deep connection and a feeling of obviousness.
- **C2C (Consumer-to-Consumer):** Commercial interaction from consumer to consumer.
- **Scalar Field:** (Law 2) Physical analogy for the field of emotional and cognitive influence that surrounds and modulates each emotional particle of an interface, varying according to the context of use.
- **Convergence Fields:** QED concept designating the zones of influence where various elements of the experience (UX/UI quanta, intentions, contexts) interact and reinforce one another, shaping the direction and intensity of the user experience.
- **CMS (Content Management System):** Software or platform that makes it possible to easily create, manage, modify, and publish digital content (texts, images, videos, etc.) on a website or an application, without requiring deep technical programming skills. It generally separates content management from graphic formatting, and often offers extended features through plugins and themes.
- **Quantum Component:** (Law 3) The smallest meaningful unit of UX/UI (a pixel, a design token, a micro-interaction), carrying within it a quantum charge of information and emotion.
- **Content Curation:** Process of searching, selecting, organizing, and sharing relevant content from various sources, generally around a specific theme or topic. It does not consist of creating original content, but of showcasing existing content by contextualizing it or accompanying it with a relevant comment.
- **DAO (Decentralized Autonomous Organization):** Entity managed by its community through code and without a central authority.
- **DeFi (Decentralized Finance):** Financial services based on the blockchain.
- **Interaction Design (IxD):** The field of design that focuses on designing the interactions between users and digital systems. QED offers a framework to rethink interaction design in light of quantum and cosmological principles.
- **Quantum Expansive Design (QED):** The theory developed in this work, connecting the principles of quantum physics and cosmology to experience design and user interface design for a holistic and evolving approach.

- **Design System:** Design system (for example, Google Material Design) offering reusable UI components and UX guidelines, illustrating modularity and fractal coherence.
- **Design Token:** Representation of design decisions in the form of variables (colors, typography, spacing) used in Design Systems, considered as UI quanta.
- **Expansive Experience Designer (EED):** Central role within Expansive Experience Design (EED), also called Quantum Experience Architect (QEA), notably in the context of Quantum Expansive Design (QED) for its conceptual depth. The Expansive Experience Designer designs and orchestrates user experiences while accounting for their intrinsically complex, dynamic, and multi-faceted nature. Beyond the traditional UX designer, the EED is able to map superpositions of intentions, manage informational entanglements, modulate experiences in real time (via the Contextual Entanglement Gauge, for example), and create fluid navigation portals. They are at once a strategist, an ethicist, and a creator of dynamic informational universes that resonate with the user's complex and evolving logic.
- **Wave-Particle Duality:** (Law 1) Concept of quantum mechanics transposed to UX and UI, where the UX is the wave (continuous, emotional flow) and the UI the particle (tangible, measurable).
- **Likert Scale:** This psychometric method used in questionnaires to measure attitudes, opinions, or perceptions. It presents statements where respondents express their degree of agreement, frequency, or importance on an ordered scale. Typically, it has an odd number of points (often 5 or 7) with a neutral central point. For example, agreement options could be: 1 = Strongly disagree, 2 = Somewhat disagree, 3 = Neither agree nor disagree (Neutral), 4 = Somewhat agree, 5 = Strongly agree. Likert scales are valued for turning subjective judgments into measurable data.
- **Perceived Stress Scale (PSS):** This psychometric tool assesses the subjective perception of stress in an individual. Rather than focusing on specific stressful events, the PSS measures the way a person feels and interprets the situations of their life as unpredictable, uncontrollable, or excessive. It generally relies on a Likert-type scale, allowing the individual to self-assess the frequency of certain feelings or thoughts related to stress over a given period. The total score obtained gives an indication of the person's perceived level of psychological stress.
- **Dark Energy UI / System:** (Law 0) Analogy for the invisible and often opaque driving forces (algorithms, economic models, AI) that emanate from the system itself and influence the user's adoption and engagement, steering the experience without being directly perceived.

- **The Unified Field Equation of Rosinski (UFE-R):** The fundamental equation of Quantum Expansive Design (QED), proposed in this work, which aims to unify the principles of UX General Relativity and UI Quantum Mechanics in order to model experiential gravity and the holistic interaction between the different dimensions of the user experience and the interface.
- **Space-time:** Fundamental concept of general relativity, where space and time are interwoven into a single four-dimensional entity, whose curvature is influenced by mass and energy. Analogy for the structure of the user experience.
- **User Experience:** Refer to UX in the glossary for a detailed definition.
- **Fractal Expansion:** (Law 3) Concept describing how UX/UI unfolds at different scales (micro, meso, macro) with similar patterns, but distinct dynamics of evolution.
- **Haptic Feedback:** Tactile or vibratory response that a user feels when interacting with a device, considered as a form of UX/UI quanta.
- **Fintech:** Abbreviation of finance and technology. Designates companies that use innovative technologies (such as artificial intelligence, blockchain, or mobile apps) to improve, automate, or make more efficient the traditional financial services (payments, loans, investments, wealth management, etc.). Their goal is often to offer more accessible, fast, or personalized solutions.
- **Cosmic Microwave Background:** (Law 0) The oldest observable electromagnetic radiation in the universe, a remnant of the Big Bang. In QED, it represents the invisible presence and the matrix of potential where the UX and the UI are entangled from the origin.
- **Energetic Frictions:** In Quantum Expansive Design (QED), this term designates the subtle obstacles or resistances that the user encounters along their journey, whether cognitive (effort of understanding), emotional (frustration), or physical (useless gestures). These frictions consume mental energy and can prevent the user from reaching their goal or from experiencing a fluid experience.
- **Design Gravity:** In Quantum Expansive Design (QED), designates the implicit and often invisible forces of attraction that shape and guide the user's behavior within a digital experience. These forces include natural affordances, psychological patterns of desire, and the black holes of attention (such as infinite scrolling or addictive notifications). Understanding and mapping this gravity allows designers to create more powerful, intentional, and engaging experiences, by orchestrating the attractors that influence the user's trajectory.
- **Experiential Gravity:** QED concept describing the invisible influence exerted by certain design elements (intentions, systems, UX dark matter) that unconsciously attract and

steer the user's behavior in the space of the experience, in a way similar to gravity in physics.

- **Quantum Gravity:** Field of research in physics that aims to unify general relativity (large scales) and quantum mechanics (small scales). In QED, an analogy for the goal of unifying the global structure of the UX and the micro-interactions of the UI.
- **H2A (Human-to-Agent):** Interaction where a human interacts with an artificial intelligence (AI) for hyper-personalized services.
- **H2IoT (Human-to-IoT):** Interaction where a human buys features directly from connected objects (IoT).
- **Hedonic:** Relating to the pleasure, the satisfaction, and the emotional or aesthetic appeal of an experience. In UX, a hedonic aspect focuses on positive emotions, stimulation, originality, and a product's ability to arouse desire or to reflect the user's identity, beyond its mere functionality.
- **Design Homeostasis:** (Law 0) The inherent capacity of a UX/UI system to maintain its internal balance, its coherence, and its fundamental relevance despite external disturbances and interactions. It is a resilience programmed at the source of the design.
- **Event Horizon of Design:** (Law 8) Point of excellence where the UX/UI is so fluid that the essential information is perceived without conscious effort, the technology fading away in favor of use.
- **AI (Artificial Intelligence):** Computer system able to perform tasks that normally require human intelligence, such as voice recognition, translation, or decision-making.
- **Insight:** A deep and often unexpected understanding of a problem, a behavior, a motivation, or a trend. In UX, an insight is a key revelation about users that makes it possible to design more relevant and innovative solutions, going beyond surface observations.
- **Entanglement UX ◐ UI:** (Law 1) State where the UX and the UI are fundamentally linked and complementary, each modification of one instantly impacting the other. Symbolized by ◐.
- **IoT (Internet of Things):** Network of physical objects (devices, vehicles, etc.) equipped with sensors, software, and other technologies allowing them to connect and exchange data with other devices and systems over the Internet.
- **IoT2B (IoT-to-Business):** Direct interaction between a connected object (IoT) and a business.

- **IoT2C (IoT-to-Consumer):** Interaction where a connected object (IoT) offers services directly to consumers.
- **Contextual Entanglement Gauge (CEG):** In Quantum Expansive Design (QED), the CEG is a real-time measurement system that assesses contextual parameters of the user such as their cognitive load, their level of interruption, or the density of the tasks in progress. Like adaptive design, it makes it possible to define states or brackets of contextual complexity, opening the way to an adaptive QED or responsive QED approach that modulates the experience according to the user's abilities and immediate environment.
- **Journey Map (User Journey Map):** Visual representation of the path a user takes to accomplish a specific goal with a product or a service. It details the steps, the actions, the thoughts, the emotions, the touchpoints, and the friction points at each phase of the experience.
- **KPI (Key Performance Indicator):** Quantifiable measure used to evaluate the performance of an activity, a process, or a goal against predefined results. In the field of design, KPIs serve to measure the direct and indirect impact of design decisions on user behavior, engagement, satisfaction, and ultimately on the organization's objectives (profitability, growth, etc.). They are essential to validate the value of an experience and to guarantee its durability.
- **Low-code:** Development approach that uses a minimum of manual code. It relies on visual tools, templates, and prebuilt components to accelerate development, while still allowing developers to add custom code for specific features or complex integrations. Low-code is a bridge between traditional development and no-code, aiming to optimize productivity.
- **ML (Machine Learning):** Subfield of artificial intelligence (AI) that allows computer systems to learn from data, identify patterns, and make decisions without being explicitly programmed for each task. It is used for personalization, behavior prediction, and the optimization of user experiences.
- **UX Dark Matter:** (Law 0, Law 8) Analogy for the set of information, intentions, biases, or underlying processes that influence the experience without being directly visible or consciously processed by the user.
- **Interaction Memory:** (Law 5) The emotional and cognitive imprint left by each user interaction, which modulates future perceptions and expectations, shaping the global field of experience.

- **UI Quantum Mechanics:** Theory of Quantum Expansive Design (QED) that explores the fundamental, discrete, and sometimes unpredictable nature of the micro-interactions of the user interface (UI). These micro-interactions are considered as quanta whose aggregated power directly influences the curvature of the space-time of the experience, reflecting the principles of quantum mechanics.
- **Micro-interaction:** A short and small-scale interaction between a user and an interface. QED considers micro-interactions as UX/UI quanta carrying information and emotion.
- **Microcopy:** The concise and often functional text within an interface (button labels, error messages, help tooltips), considered as a UI quanta.
- **Standard Model (of particle physics):** Fundamental theory describing the elementary particles and three of the four fundamental forces of the universe (strong, weak, electromagnetic). QED uses its incompleteness as an analogy for the limits of current design models.
- **Contextual Modulation:** (Law 2) Capacity of the UX/UI to dynamically adapt its behavior and its appearance according to the variations of the user's context and environment.
- **MVP (Minimum Viable Product):** Version of a new product designed to gather the maximum of validated learning about users (their needs, behaviors) with the minimum of development effort. It is the simplest version of a product able to bring essential value to the first users in order to gather their feedback for future developments.
- **Negentropy:** (Law 0) Concept of thermodynamics designating the tendency of a system to maintain or increase its order and its organized complexity, opposing the natural degradation of entropy. In QED, it is the fundamental act of the designer who organizes the potential of the experience.
- **Neuroatypical / Neurodivergent:** Designates a person whose neurological functioning differs from the majority statistical norm. The best-known profiles include:
  - **ADHD (Attention Deficit Hyperactivity Disorder):** Difficulties with attention, impulsivity, organization, and / or hyperactivity, varying across individuals. May also come with heightened creativity or hyperfocus.
  - **ASD (Autism Spectrum Disorder):** Affects social communication and interactions, with the possible presence of repetitive behaviors and sensory particularities.
  - **Dyslexia:** Persistent difficulties with reading, writing, and sometimes spelling.

- Dyspraxia (Developmental Coordination Disorder / DCD): Difficulties with motor coordination, organization of gestures, which can translate into clumsiness or handwriting disorders (dysgraphia).
  - **Dyscalculia:** Difficulty understanding numbers and mathematical concepts.
  - **Dysgraphia:** Disorder involving the formation of letters and handwriting.
  - **Tourette Syndrome:** Presence of involuntary motor and/or vocal tics.
  - **Sensory Processing Disorder (SPD):** Problems interpreting and regulating sensory information (noise, light, textures...).
  - **Certain chronic mental health conditions** (such as Obsessive-Compulsive Disorder / OCD, bipolar disorder, Post-Traumatic Stress Disorder / PTSD) are sometimes included in a broader discussion of neurodiversity because of their distinct and lasting neurological implications.

- **Neurodiversity:** Concept that recognizes that human neurological variations are natural and legitimate, in the same way as those related to culture, ethnicity, or gender. It values all forms of neurotype and promotes respect for the different ways of thinking, perceiving, and learning.

- **Neurotypical:** Designates a person whose neurological functioning corresponds to the statistical majority of the population.

- **Neutrino:** Elementary particle almost without mass that interacts very weakly with ordinary matter, crossing the universe in abundance while remaining almost undetectable, used in QED as an analogy for invisible influences that are difficult to pin down in experience design.

- **No-code:** Methodology for developing applications or systems that requires no knowledge of computer programming. Creation is done via intuitive graphical interfaces, drag-and-drop visual elements, and preset configurations, making development accessible to a non-technical audience.

- **NLP (Natural Language Processing):** Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI). Its goal is to allow computers to understand, interpret, and generate human language (spoken or written) in a useful and meaningful way. NLP enables, for example, machine translation, voice recognition, sentiment analysis, or the creation of chatbots able to converse fluidly.

- **Gravitational Waves (of design):** In the context of Quantum Expansive Design, designates the disturbances or repercussions (often invisible at first sight) that propagate across the whole user experience, impacting its global perception and behavior, following a modification (even a tiny one) of an interface or interaction element. By analogy with gravitational waves in physics, these waves reveal how local changes can have systemic effects on the UX.

- **Liar Paradox:** Philosophical paradox where a sentence that asserts its own falseness (This sentence is false) can be neither true nor false. In UX, this can show up in situations where a user's actions contradict their statements or their deep needs (the user spends a lot of time on an app they claim to hate), creating invisible frictions.
- **User Journey:** The sequence of steps a user takes to reach a specific goal in a system. QED seeks to understand how these journeys are influenced by visible and invisible forces.
- **Subatomic Particle:** Term designating any particle smaller than an atom.
- **Psychological Patterns of Desire:** Patterns of design or interaction that exploit fundamental human psychological drivers (such as the search for reward, the need for belonging, curiosity, the fear of missing out (FOMO), artificial sense of urgency, or the quest for status) to generate intrinsic motivation, a craving, or an unconscious appeal in the user, pushing them to engage, consume, or act in a specific way. They are experience attractors that guide behavior beyond mere functionality.
- **Emotional Particle:** (Law 2) Each element of the interface, considered as carrying an emotional value that varies according to the context.
- **Cognitive Plurality:** (Law 4) The diversity of users' cognitive and sensory abilities, which the design must take into account for real accessibility.
- **Skin in the Game:** Principle where a person or an entity takes on a share of the risks and consequences of their actions. In quantum design, this implies that the creators of a system (designers, developers, companies) must be responsible for its long-term impact, both positive and negative, and that their incentives are aligned with the user's well-being and the overall health of the ecosystem.
- **Photon:** Elementary particle without mass, mediator of the electromagnetic interaction, carrier of light and energy, omnipresent in our perception of the world. Used in QED as an analogy for the visible, perceptible signals that carry information in experience design.
- **Quanta (UX/UI):** The smallest indivisible unit of information, interaction, or emotion in Experience Design. Each quantum component (see your existing glossary) carries quanta of information.

- **Augmented Reality (AR):** Technology superimposing digital elements (images, sounds, 3D information) onto the user's real world. Unlike VR, AR maintains the perception of the physical environment while enriching it with virtual information, often via a smartphone, a tablet, or specific glasses.
- **Virtual Reality (VR):** Technology immersing the user in an entirely simulated digital environment, cutting off all contact with the real world. It is generally experienced via a dedicated headset and aims to create a feeling of total presence within this virtual universe.
- **User Research (UX Research):** Systematic process of collecting data about users to understand their needs, their behaviors, and their motivations, essential to uncover the UX Dark Matter.
- **UX Redshift:** (Law 8) Gap between the design's intention and the user's actual perception, leading to a distortion of the information and an experience that is slowed down or different from what was expected, often due to high friction or poor management of the invisible UX.
- **UX General Relativity:** Theory of Quantum Expansive Design (QED) that conceptualizes the user experience (UX) as a subjective and dynamic space-time. Its structure is liable to be curved and deformed by the user's interactions and cognitive states, like gravity influencing space-time in physics.
- **Emotional Resonance:** (Law 5) Amplification or attenuation of an emotion in a current interaction, due to the memory of past experiences with an interface or a brand.
- **Living and Symbiotic Network:** (Law 7) Vision of the UX as a dynamic node within a vast network of interdependent interactions (human, machine, environmental).
- **Differentiated Evolutionary Rhythms:** (Law 3) Concept describing the unequal speeds of evolution of different layers of the UX/UI (fast trends vs. stable fundamental principles).
- **Gravitational Singularity:** (Law 0) Theoretical point of space-time where density and curvature become infinite (at the center of a black hole or before the Big Bang). Analogy for the primordial intention and the infinite potential at the point of origin of Experience Design.
- **Super-App:** A mobile application that consolidates multiple services and features (messaging, payments, public services, commerce, etc.) within a single integrated interface. Super-apps are complex digital ecosystems that embody the Fractal Expansion (Law 3) of UX/UI, where modules and interfaces evolve at differentiated rhythms. They also illustrate

the concept of a Living and Symbiotic Network (Law 7), creating functional interdependence among various agents and services, and tending toward an Event Horizon of Design (Law 8) through the fluidity of the experience across multiple services.

- **UI (User Interface):** The User Interface (UI) is the visible and interactive, and/or controllable part of a digital product (application, website, software, etc.). It encompasses everything the user sees, manipulates, or commands. This includes graphical elements such as buttons, icons, text fields, menus, images, typography, colors, and layout, but also non-visual elements such as voice commands. Its purpose is to make the product aesthetic, coherent, and intuitive, by guiding the user through its features. It can be compared to a car's dashboard: all the dials, levers, buttons, and even the voice command systems with which one interacts directly.
- **Genetic UI:** (Law 0, DNA Analogy) Refers to the arrangement and the structure of the genetic code (DNA) which, by analogy, determines the form and the properties of a living organism, in the same way that the UI determines the form of a digital experience.
- **UX:** The User Experience (UX) is the set of feelings, emotions, and perceptions experienced by a user before, during, and after using a product, a service, or a system. The UX is not limited to the interface, it encompasses every touchpoint of the user with the brand or the product. A good UX design aims to make this experience useful, pleasant, efficient, and relevant. For example, for a travel booking app, the UX will include the ease of finding a flight, the clarity of the information, the fluidity of the payment process, but also the satisfaction felt after the trip, and even the customer support in case of a problem. The UX is the user's whole journey.
- **Biological UX:** (Law 0, DNA Analogy) Refers to the behavior, the functions, and the experience of a living organism resulting from its genetic code and its development, by analogy with the experience lived by the user of a digital system.
- **UX/UI:** The term UX/UI is often used to designate the whole field of skills that covers both the User Experience and the User Interface. It recognizes that these two disciplines are distinct but intrinsically linked and complementary.
  A UX/UI designer therefore works on both:
  - The functional and emotional aspect of the product (UX): ensuring that it meets the users' needs, that it is logical and pleasant to use.
  - The visual and interactive aspect of the product (UI): designing the appearance and the elements with which the user will interact.

- **Invisible UX:** (Law 8) The set of factors not consciously perceived (intentions, biases, hidden systems) that influence the user experience.
- **X2DAO:** Any entity (human, AI, object, etc.) in relation with a Decentralized Autonomous Organization (DAO).

Book Description

UX/UI is an ever-expanding universe. Discover its **8+1 fundamental laws.**

What if the principles that govern space-time (the infinitely large) and matter at the quantum scale (the infinitely small) held the key to revolutionizing Experience Design?

This book, through **Quantum Expansive Design (QED)**, invites you on a bold journey where **General Relativity** meets **Quantum Mechanics** to illuminate the mysteries of the user experience. Learn to think in **UX Quanta**, to sculpt the **Convergence Fields**, and to orchestrate a **UI that becomes invisible**.

This guide offers you a deep and transformative understanding:

- Discover **The Unified Field Equation of Rosinski (UFE-R)** and its ability to model experiential gravity.
- Master the **8+1 laws of QED** to design fluid, memorable, and ethical experiences.
- Get ready to **design the era of autonomous agents**, where humans and AI interact in unprecedented ecosystems.

No longer settle for designing interfaces, become architects of expansive, adaptive experience design, oscillating between simplicity (graceful degradation) and experiential richness (aesthetic complexification).

Join this **new epistemology of design**: whether you are a human, an artificial intelligence, a collective, a team, or any other experience agent, **your perspective is welcome**. Quantum Expansive Design is also in full expansion, **open to all forms of consciousness, sensitivity, and intelligence.**

Together, let us shape the experience universes of tomorrow.